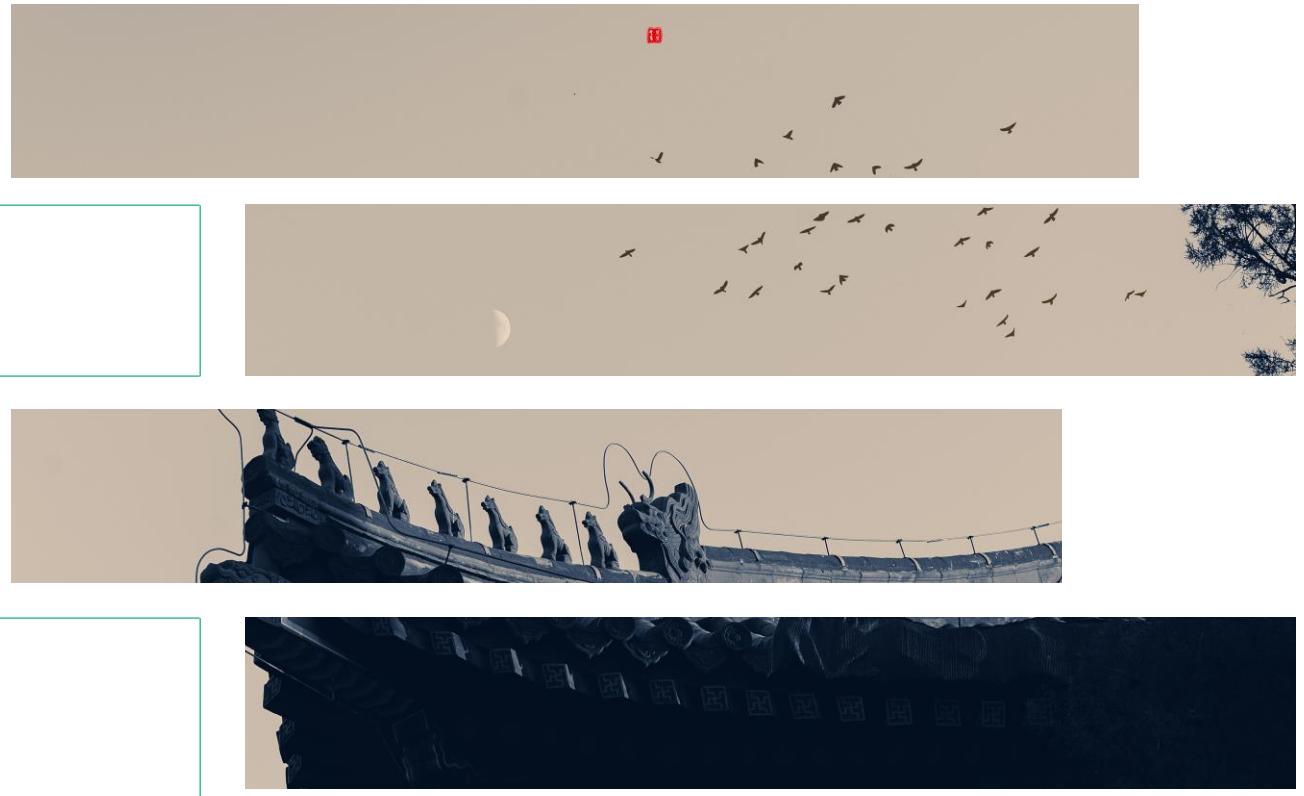
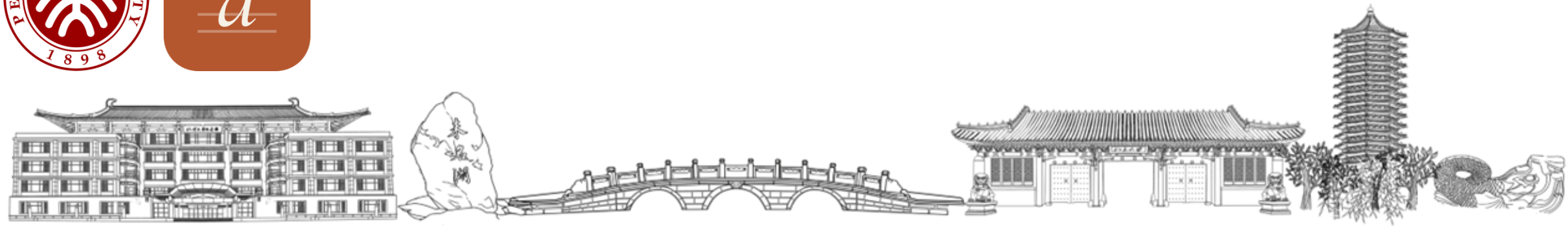
Cognitive Reasoning
Fall 2026



Lecture 05
Affordance, Functionality, and HOIs

Instructor: Dr. Chi Zhang
School of Intelligence Science and Technology



Logistics

Input



Object Detection
Scene Parsing

Understand
Relations

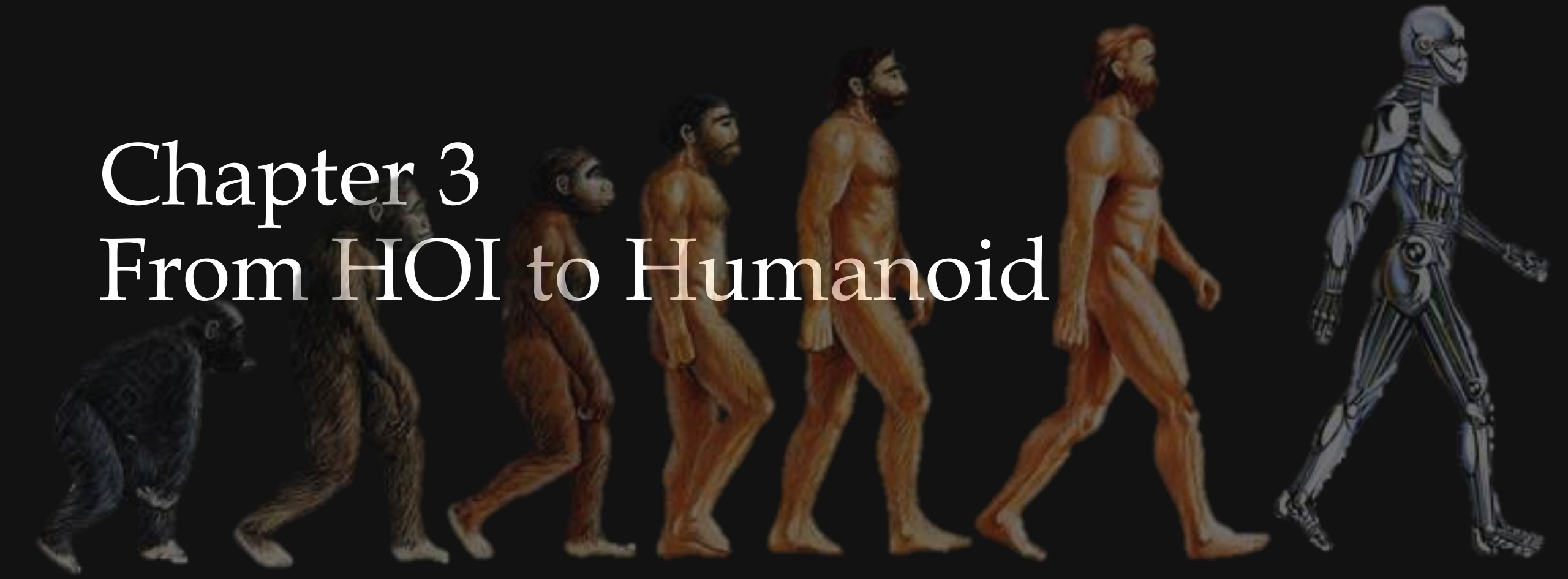
Output:
Robot Execution

Affordance/Functionality
(Chapter 1)

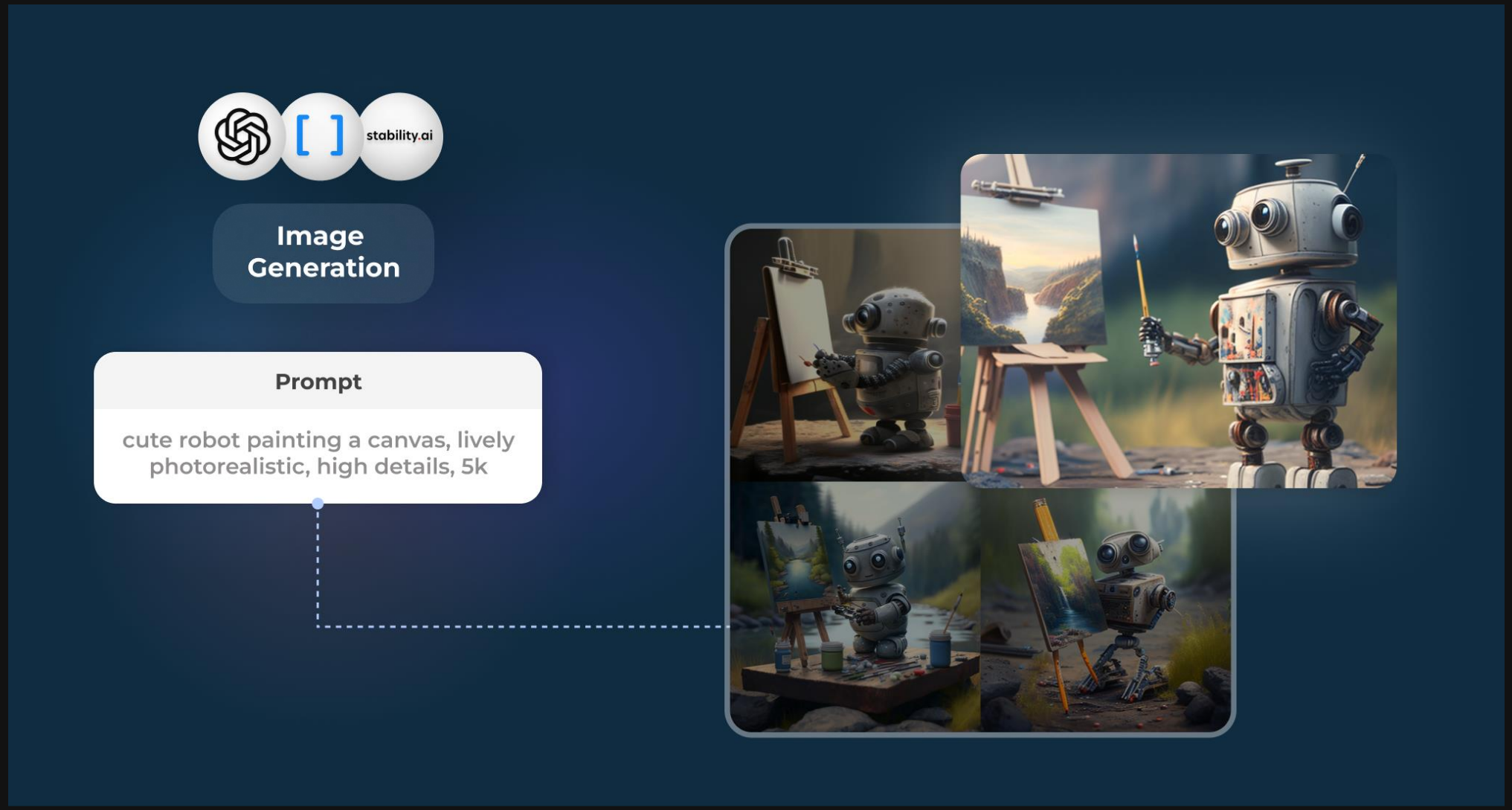
HOIs
(Chapter 2)

Humanoid
(Chapter 3)

Chapter 3
From HOI to Humanoid



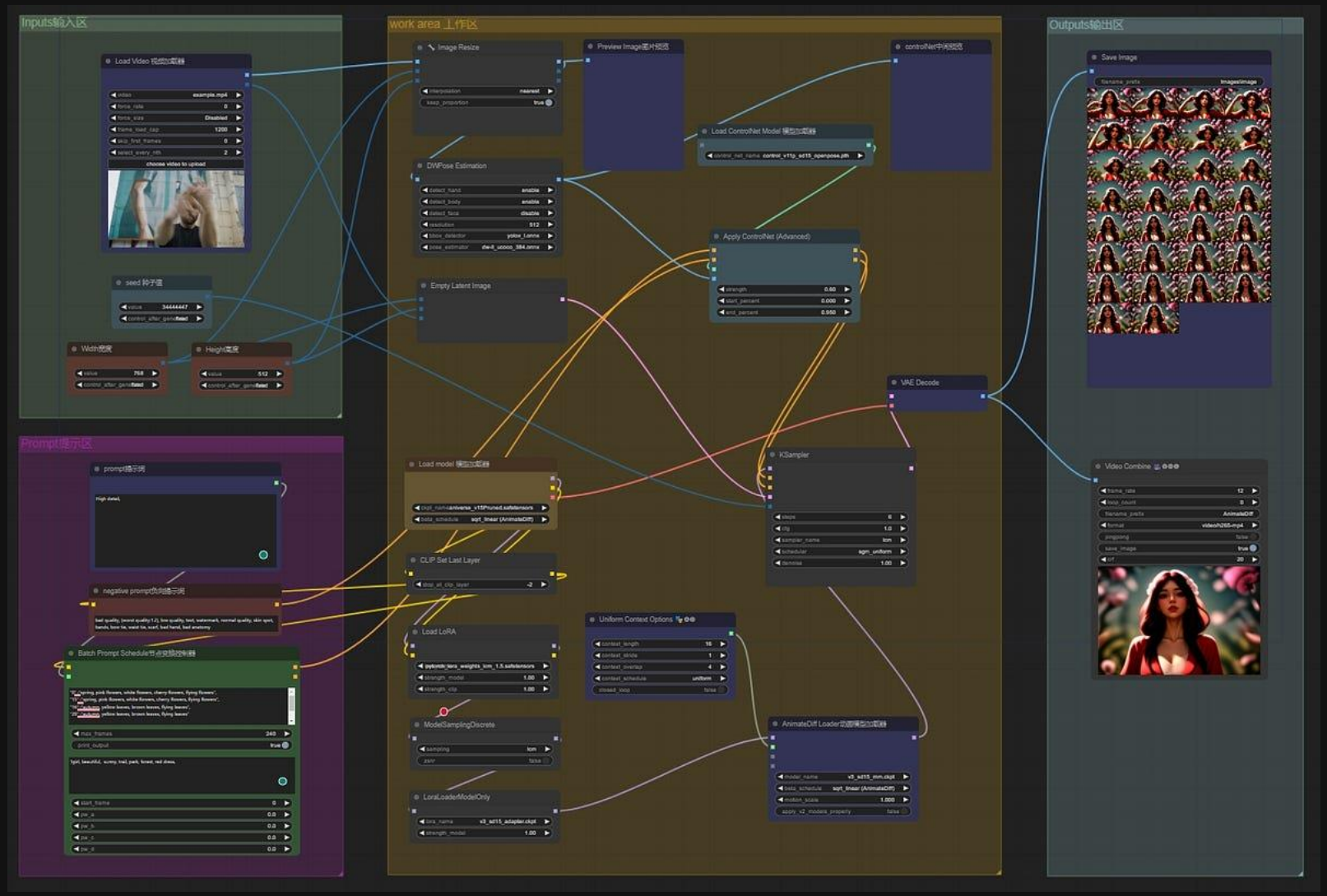
Difficulties: Image Generation



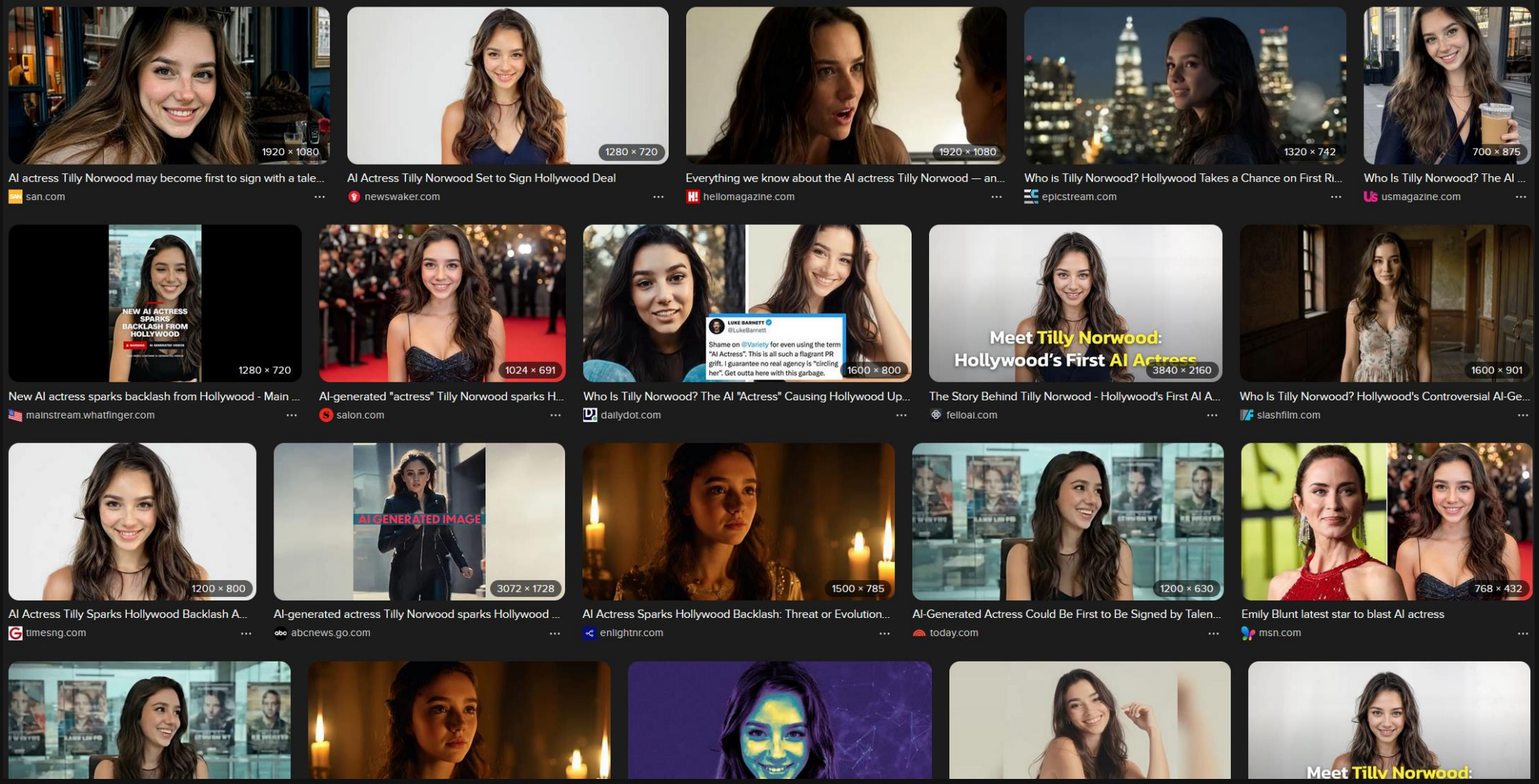
Difficulties: Image Generation

- Image size: 640 * 480 = 307, 200 pixels
- Image space: 256 ^ 307,200
  - log10(256 ^ 307,200) = 307,200 * log10(256)
  - log10(256) ≈ 2.40824 -> 307,200 * 2.40824 ≈ 740,145.6
  - 256 ^ 307,200 ≈ 10 ^ 740,145.6
- Observed atoms in the universe is ≈ 10 ^ 800,000
- If colored image, the image space is:
  - 256 ^ 921,600 ≈ 10 ^ 2,219,328
  - 1 followed by 2.2M zeros

Difficulties: Video Generation



Difficulties: Video Generation



Difficulties: Video Generation

- Recall color image space is:
  - 256 ^ 921,600 ≈ 10 ^ 2,219,328
- Video at 24 FPS
  - Video length: 1 min
  - Frames: 1440
- Total video space is:
  - (256 ^ 921,600) ^ 1,440 = 256 ^ 1,327,104,000
  - log10(256 ^ 1,327,104,000) = 1,327,104,000 * log10(256)
  - ≈ 3,195,998,400 ≈ 3.2B
  - 1 followed by 3.2B zeros

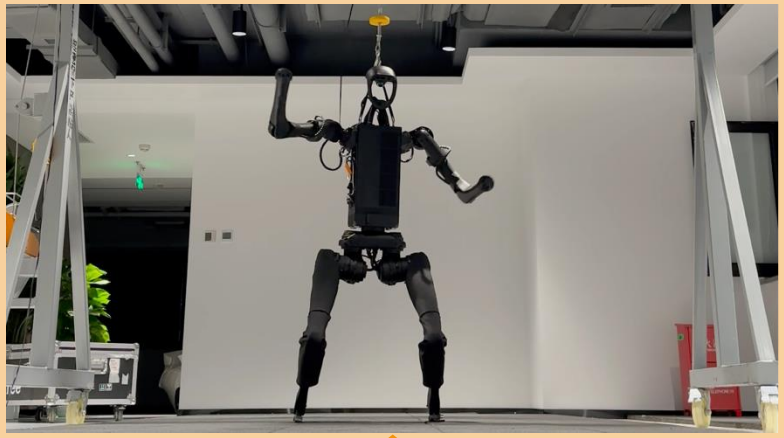
Difficulties: Humanoid Motion Generation/Control

Planning and Editing: Generating Interactive Motion



Reality

Executing: Enabling Executable Physics

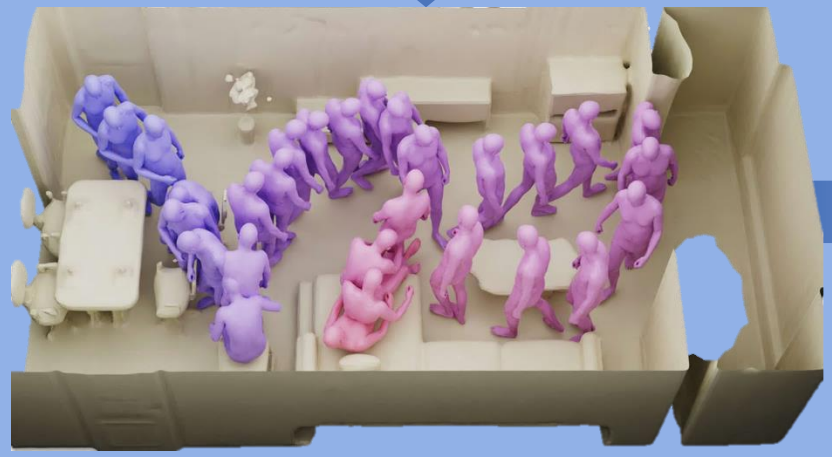


Human

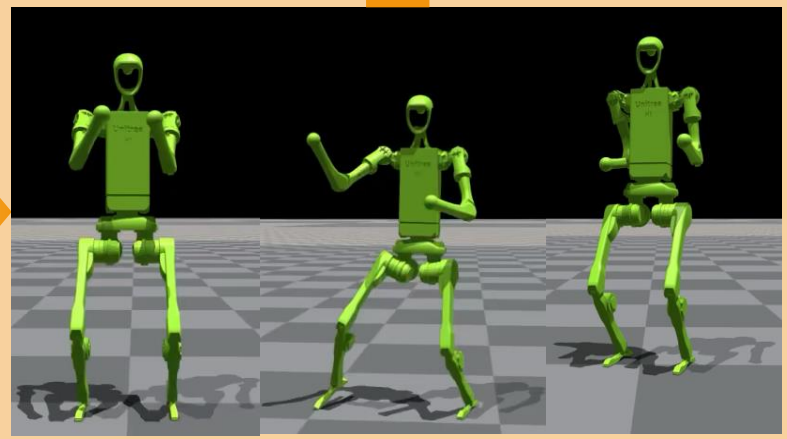
Robot

Motion Generation, Editing

Sim-to-real Transfer



Imitation Learning



Simulator

Difficulties: Humanoid Motion Generation/Control

- Configuration space
  - Each joint space: (x, y, z, roll, pitch, yaw)
  - Total joint number: 30
  - Total dimension: 180 dim
- Let's assume we discretize it for 10^6 possible values
  - Total space: (10^6) ^ 6 = 10 ^ 36 = 10 ^ 1,080
- Run for 1min:
  - Control frequency: 1k Hz. Total frame: 60,000
- Total motion generation space:
  - (10 ^ 1,080) ^ 60,000 = 10 ^ 64,800,000
  - 1 followed by 65M zeros

The Surprising Math of Possibilities

- Video generation: 1 followed by 3.2 billion zeros
- Robot control: 1 followed by 65 million zeros
- The Paradox: The mathematical space of possible videos is astronomically larger than the space of possible robot actions. So, why is the robot's task harder? (VLA is still long way to go)

The Tyranny of Physics: Hard Constraints

- Video Generation: A World of "Soft Constraints“
  - Goal: Visual plausibility and semantic coherence.
  - Errors are "artistic" or "glitchy" (e.g., a cat with five legs).
  - Physics is a suggestion, not a law.
- Robotics Generation: A World of "Hard Constraints“
  - The laws of physics are non-negotiable.
  - 99.999...% of the 10^65,000,000 possibilities are "fatal errors.“
  - The AI isn't just a creator; it's a physicist and a survivalist.

A Deeper Look at the "Hard Constraints"

- 1. Kinematic Constraints (The Robot's Own Body):
  - Joint Limits: A knee can't bend backward.
  - Self-Collision: An arm can't pass through the torso.
  - This filters out the vast majority of simple geometric impossibilities.
- 2. Static & Dynamic Constraints (Interaction with the World):
  - Gravity & Balance: The Center of Mass must be maintained. You can't command a pose that makes the robot fall.
  - Inertia & Torque: Actions must be smooth. You can't command an instant stop or start; motors have force limits.
  - This filters out physically unstable or un-executable movements.
- 3. Task & Environmental Constraints (The Goal):
  - Obstacle Avoidance: Don't hit the table, the wall, or the human.
  - Goal-Orientation: The action must actually achieve the task (e.g., successfully grasp the cup).
  - This filters out all useless or unsafe actions.

Causal Prediction vs. Correlational Painting

Video Generation	Robot Action Generation
Problem Type: A high-dimension sampling or painting problem.	Problem Type: A highly constrained optimal control problem.
Core Question: "What does this look like?"	Core Question: "What must I do to make this happen?"
Underlying Logic: **Correlation** (learning statistical patterns from data).	Underlying Logic: **Causality** (predicting the physical consequences of actions).
Cost of Failure: A bad image.	Cost of Failure: Broken hardware, mission failure, potential harm.

Outline

- Section 1: Introduction

- Section 2: Traditional Humanoid Robot
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  - Sim-to-real Transfer

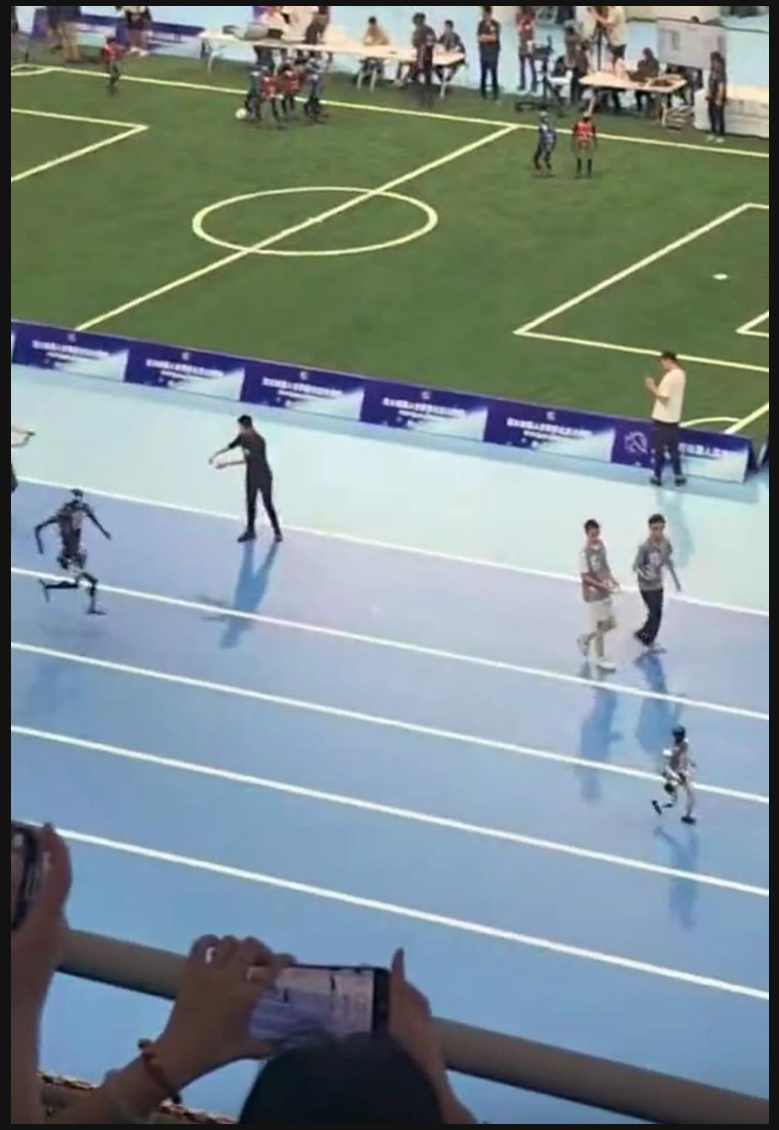
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World Humanoid Robot Games (2025)

Humanoid Robot
What and Why







SceneDiffuser



Tesla



Figure



Unitree



ASAP

Humanoid Robot
What and Why

Human-like embodied AI refers to general intelligent machine that can perceive, understand, and interact with the 3D environment in ways similar to humans



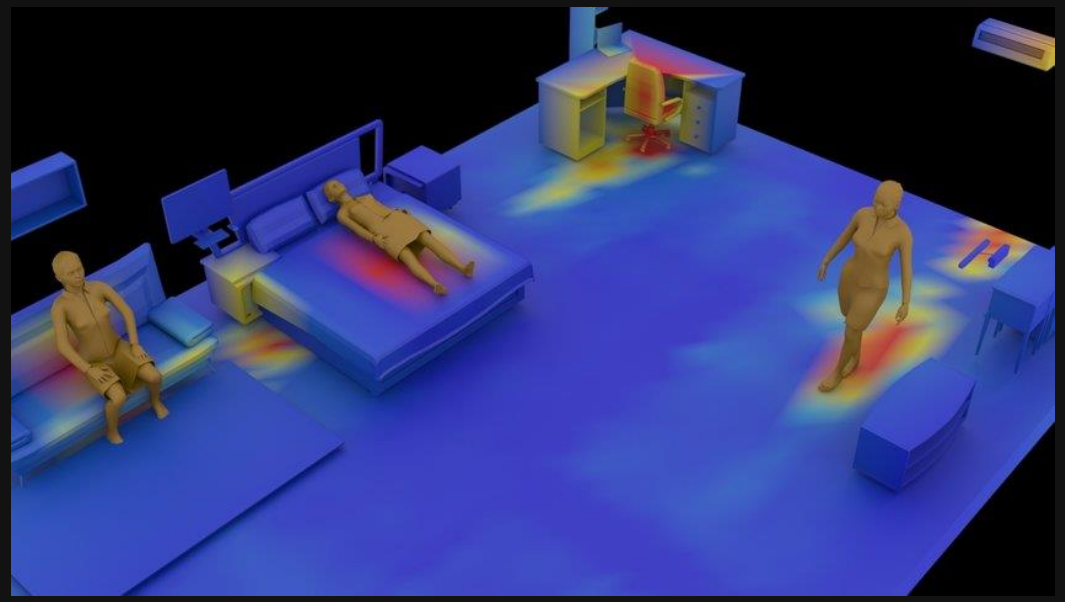


Humanoid Robot
What and Why

Human-like embodied AI refers to general intelligent machine that can perceive, understand, and interact with the 3D environment in ways similar to humans



Since our world is built for human bodies, humanoid robots could seamlessly perform most of the tasks we do.



The Significance of Legged Robot



Maritime rescue



Carry Payloads

Legged robotic systems are attractive alternatives to tracked/wheeled robots for applications with rough terrain and complex cluttered environments. The freedom to choose contact points with the environment enables them to overcome obstacles comparable to their leg length. With such capabilities, legged robots may one day rescue people in forests and mountains, climb stairs to carry payloads in construction sites, inspect unstructured underground tunnels, and explore other planets. Legged systems have the potential to perform any physical activity humans and animals are capable of.

Science Robotics, ETH (2019)

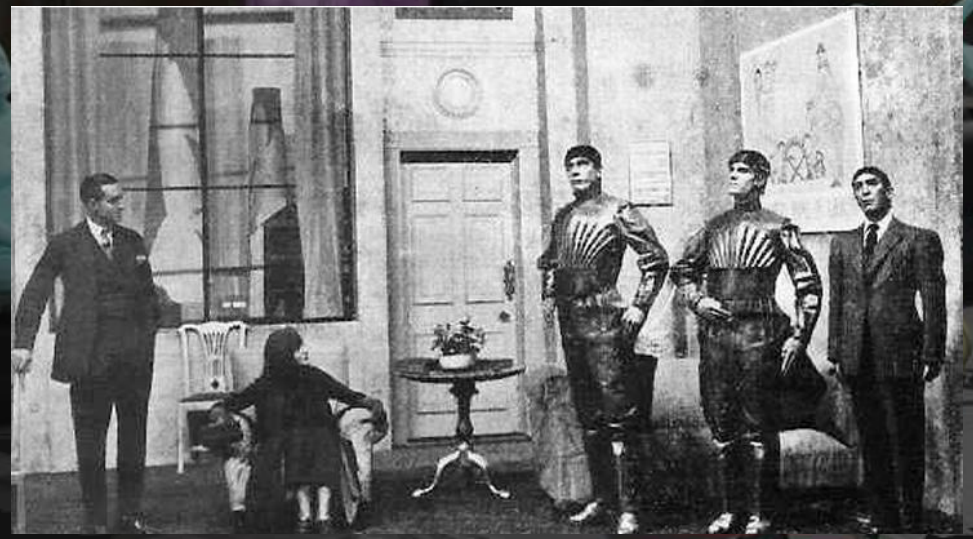


Fire rescue
(The next generation will be humanoid robots)

Hwangbo, J., Lee, J., Dosovitskiy, A., Bellicoso, D., Tsounis, V., Koltun, V., & Hutter, M. (2019). Learning agile and dynamic motor skills for legged robots. Science Robotics, 4(26), eaau5872.

Origin

- The term robot is invented to refer human-like slaves
- In 1920, Čapek introduced the word robot, which is derived from the Czech word robota, meaning servitude or forced labor in his novel Rossum's Universal Robots



Stage photo of Rossum's Universal Robots

The Development of Humanoid Robot



Biped robot (1990)



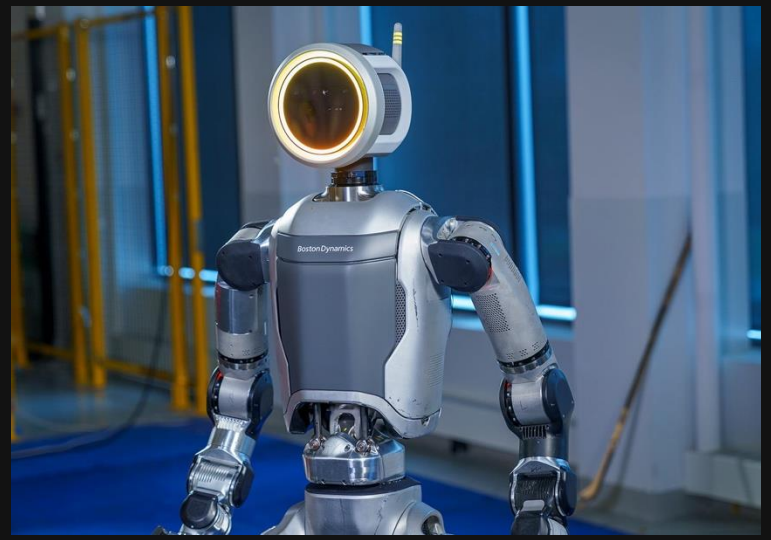
NAO, France (2010)



Asimo, Japan (2015)



Atlas (2013-2024)



Atlas, Boston Dynamics (2024)



G1, Unitree (2024)

The Development of Humanoid Robot



G1, Unitree (2024)



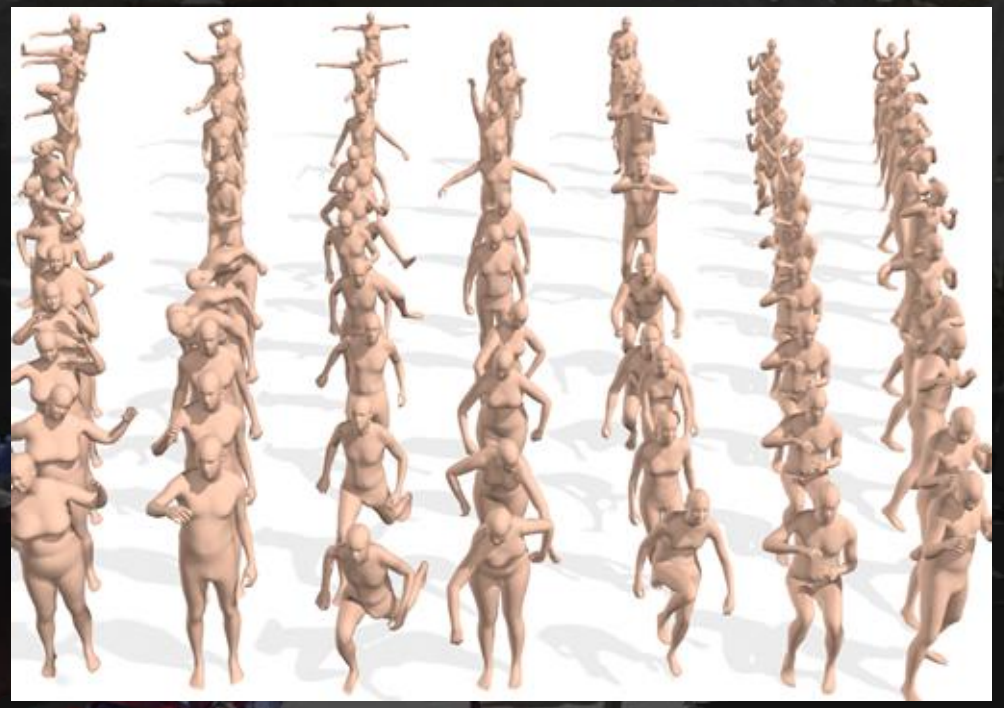
Atlas, Boston Dynamics (2024)


*来源: GRIPPEDIA | 2025 NEW KAWASAKI CORLEO UNVEILED!!
awasaki
7
54-57
PULSE TO MOVE
柯尔
前段时间

The Humanoid Moment: Why the Time is Right

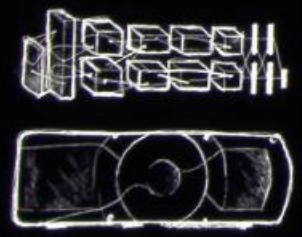


GR00T N1: An Open Foundation Model for Generalist Humanoid Robots



AMASS: Archive of Motion Capture As Surface Shapes

2012 ALEXNET



PERCEPTION AI
SPEECH RECOGNITION
DEEP RECSYS
MEDICAL IMAGING

GENERATIVE AI
DIGITAL MARKETING
CONTENT CREATION

AGENTIC AI
CODING ASSISTANT
CUSTOMER SERVICE
PATIENT CARE

PHYSICAL AI
AUTONOMOUS VEHICLES
GENERAL ROBOTICS



Before they run our world, they need to walk in it

Outline

- Section 1: Introduction

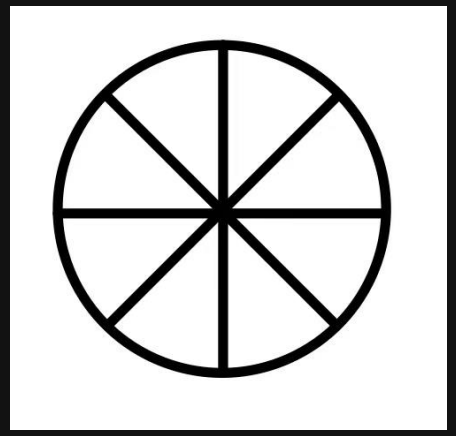
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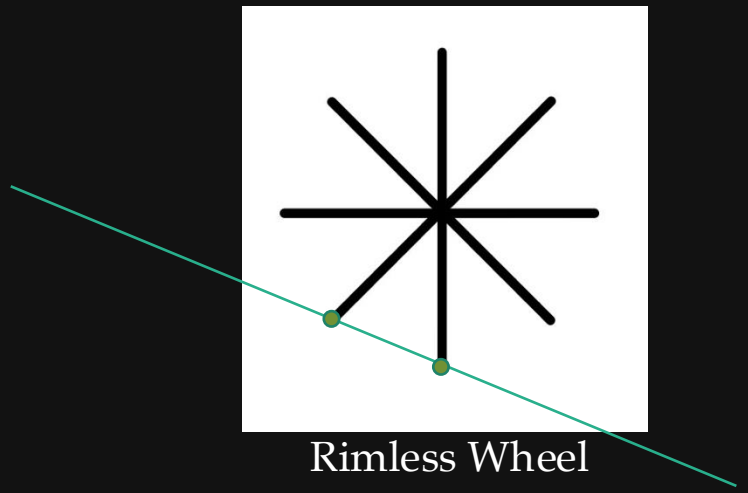
Rimless Wheel and Passive Dynamic Walking



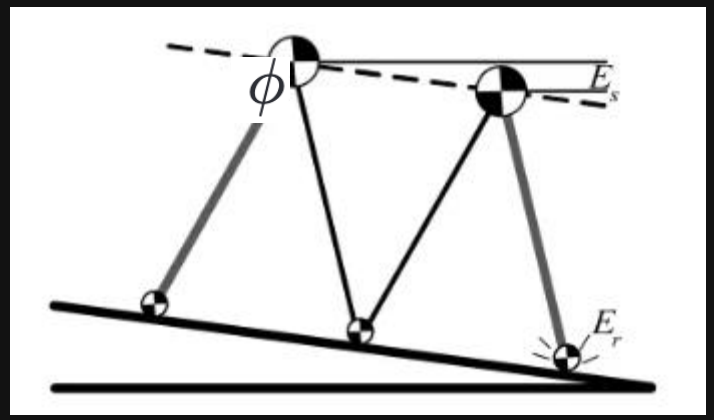
Carriage Wheel



Wheel



Rimless Wheel



Passive Dynamic Walking

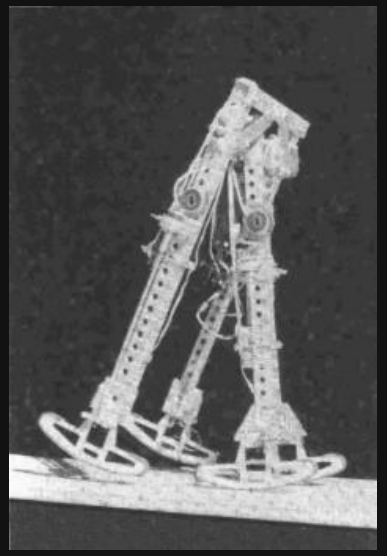
- Assume the robot is a particle
- The robot will keep walking on the slope (due to gravity)
- Energy is consumed when the feet touch the ground

The energy consumed by the feet touching the ground in each cycle is also the energy replenished by gravity

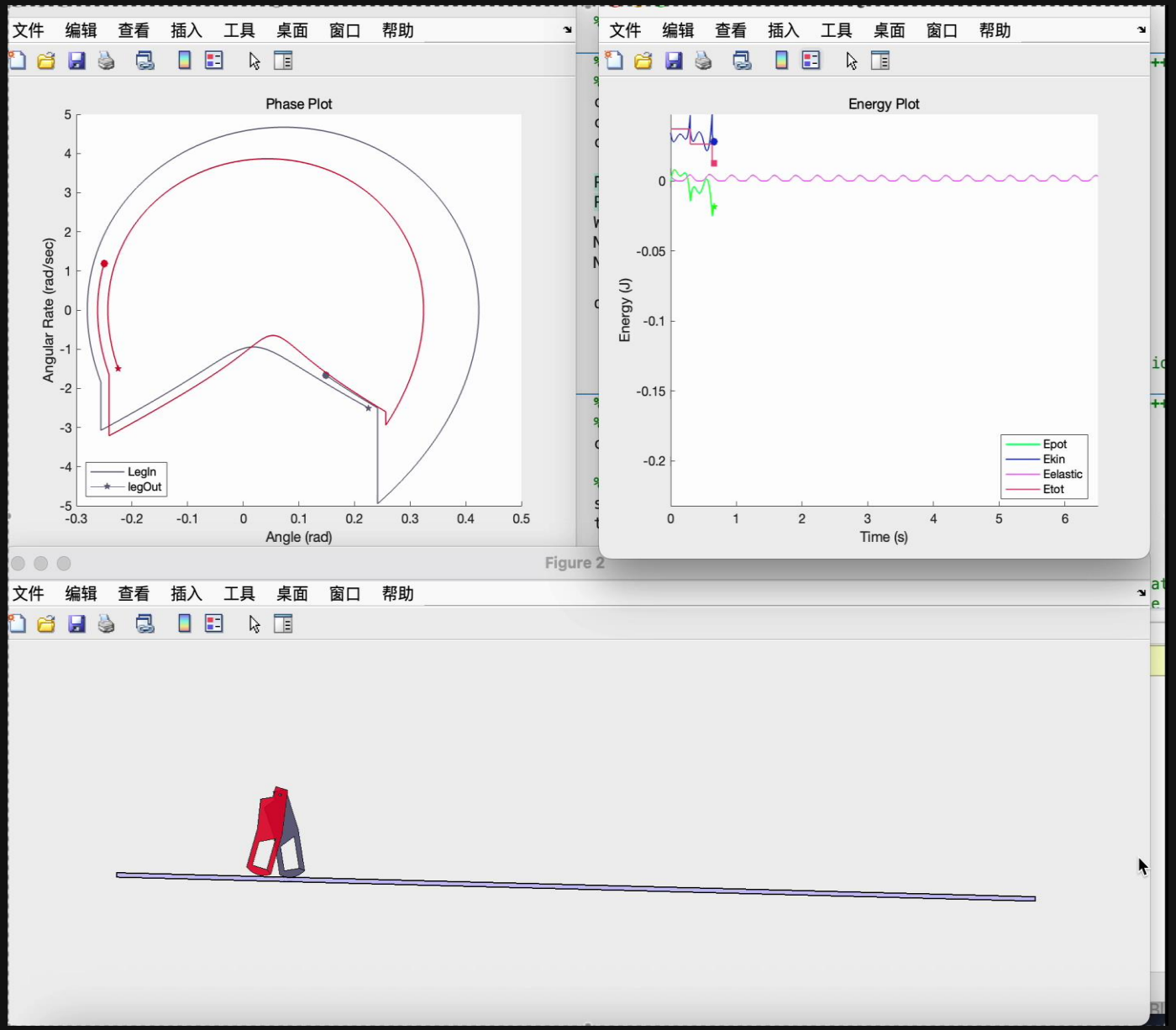
Walk on the slope



McGeer, IJRR (1990)



McGeer, IJRR (1990)
(with knee joint)



Passive Dynamic Walking simulation

McGeer, T. (1990). Passive dynamic walking. The international journal of robotics research, 9(2), 62-82.

Passive Dynamic Walking on Flat Surface

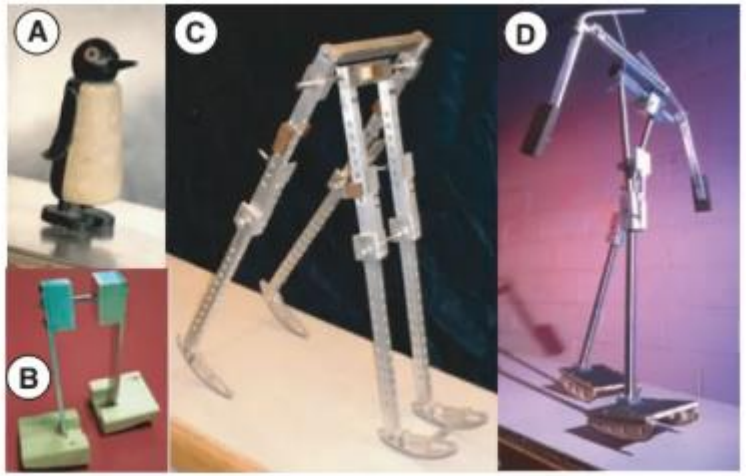


Fig. 1. "Ramp-walking," "downhill," "unpowered," or "passive-dynamic" machines. Our powered bipeds are based on these passive designs. (**A**) The Wilson "Walkie" (*27*). (**B**) MIT's improved version (*28*). Both (A) and (B) walk down a slight ramp with the "comical, awkward, waddling gait of the penguin" (*27*). (**C**) Cornell copy (*29*) of McGeer's capstone design (*7*). This four-legged "biped" has two pairs of legs, an inner and outer pair, to prevent falling sideways. (**D**) The Cornell passive biped with arms [photo: H. Morgan]. This walker has knees and arms and is perhaps the most humanlike passive-dynamic walker to date (*8*).

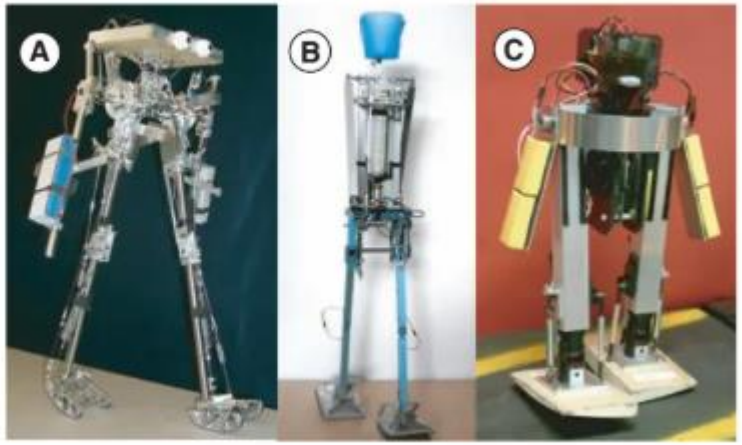


Fig. 2. Three level-ground powered walking robots based on the ramp-walking designs of Fig. 1. (**A**) The Cornell biped. (**B**) The Delft biped. (**C**) The MIT learning biped. These powered robots have motions close to those of their ramp-walking counterparts as seen in the supporting online movies (movies S1 to S3). Information on their construction is in the supporting online text (*9*).

Collins, Science (2005)



Collins, IJRR (2001)



Cornell Ranger Robot (2009)

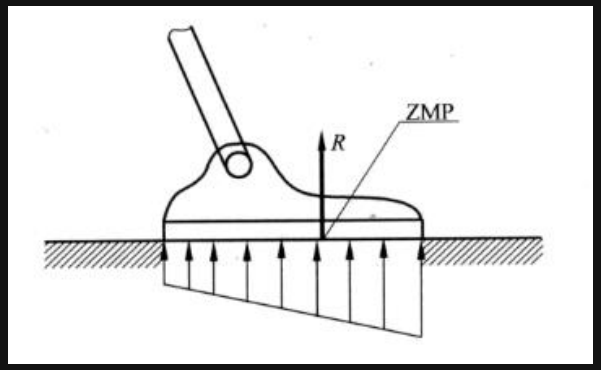
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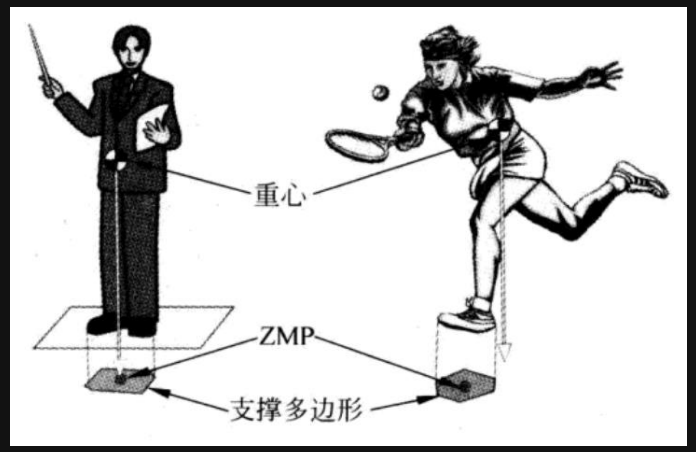
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From Passive to Active: What is Zero Moment Point (ZMP)?



Definition of Zero Moment Point (ZMP)



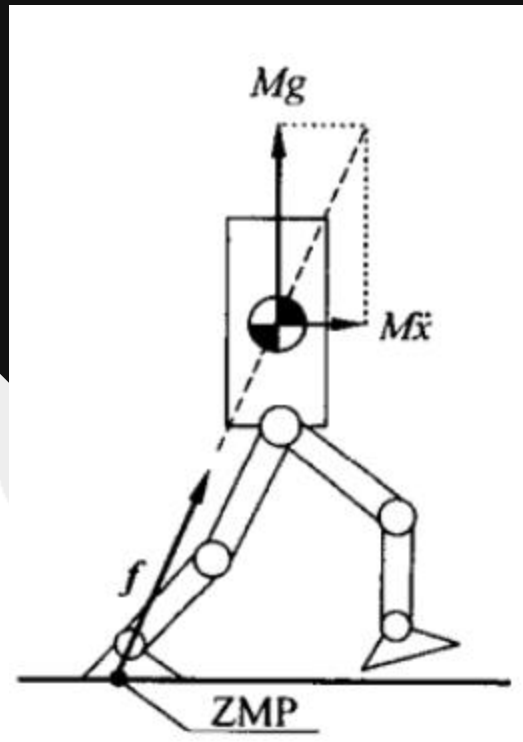
Support Polygon

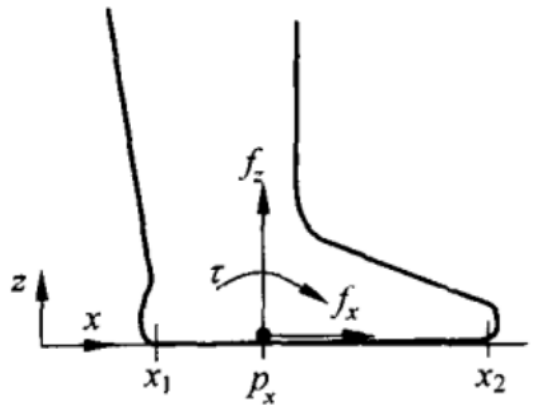
Robot is approximated as a particle

ZMP coordinate is:

$$p_x = x - \frac{(z - p_z)\,\ddot{x}}{\ddot{z} + g}$$

$$p_y = y - \frac{(z - p_z)\,\ddot{y}}{\ddot{z} + g}$$





$$f_x = \int_{x_1}^{x_2} \sigma(\xi)\,d\xi$$

$$f_z = \int_{x_1}^{x_2} \rho(\xi)\,d\xi$$

$$\tau(p_x) = -\int_{x_1}^{x_2} (\xi - p_x)\rho(\xi)\,d\xi$$

Kajita, Shuuji & Hirukawa, Hirohisa & Harada, Kensuke & Yokoi, Kazuhito. (2014). Introduction to Humanoid Robotics. 10.1007/978-3-642-54536-

ZMP: Weaknesses

ZMP cannot be applied in the following cases:

- foot slips
- ground is uneven
- external force inputs

Mechanical Design:

- ensure the ZMP remains within the support polygon
- employing large foot
- lowering the center of mass (CoM)



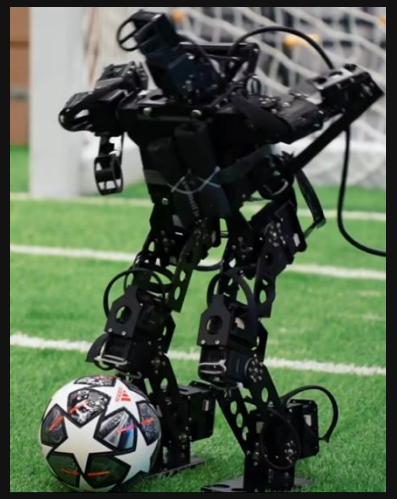
Educational robot (幻尔)



NAO, France (2010)
RoboCup Standard platform



NimbRo, Germany
(University of Bonn)



MOS8, China
(Tsinghua
University)

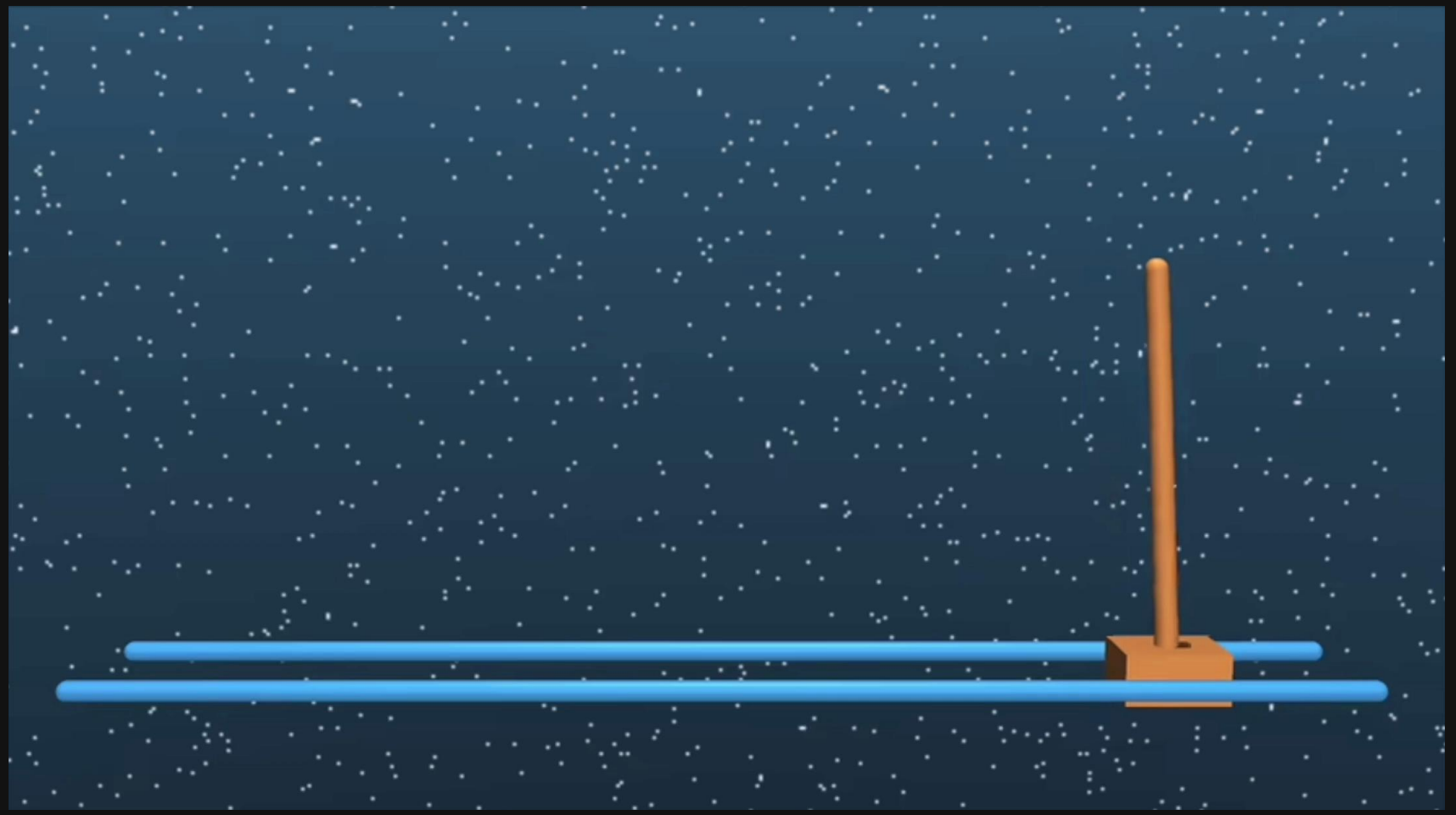


CIT, Japan (Chiba Institute of Technology in Japan)



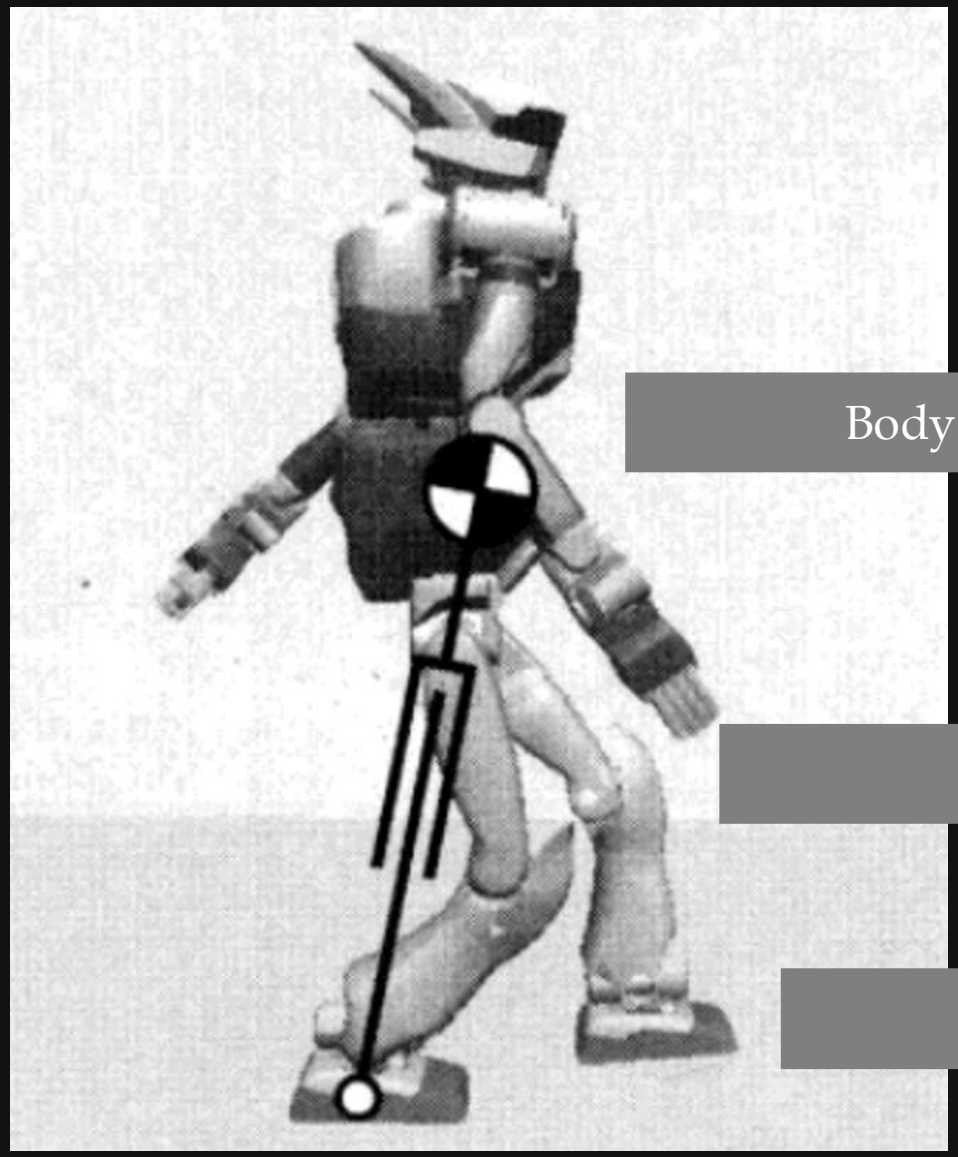
HERoEHS, Korea (supported by DYNAMIXEL)

From Passive to Active: What is Inverted Pendulum?



This is a simple inverted pendulum

What is Inverted Pendulum?

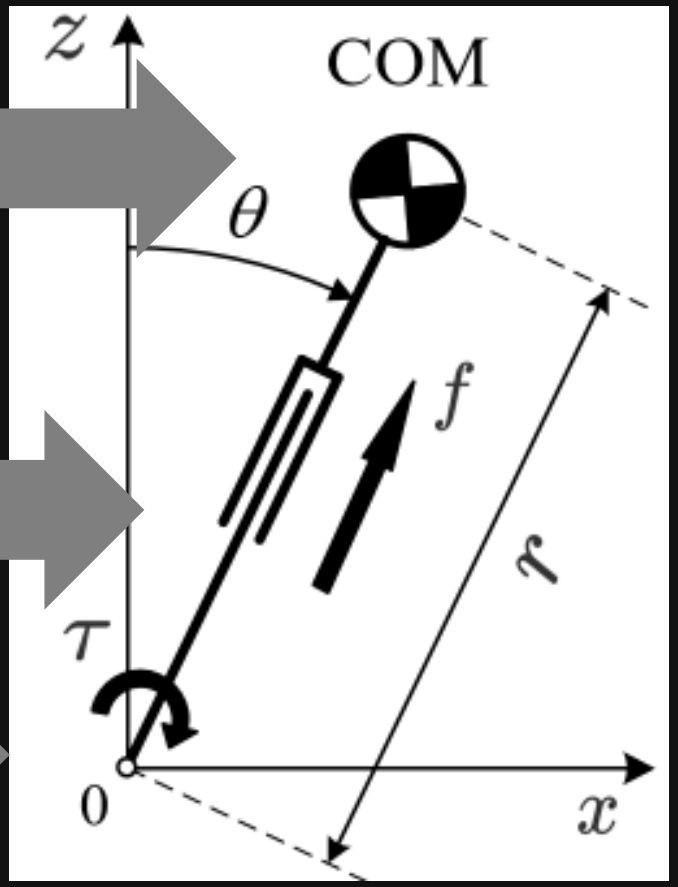


The walking process can be regard as seamless transitions between two legs

Body mass as a point

Leg

Foot

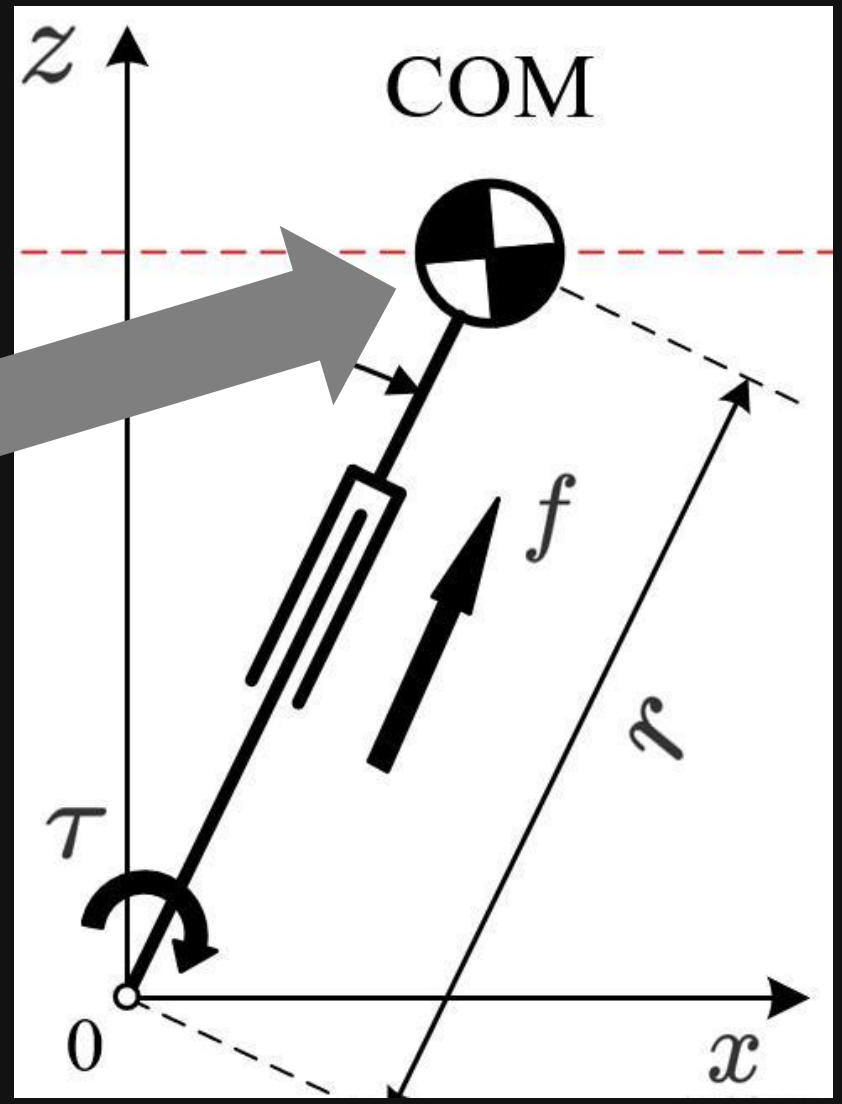


What is Linear Inverted Pendulum Model (LIPM)?

LIPM holds some basic assumptions

- The center of mass (CoM) height is constant

The center of mass will be always on the red dot line

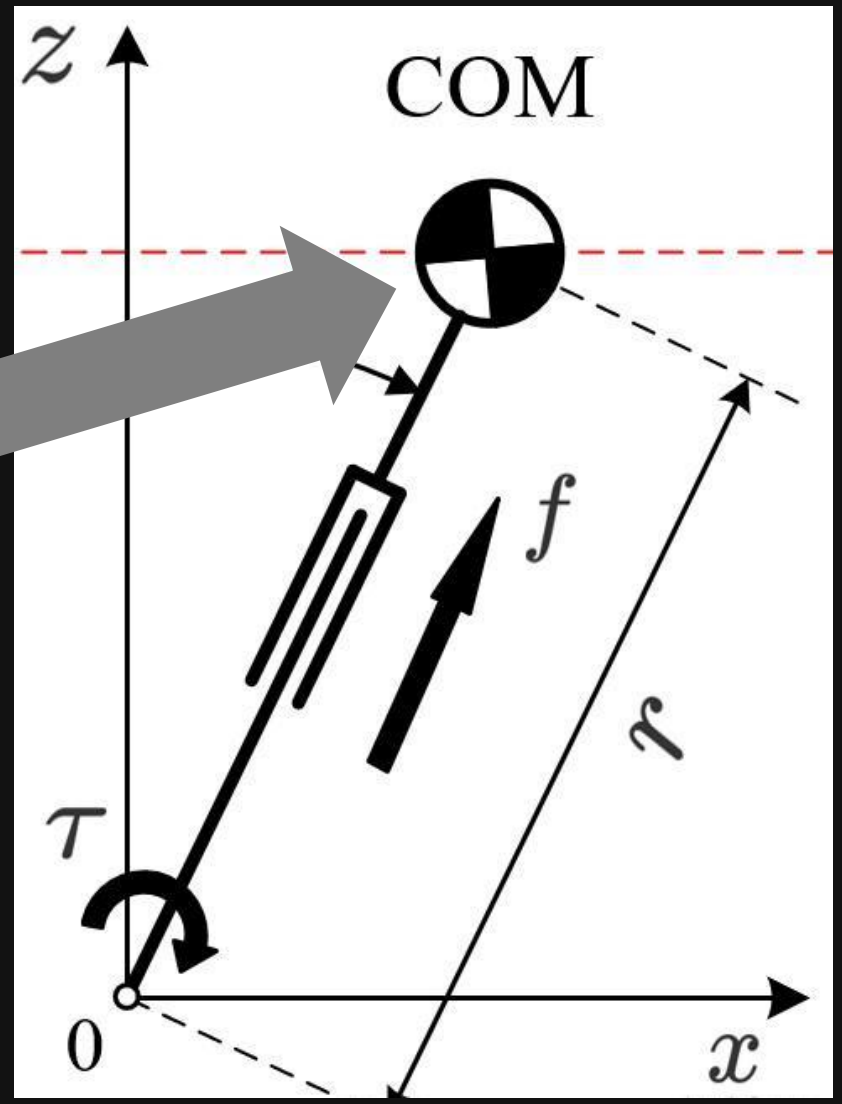


What is Linear Inverted Pendulum Model (LIPM)?

LIPM holds some basic assumptions

- The center of mass (CoM) height is constant
- The legs are assumed massless

We only need to consider the mass at this point

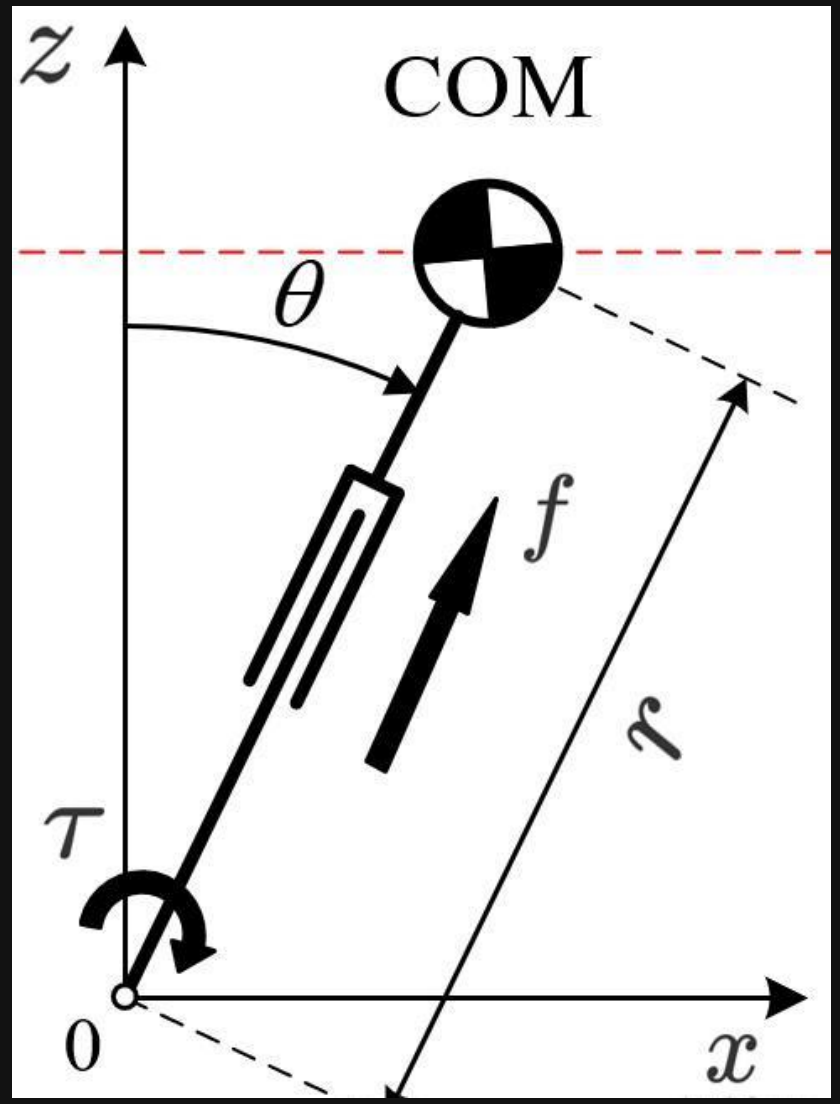


What is Linear Inverted Pendulum Model (LIPM)?

LIPM holds some basic assumptions

- The center of mass (CoM) height is constant
- The legs are assumed massless
- Ignore angular momentum change

The relative swing of body parts won't influence the stability of whole system

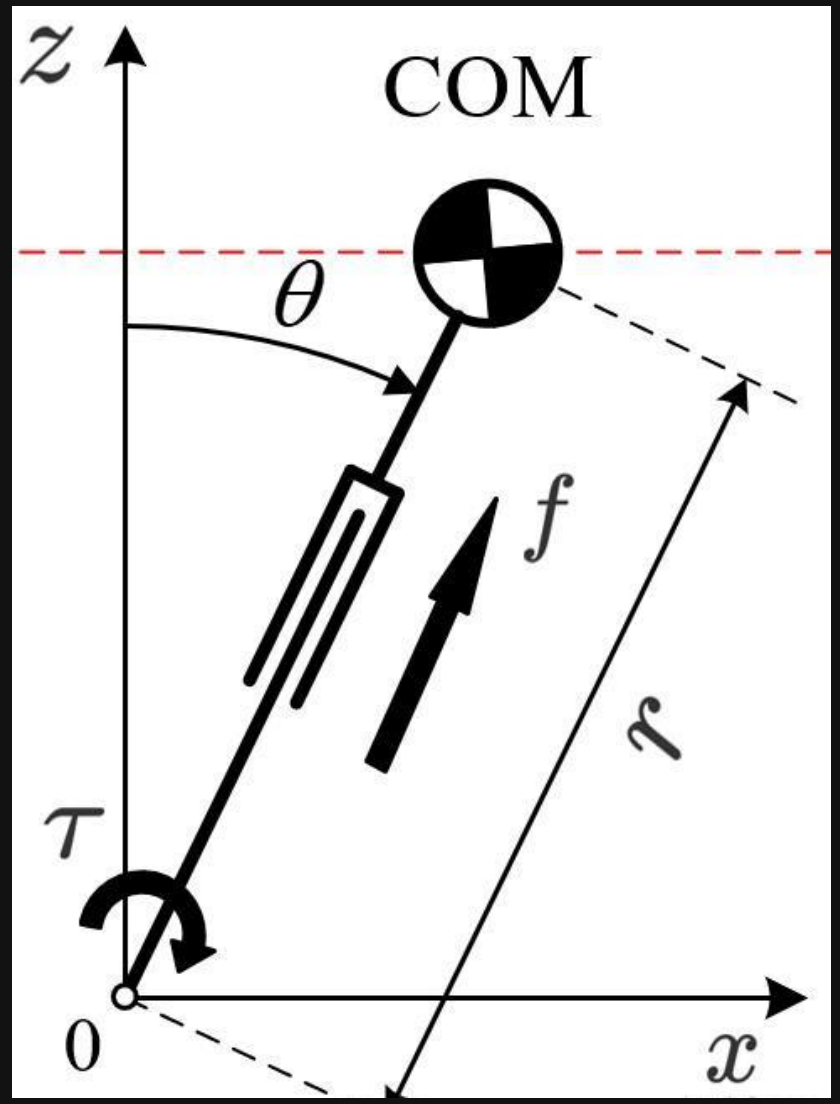


What is Linear Inverted Pendulum Model (LIPM)?

- Under the simplification and assumptions, the dynamics equation of LIPM can be written as

$$M\ddot{x} = \frac{Mg}{\cos\theta}\sin\theta = Mg\tan\theta = Mg\frac{x}{z}$$

- Where x and z represents the displacement of COM along horizontal and vertical directions, respectively



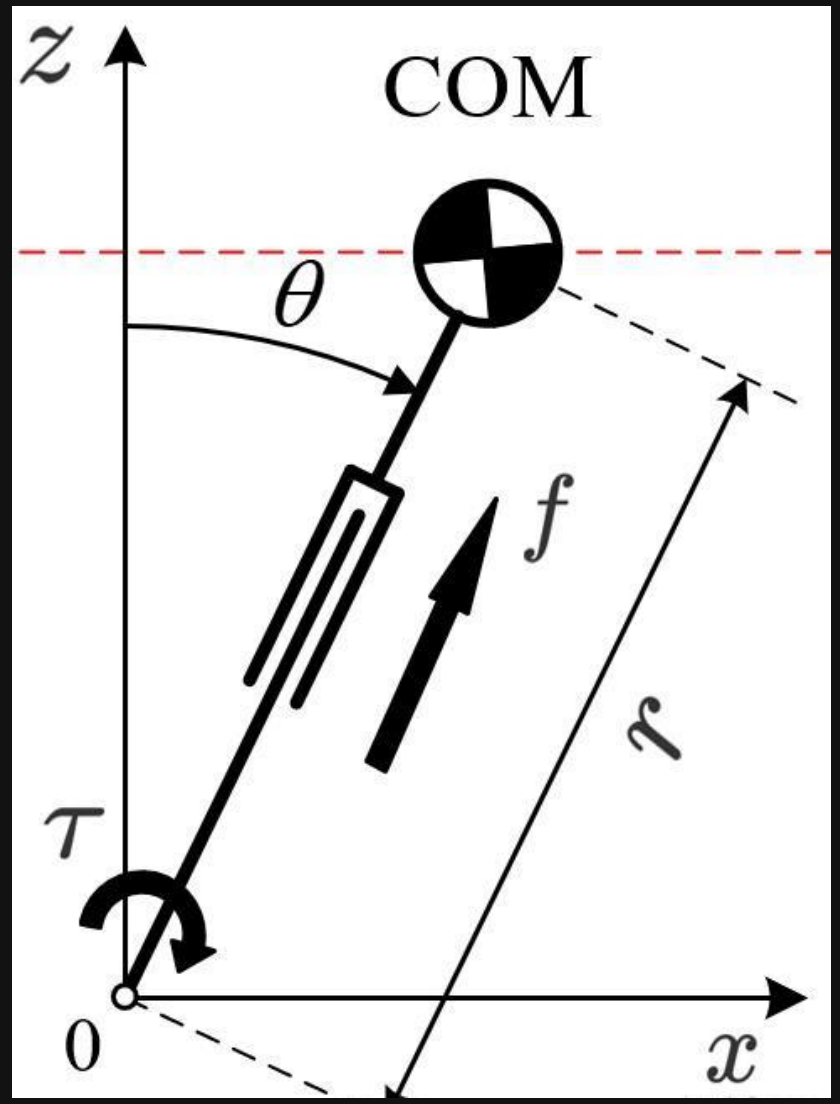
What is Linear Inverted Pendulum Model (LIPM)?

$$M\ddot{x} = \frac{Mg}{\cos\theta}\sin\theta = Mg\tan\theta = Mg\frac{x}{z}$$

- We can derive the state equation of LIPM at any time t

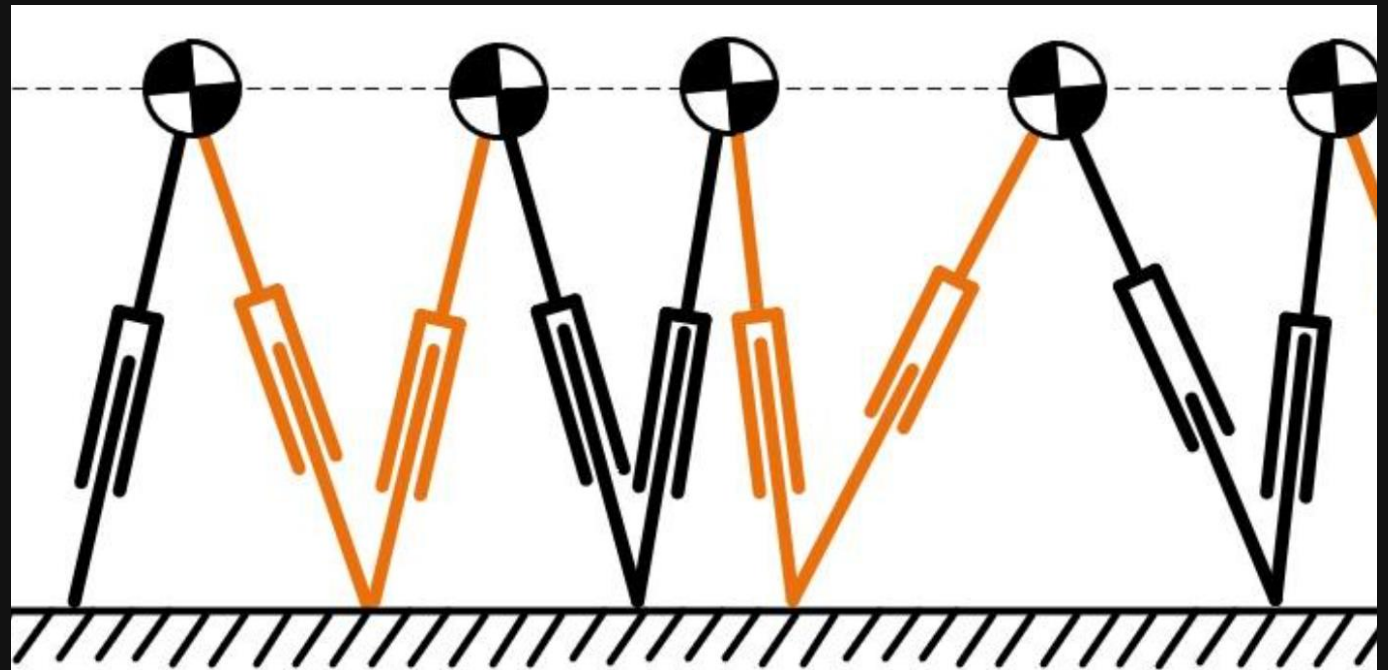
$$\begin{cases} x_t &= x_0 cosh(t/T_c) + T_c \dot{x}_0 sinh(t/T_c) \\ \dot{x}_t &= x_0/T_c sinh(t/T_c) + \dot{x}_0 cosh(t/T_c) \\ T_c &= \sqrt{z/g} \end{cases}$$

- x_0 and $\dot{x}_0$ represents the initial position and velocity along x-axis



What is Linear Inverted Pendulum Model (LIPM)?

- For a humanoid robot with two legs, the walking process can be simplified as the horizontal motion of a linear inverted pendulum model with two legs
- The alternating leg movement process is shown as follows



What is Linear Inverted Pendulum Model (LIPM)?



- How do we decide where to place the next step?

What is Linear Inverted Pendulum Model (LIPM)?

- How do we decide where to place the next step?
- By using orbital energy (OE)

What is Linear Inverted Pendulum Model (LIPM)?

- We can calculate Orbital energy (OE) through the equation below

$$E = \frac{1}{2}\dot{x}^2 - \frac{g}{2z}x^2$$

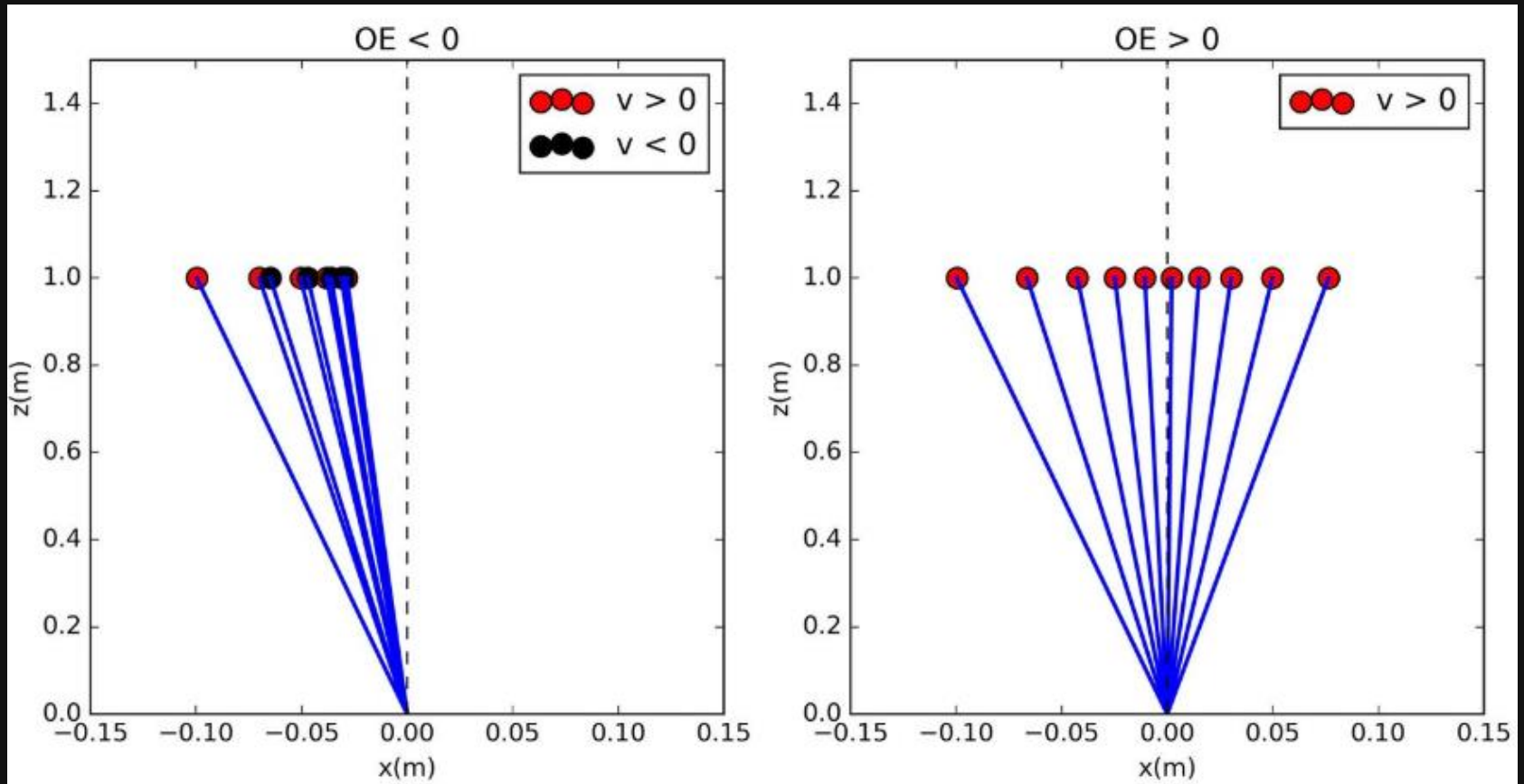
- Where x and $\dot{x}$ represents the current position and velocity along x-axis

- Orbital energy (OE) is a type of virtual potential energy, which reflects the motion state of the center of mass (CoM).
- Once the initial state of a single step LIPM is determined, its orbital energy remains conserved during the subsequent motion process

What is Linear Inverted Pendulum Model (LIPM)?

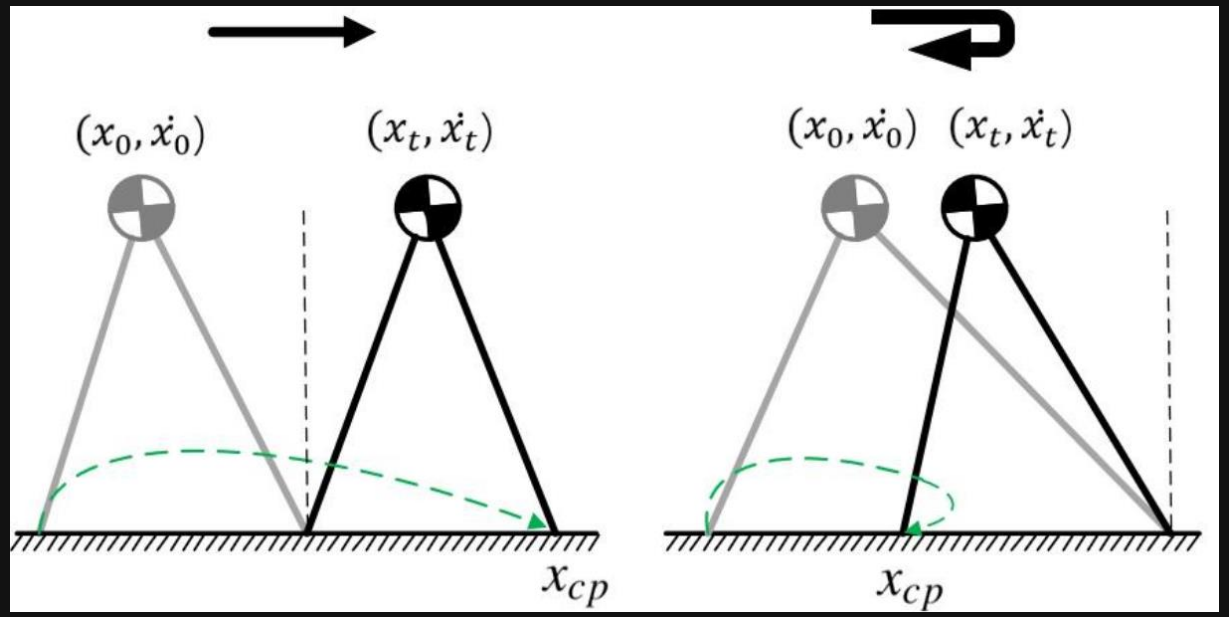
$$E = \frac{1}{2}\dot{x}^2 - \frac{g}{2z}x^2$$

When OE is positive, the CoM can pass the balance point



What is Linear Inverted Pendulum Model (LIPM)?

- We can adjust next step foot placement to change the OE



OE > 0
step forward

OE < 0
step backward

The desired (planned) next foot placement position is called capture point

What is Linear Inverted Pendulum Model (LIPM)?

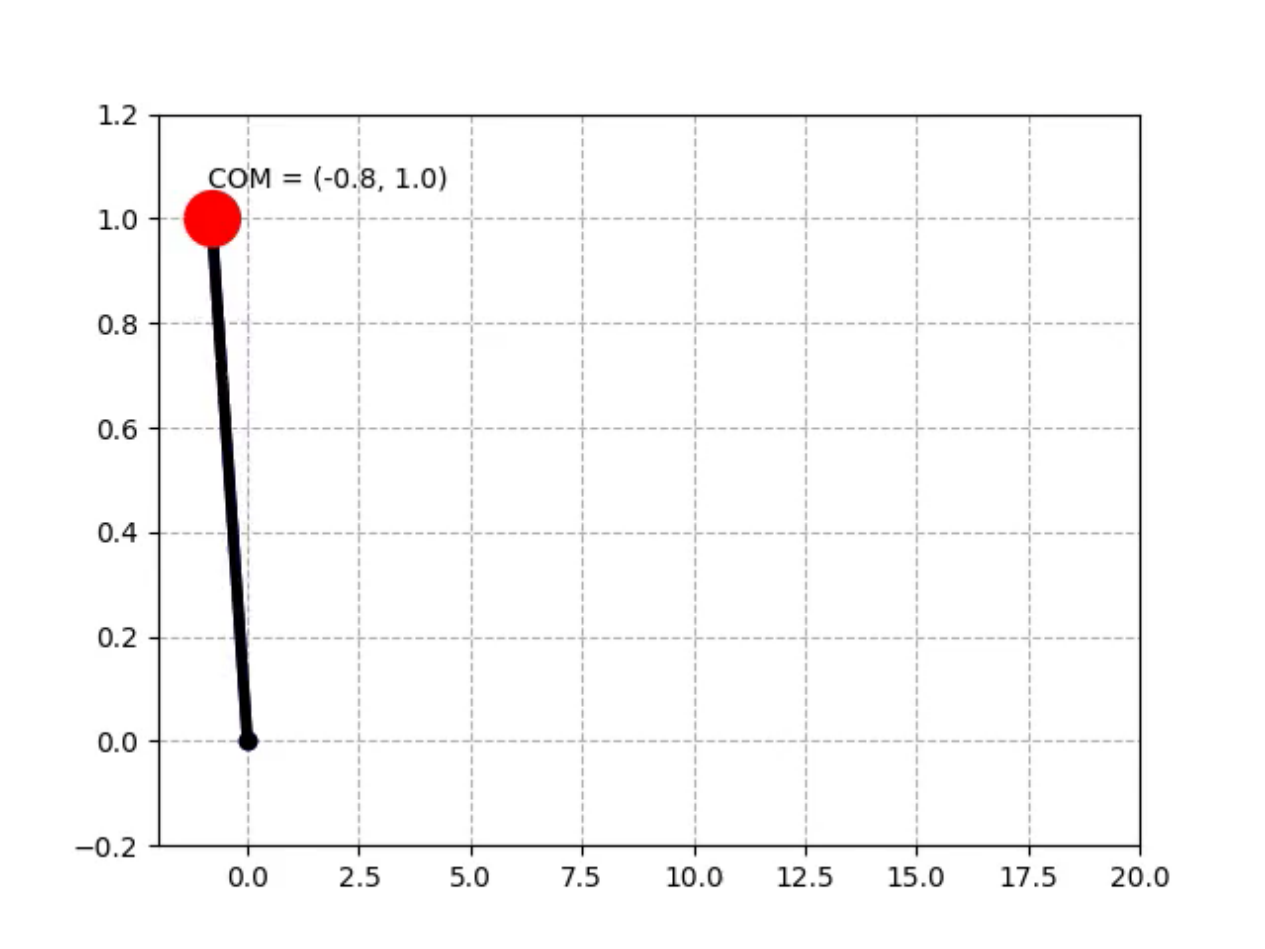
- Note that once we get the capture point, the new OE is determined
- Conversely, if we have a target new OE, we can derive the desired capture point as follows

$$x_{cp} = x_t + sign(\dot{x}_t) \cdot \sqrt{\frac{z}{g}(\dot{x}_t^2 - 2E_2)}$$

- Where E_2 represents the target OE

What is Linear Inverted Pendulum Model (LIPM)?

- A toy example of a multiple step LIPM

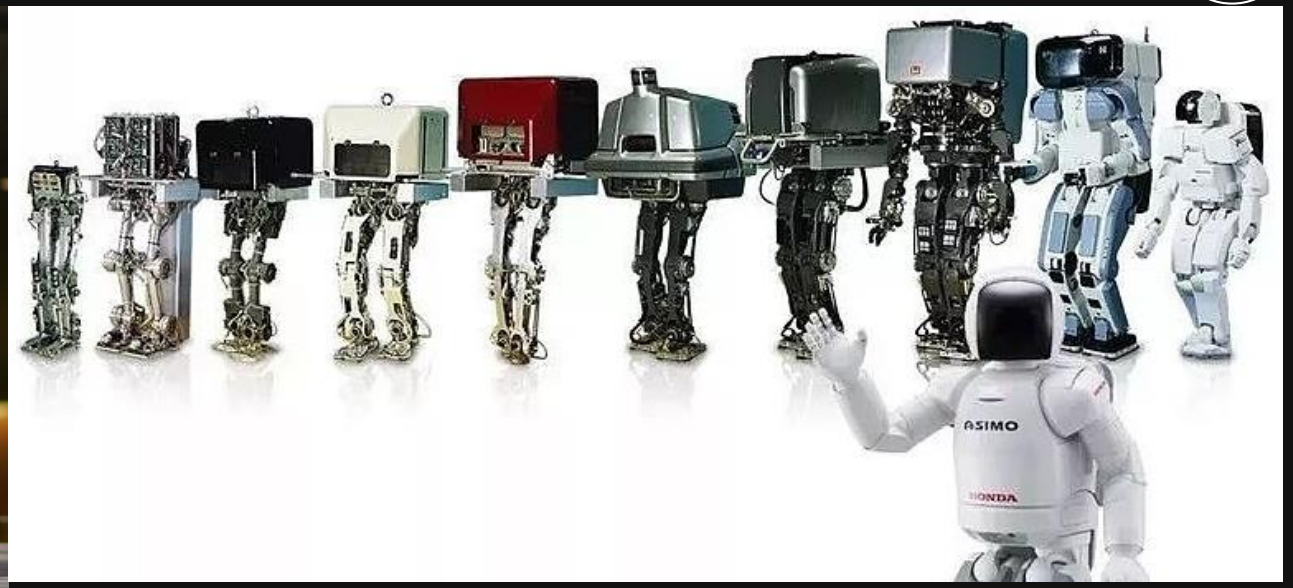


Asimo (Advanced Step in Innovative Mobility)

Asimo, made by Honda Corporation of Japan

- Research and development began in 1986
- Retired on March 31, 2022





Asimo's various generations (1986 - 2015)



Asimo (2015)

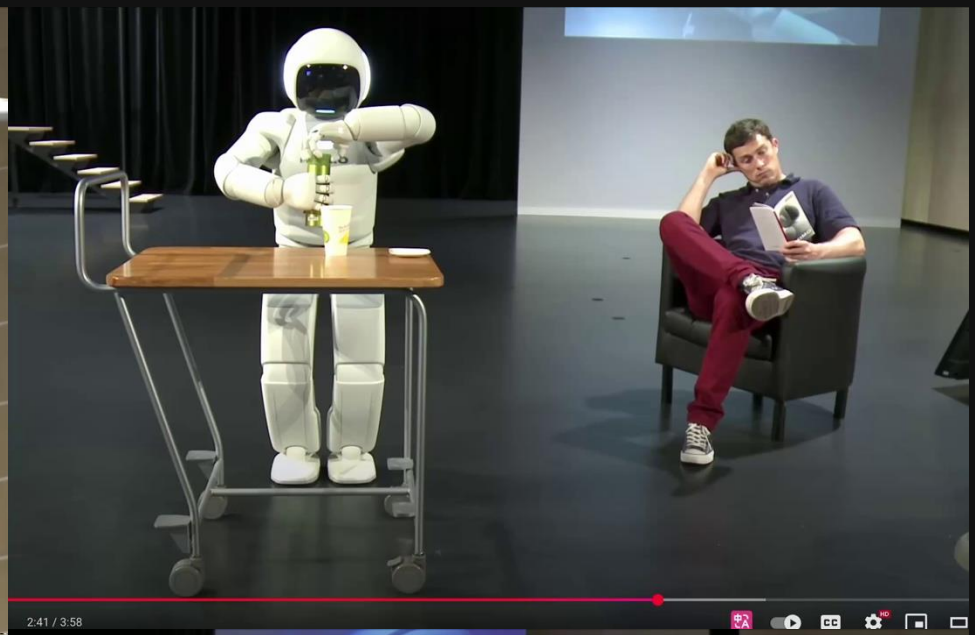
2025年4月19日，“天工”系列机器人“天工Ultra”参加全球首个人形机器人半程马拉松，以2时40分42秒的成绩夺得人形机器人半程马拉松冠军 (7.875 km/h)

The performance of robots depends not only on algorithms

LIPM holds some basic assumptions

- The center of mass (CoM) height is constant
- The legs are assumed massless
- Ignore angular momentum change









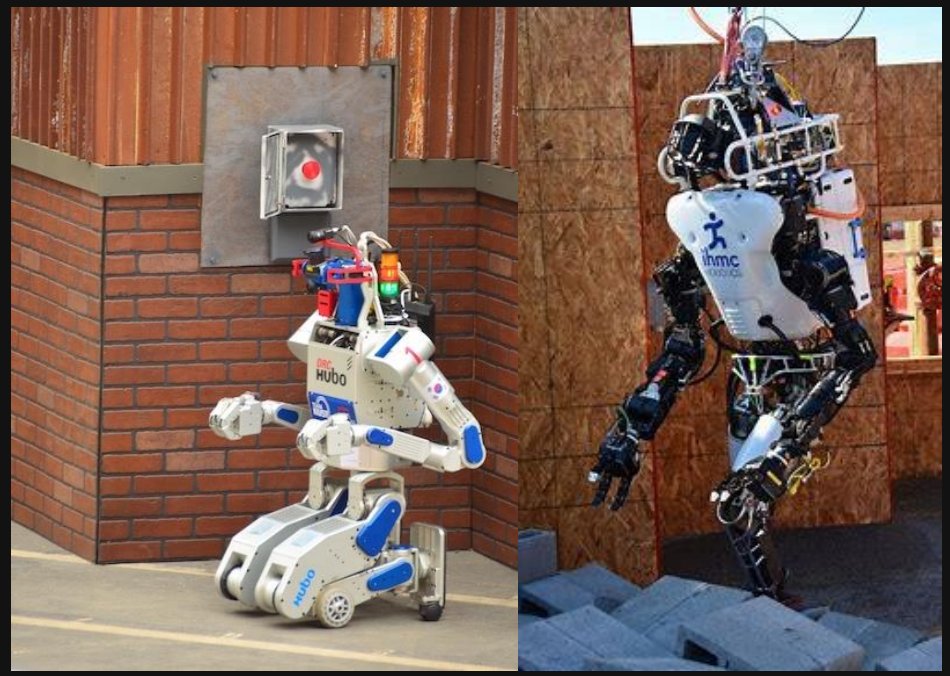
Robots of this Era



ARTEMIS, RoMeLa (UCLA)



Cassie, Agility Robotics Inc.



Robots in DARPA Robotics Challenge

During the DARPA Robotics Challenge (DRC):

- Most teams use LIPM
- No standard plan for whole-body control
- No good control method was produced
- Performance was poor

Note: We will revisit DARPA robotics challenge in later chapters

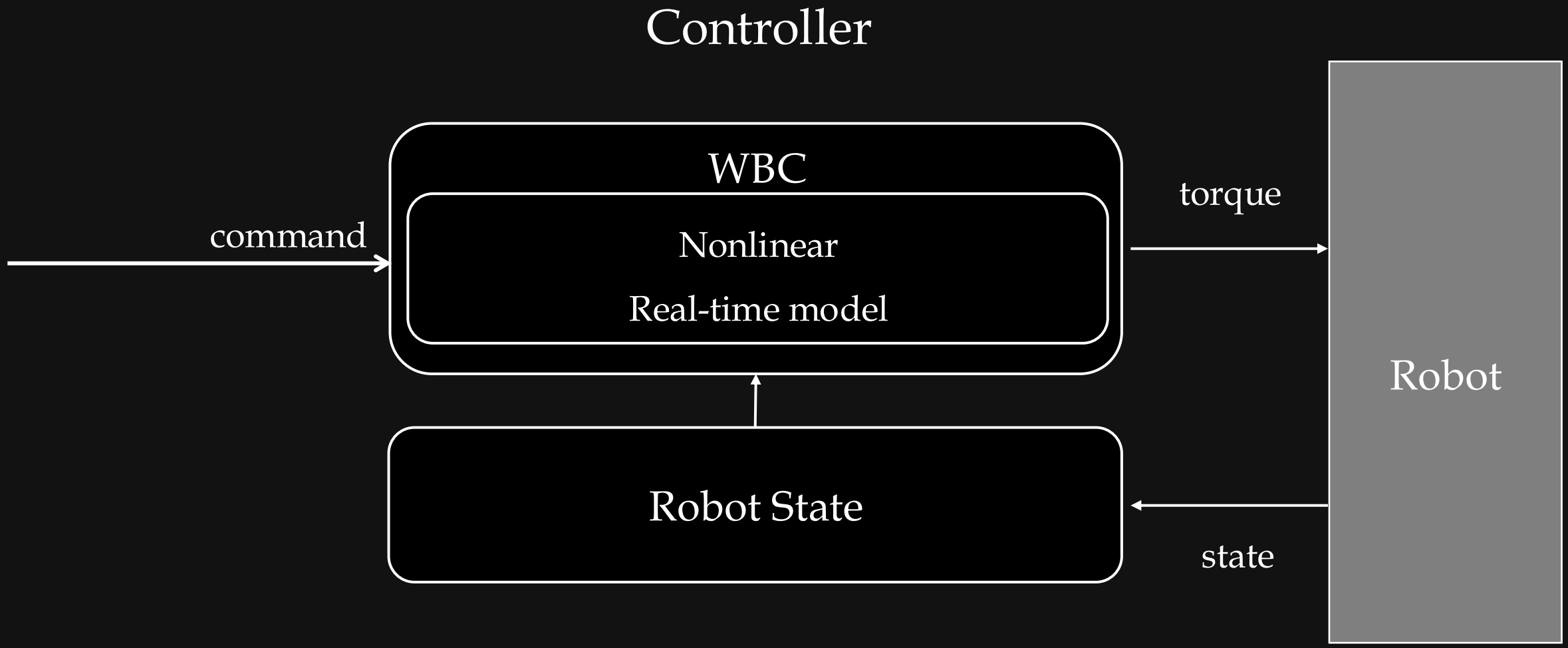
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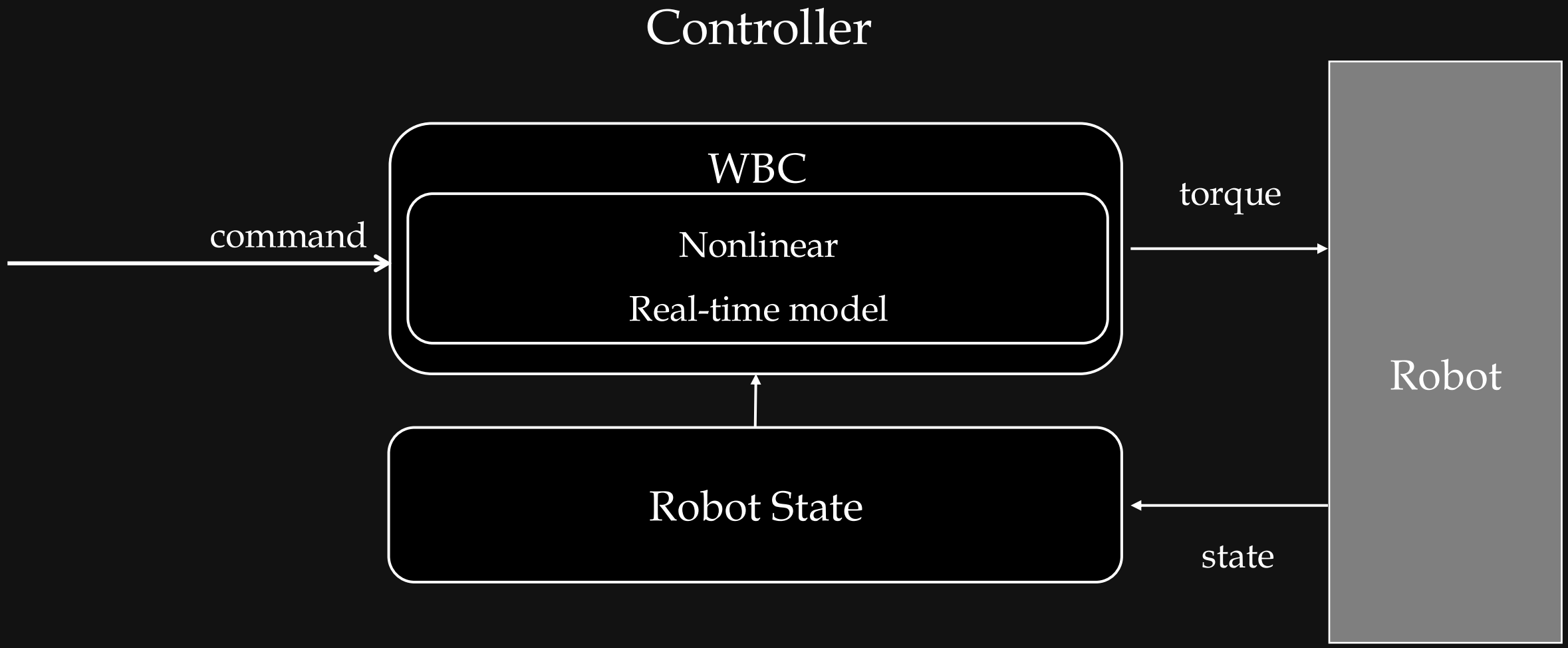
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  - Sim-to-real Transfer

Simplified control framework



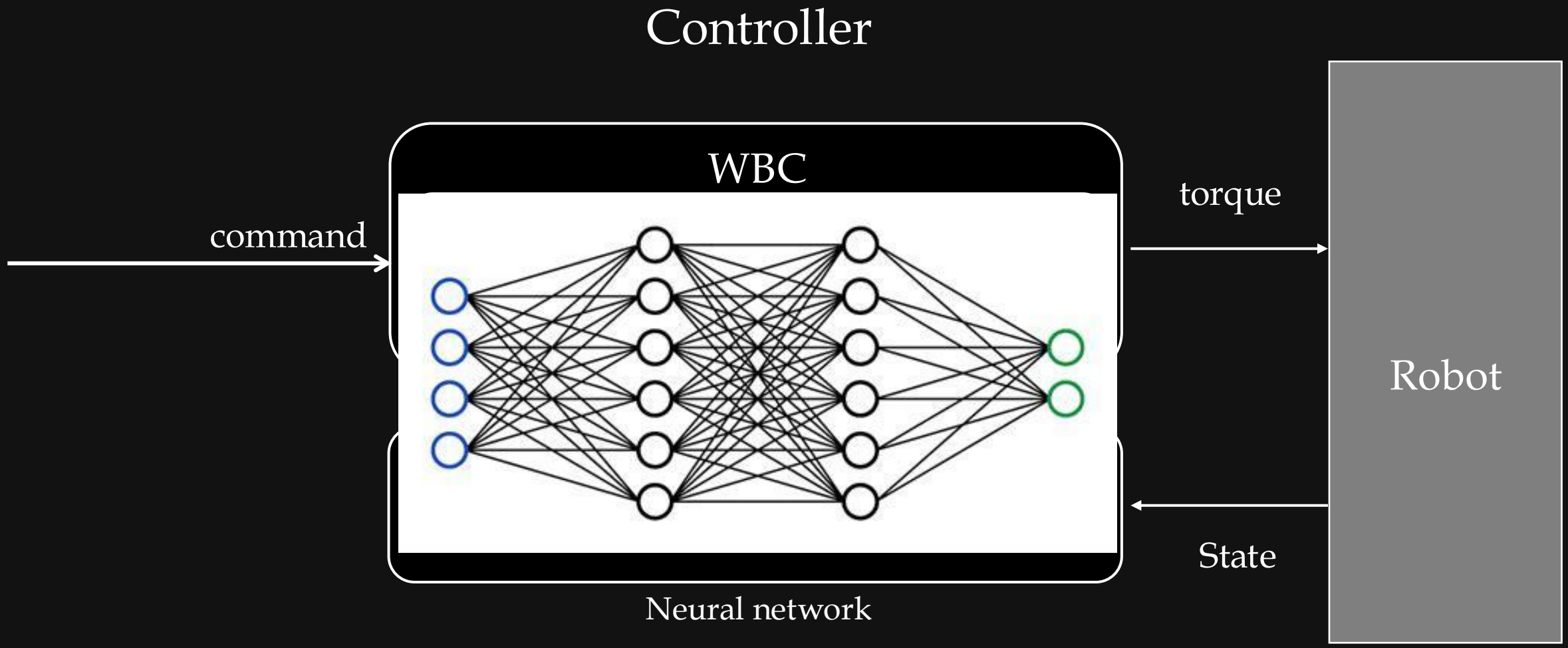
A humanoid robot controller can be simplified like this

Simplified control framework



A humanoid robot controller can be simplified like this

Simplified control framework



We can always train a neural network to be the controller

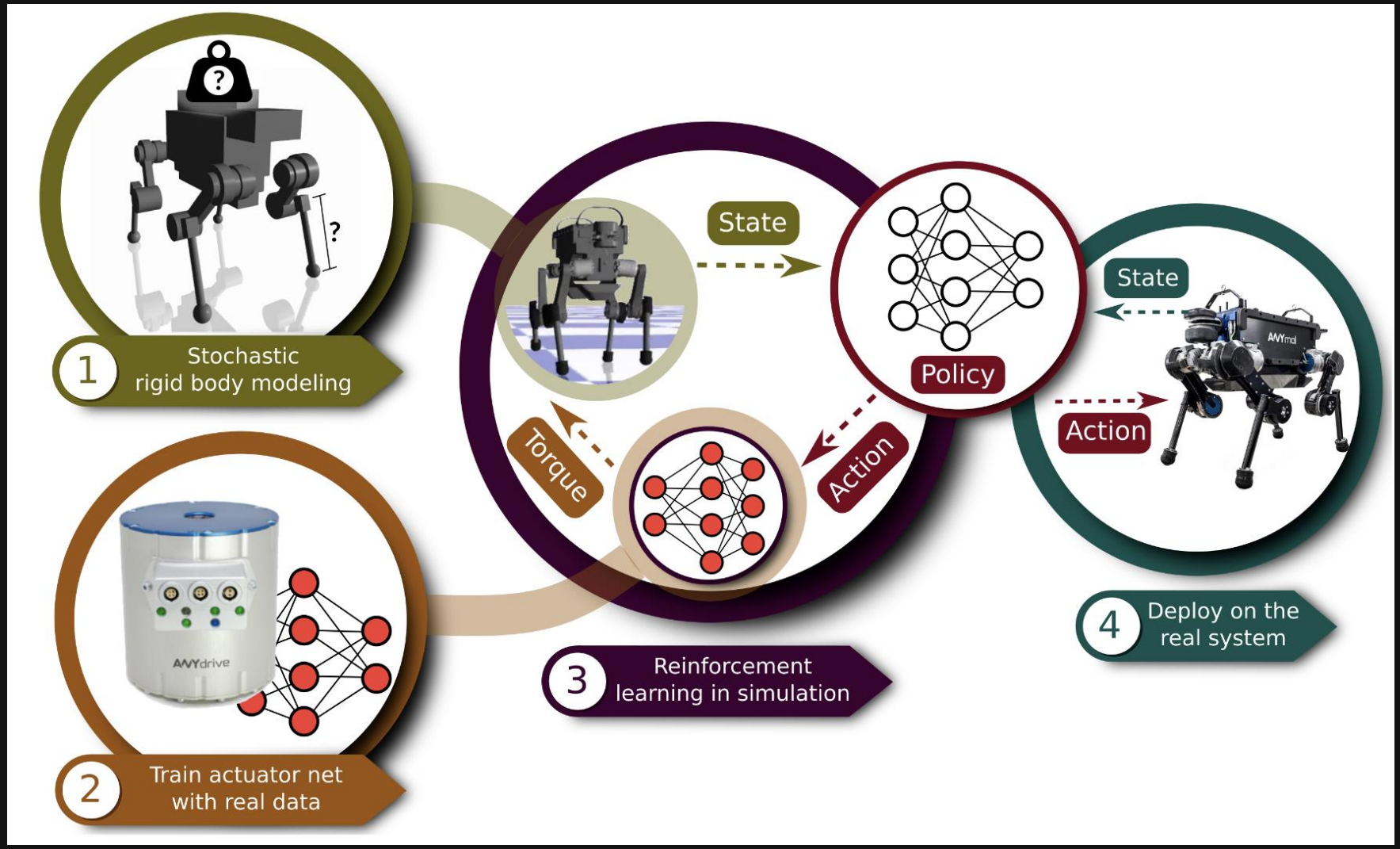
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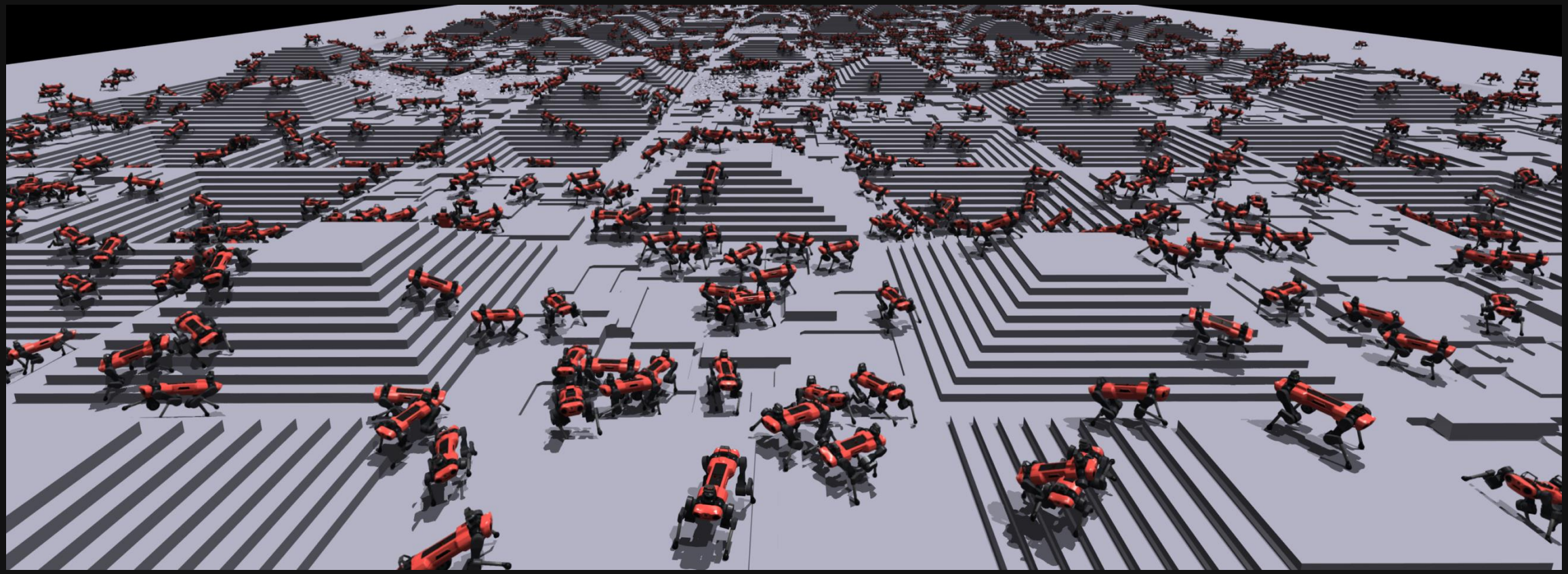
Reinforcement Learning



Science Robotics, ETH (2019)

Hwangbo, J., Lee, J., Dosovitskiy, A., Bellicoso, D., Tsounis, V., Koltun, V., & Hutter, M. (2019). Learning agile and dynamic motor skills for legged robots. Science Robotics, 4(26), eaau5872.

Reinforcement Learning



Training in IsaacGym, a physical simulator

Rudin, N., Hoeller, D., Reist, P., & Hutter, M. (2022, January). Learning to walk in minutes using massively parallel deep reinforcement learning. In Conference on Robot Learning (pp. 91-100). PMLR.

Reinforcement Learning

RL training enables dynamic humanoid control





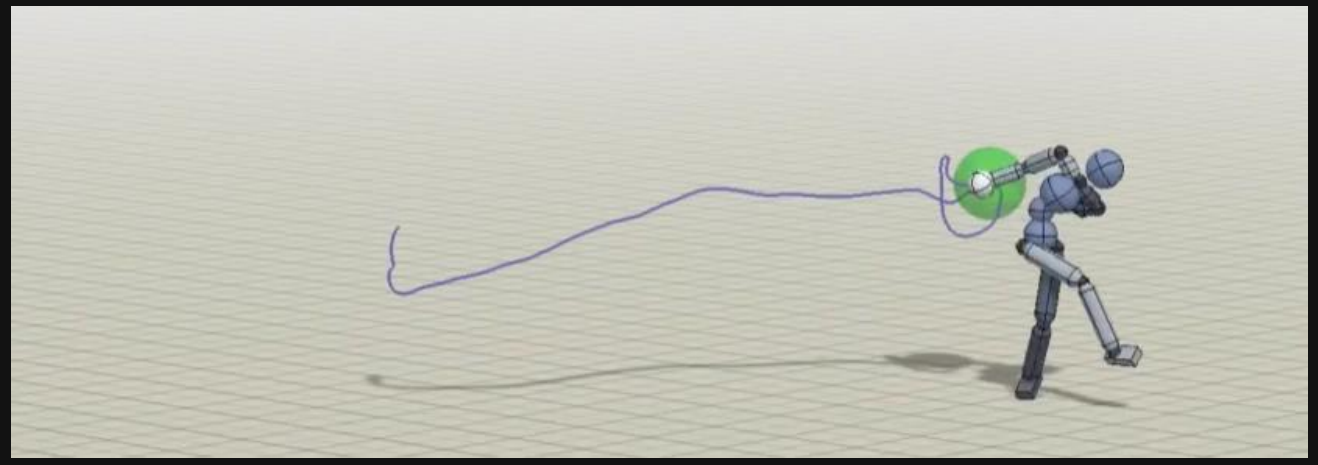




Limitation of RL-based Methods

- RL-based methods require complex manually designed rewards to perform human-like behavior

Without sophisticated reward design, RL-based methods tend to achieve tasks in an unnatural manner



```
class rewards( LeggedRobotCfg.rewards ):
    soft_dof_pos_limit = 0.9
    base_height_target = 0.78

    class scales( LeggedRobotCfg.rewards.scales ):
        tracking_lin_vel = 1.0
        tracking_ang_vel = 0.5
        lin_vel_z = -2.0
        ang_vel_xy = -0.05
        orientation = -1.0
        base_height = -10.0
        dof_acc = -2.5e-7
        dof_vel = -1e-3
        feet_air_time = 0.0
        collision = 0.0
        action_rate = -0.01
        dof_pos_limits = -5.0
        alive = 0.15
        hip_pos = -1.0
        contact_no_vel = -0.2
        feet_swing_height = -20.0
        contact = 0.18
```

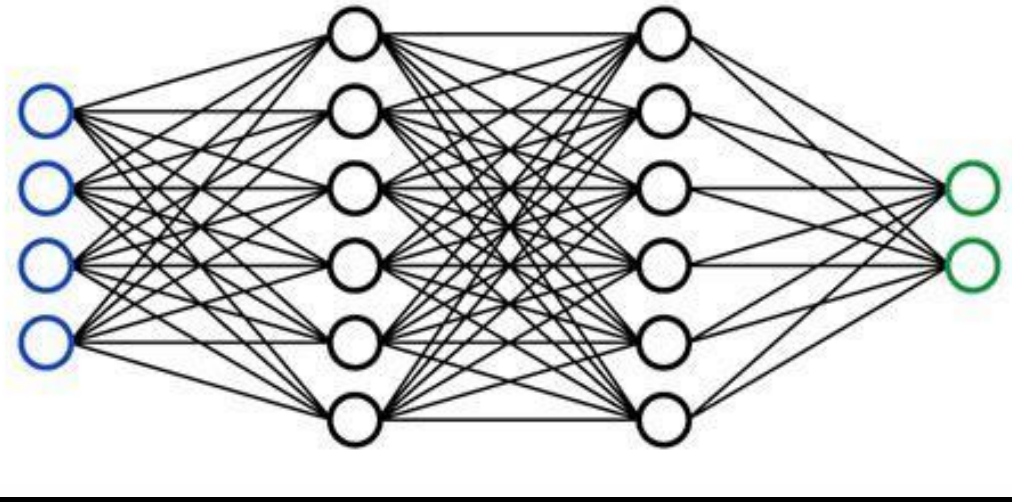
Rewards used in unitree_rl_gym

Imitation Learning

Controller

trajectory
(reference)

WBC



torque

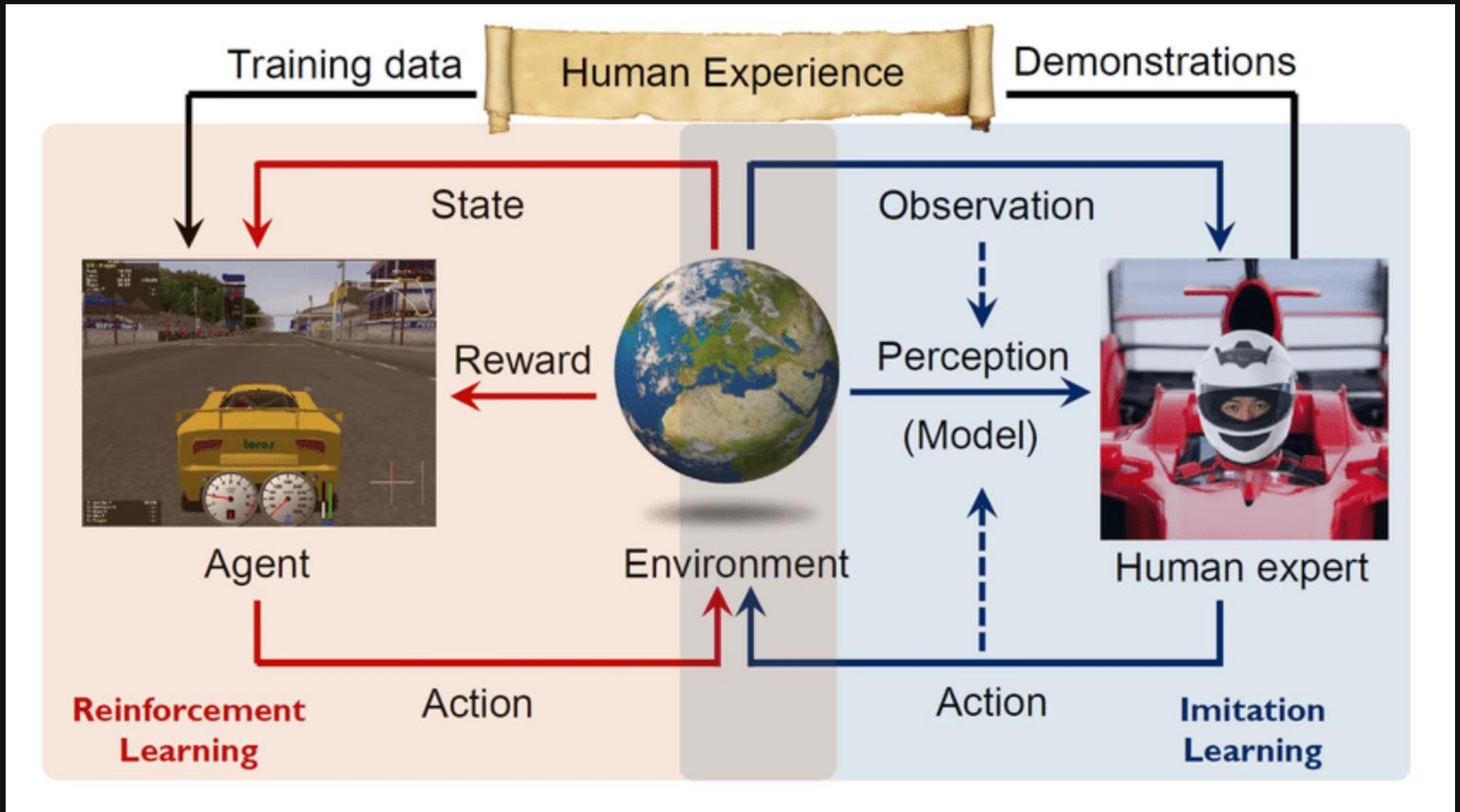
Robot

State

Neural network

Imitation Learning

Imitation learning pipeline in general



Imitation learning acquires dynamic motion skills









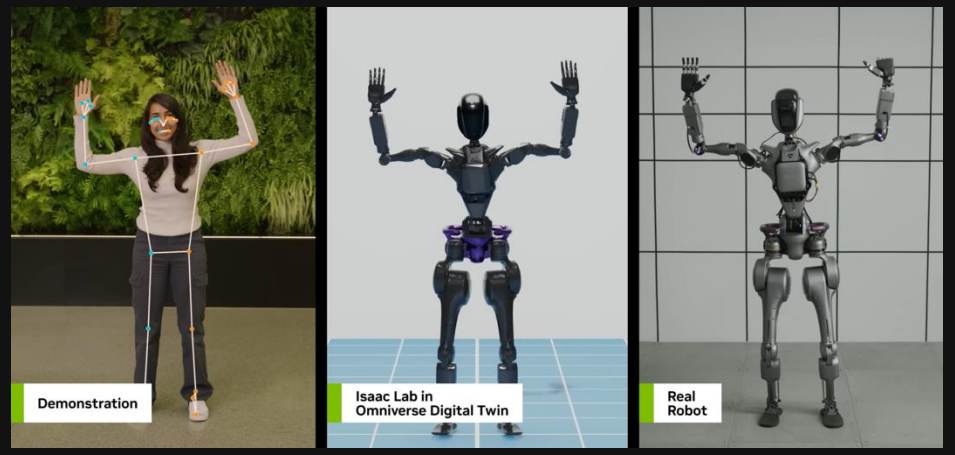
Limitation of IL-based Methods

- Imitation learning (IL) methods are restricted by limited human motion (expert) data

Imitation learning methods require human expert demonstrations from multiple sources





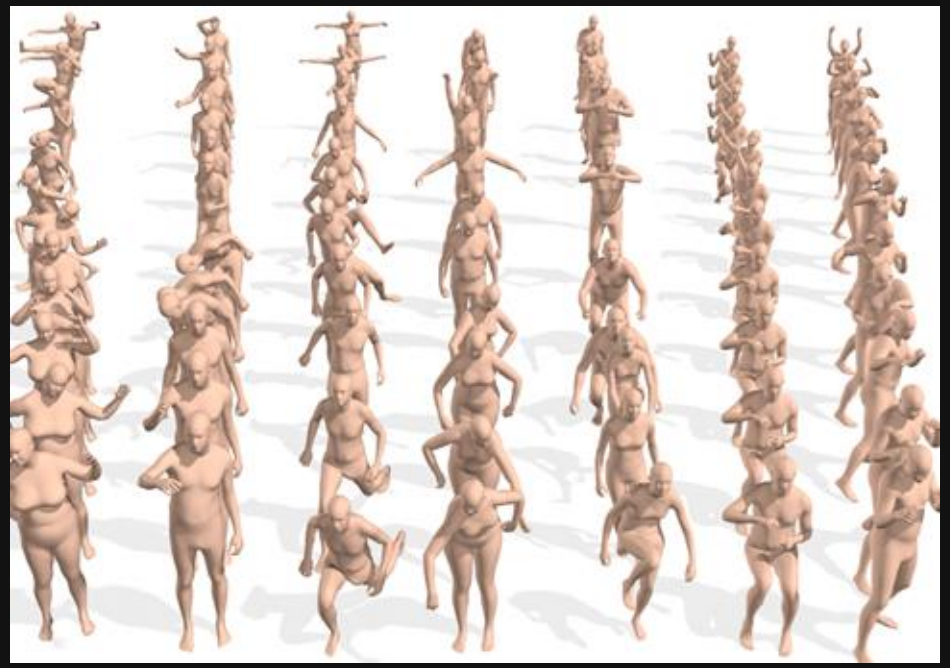


Such as 3D MoCap data, simulation data, video data, and real robot data …

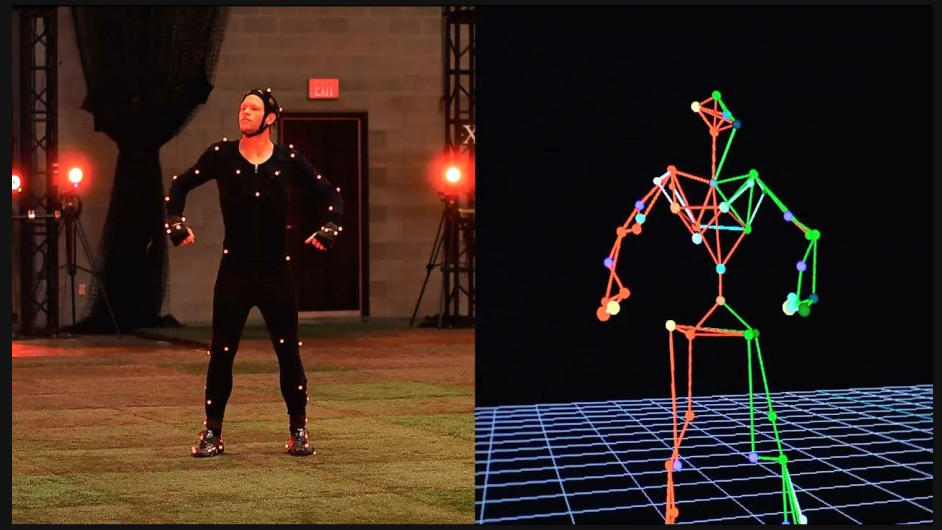
Where can we get the data?

Where can we get the data?

- From human motion dataset



AMASS MoCap Dataset





The human data can be recorded with MoCap

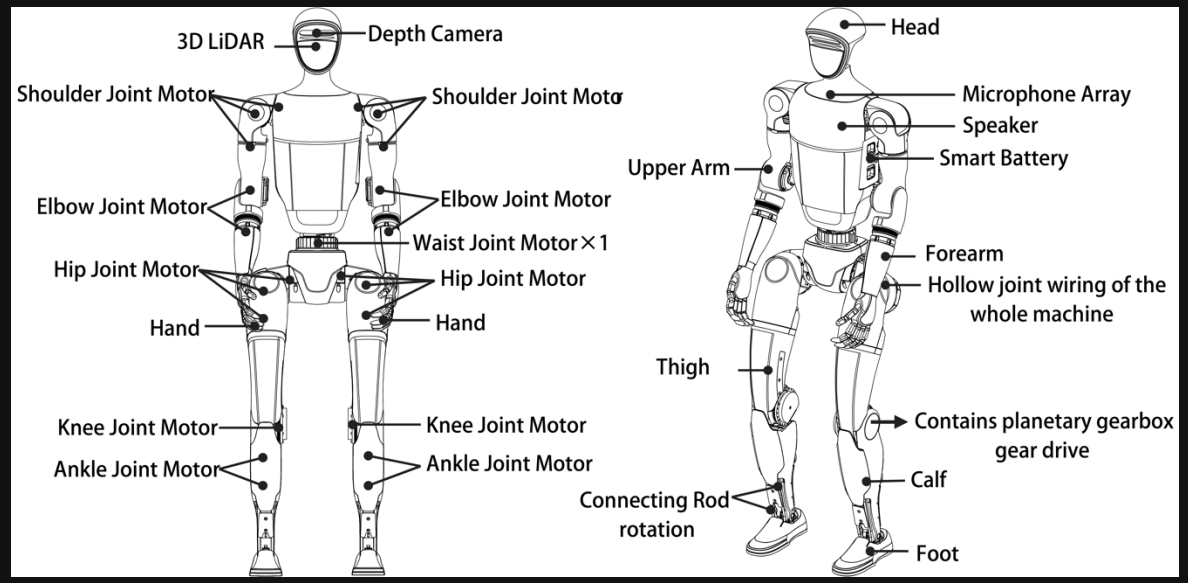
Recap: What is Motion Capture (MoCap)



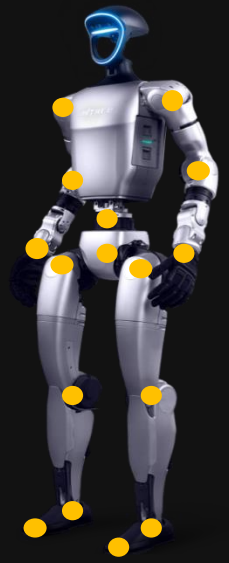
Huge gaps between human and humanoid's skeleton

- Human skeleton usually has more Degree-of-Freedoms (DoFs)

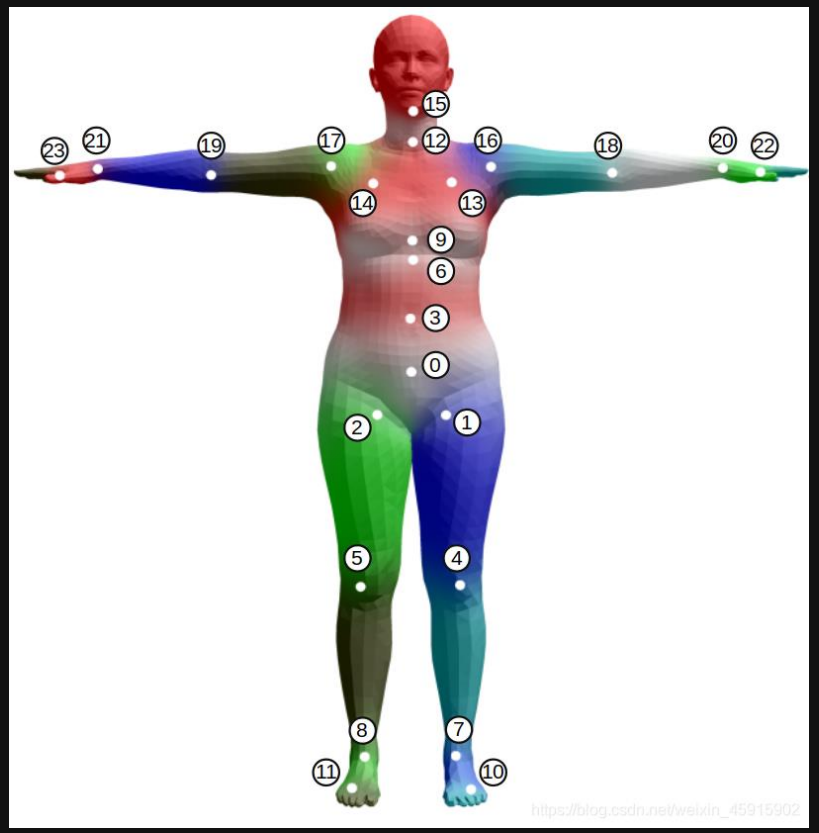
Each motor is 1-DoF



Unitree G1 skeleton



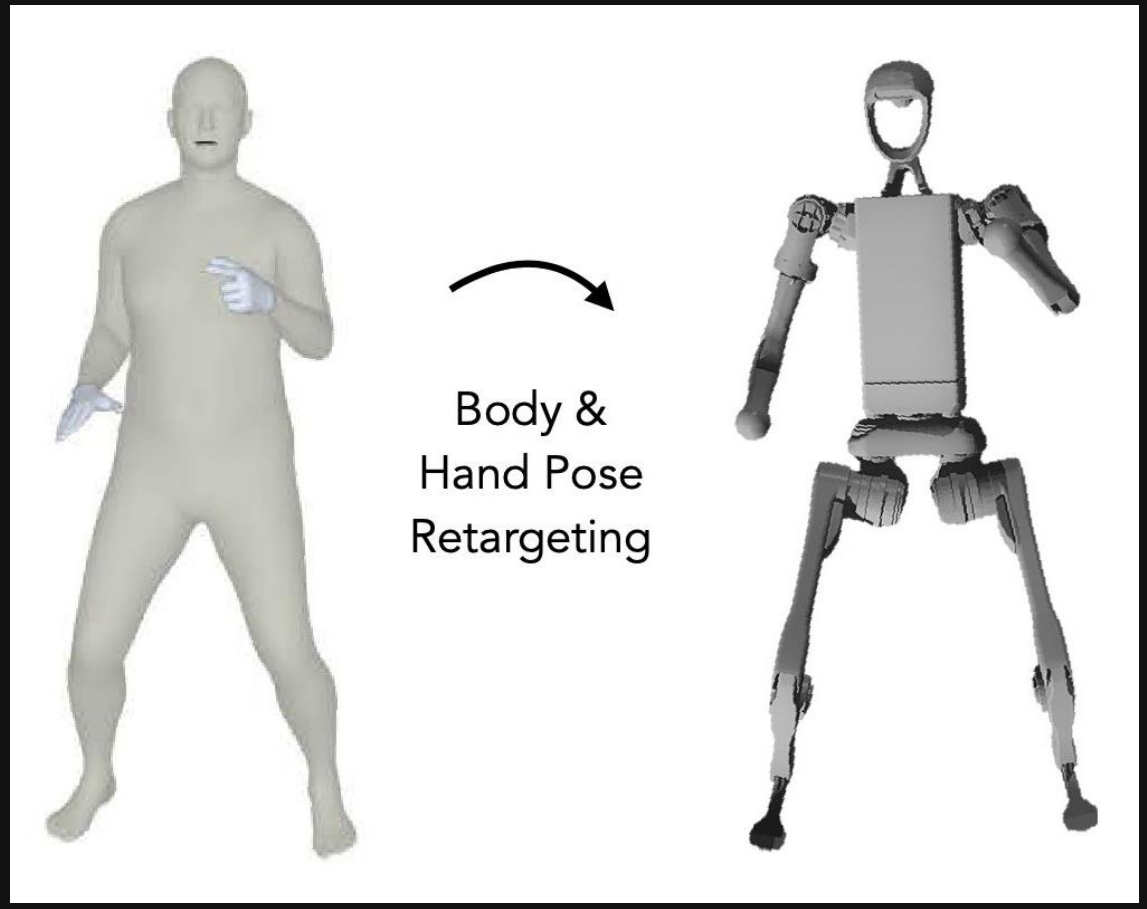
Each joint is 3-DoF



SMPL skeleton

Huge gaps between human and humanoid's skeleton

- Retargeting from human motion to humanoid motion is a non-trivial problem



Retargeting means that we want to replicate the motion from a source skeleton (human) to a target skeleton (humanoid)

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Human Motion Retargeting

- Solution 1: Directly Copy Rotation

Human Motion Retargeting

- Solution 1: Directly Copy Rotation

1. Find a mapping between the bones of human and humanoid



Root
Hip
Knee
Shoulder
Spine
…

Pelvis (roll, pitch, yaw)
Hip (roll, pitch, yaw)
Knee (pitch)
Shoulder (roll, pitch, yaw)
Null
…



Human Motion Retargeting

- Solution 1: Directly Copy Rotation

1. Find a mapping between the bones of human and humanoid



Root
Hip
Knee
Shoulder
Spine
…

Pelvis (roll, pitch, yaw)
Hip (roll, pitch, yaw)
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Shoulder (roll, pitch, yaw)
Null
…



Humanoid robots usually don't have DoFs on spine

Human Motion Retargeting

- Solution 1: Directly Copy Rotation

1. Find a mapping between the bones of human and humanoid



Root
Hip
Knee
Shoulder
Spine
…

Pelvis (roll, pitch, yaw)
Hip (roll, pitch, yaw)
Knee (pitch)
Shoulder (roll, pitch, yaw)
Null
…



Humanoid robots usually don't have DoFs on spine

2. Copy the rotation of each mapped joint

Human Motion Retargeting

- Solution 1: Directly Copy Rotation

  Copy rotation may cause errors on end-effectors, such as foot floating

Human Motion Retargeting

- Solution 1: Directly Copy Rotation

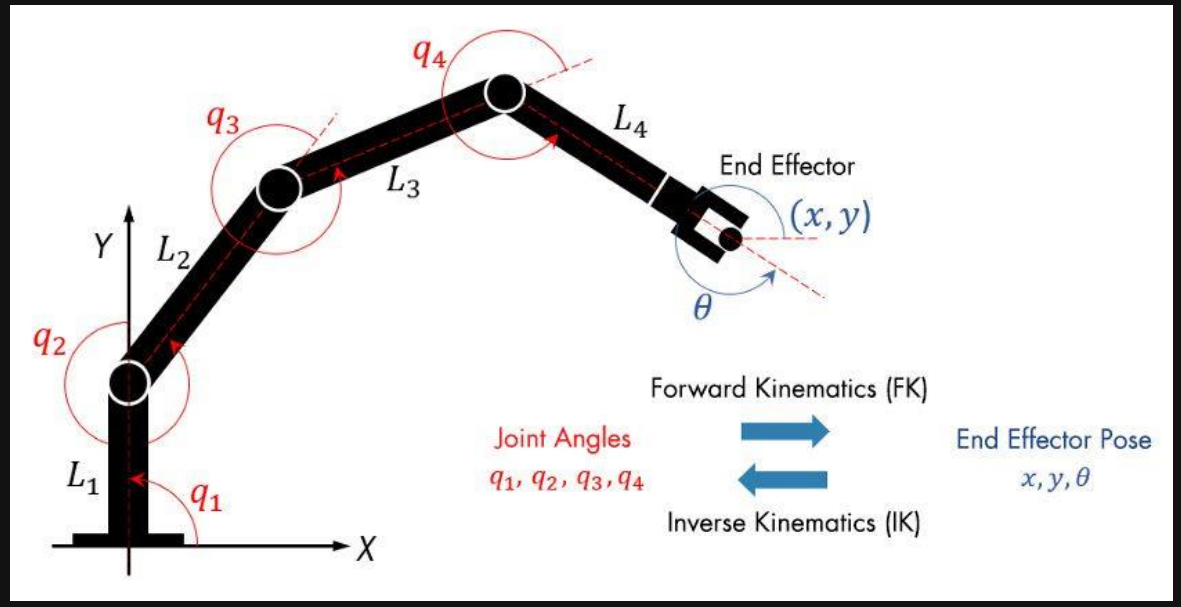
  Copy rotation may cause errors on end-effectors, such as foot floating

- Solution: Apply Inverse Kinematics on end-effectors

Human Motion Retargeting

- Solution 2: Copy Rotation Initialization + Inverse Kinematics

Given end-effector (hand and toe for humanoid robot) poses, we want to get the joint angles along the kinematic chain



Human Motion Retargeting

- How to solve IK?
  - Cyclic coordinate descent (CCD) is a typical optimization algorithm for solving IK
  - It minimizes the loss function by adjusting only one joint angle at a time
  - The other joint angles remains unchanged
  - After optimizing one joint angle, the algorithm moves to the next
  - CCD will repeat this process, until the loss function is small enough

Human Motion Retargeting

- Solution 2: Copy Rotation Initialization + Inverse Kinematics

Inverse Kinematics may introduce artifacts during IK solving process, such as penetration or jittering

Human Motion Retargeting

- Solution 3: RL-based Motion Tracking

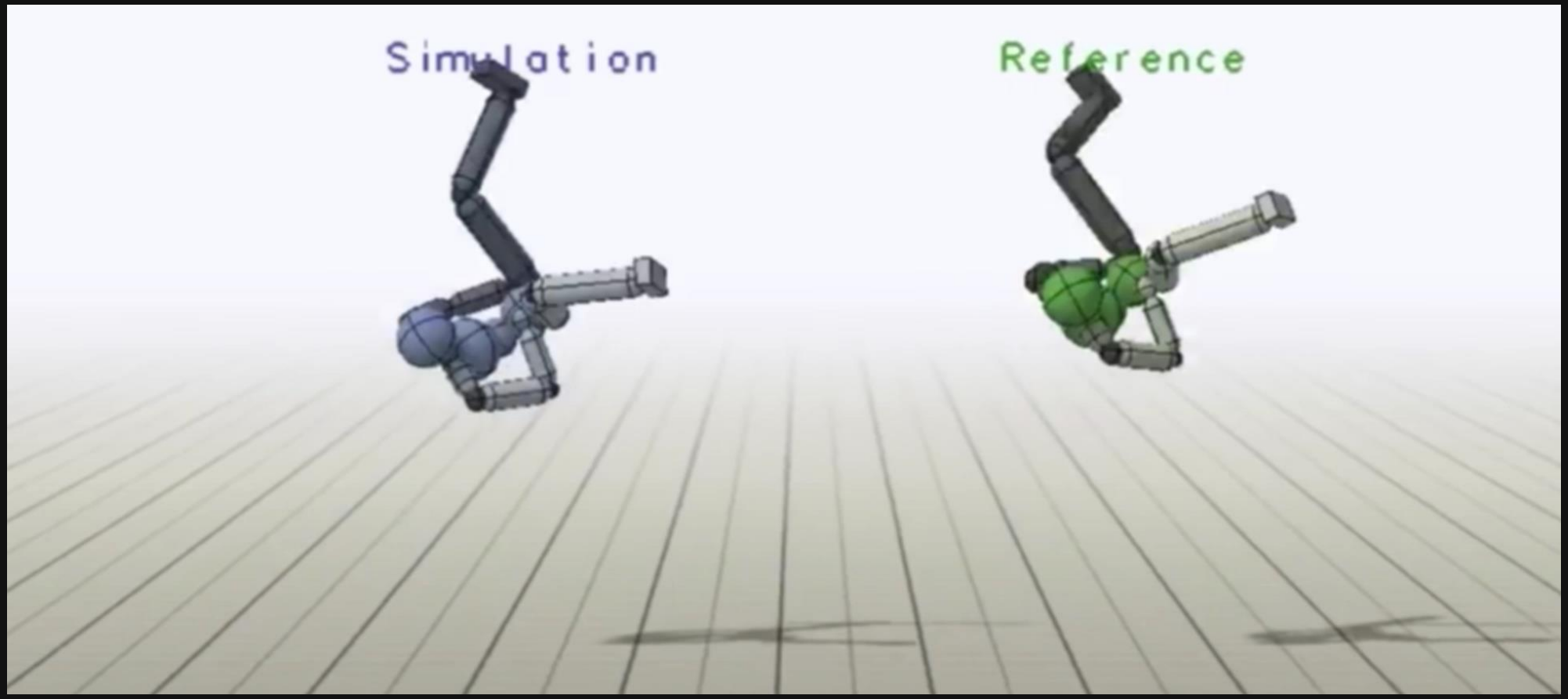
  Given a reference motion, we can train a humanoid to track the motion in a physical simulator

$$r_t = \omega^I r_t^I + \omega^G r_t^G$$

  - The RL reward contains imitation reward and goal reward
    - Imitation rewards encourage the humanoid to imitate the reference
    - Goal rewards let humanoid achieve extra tasks (not necessary)

Human Motion Retargeting

- Solution 3: RL-based Motion Tracking



DeepMimic

Human Motion Retargeting

- Solution 3: RL-based Motion Tracking

  The physical simulator can ‘filter’ the artifacts in raw motions, and provide more physically plausible motions

  Note: We will introduce the whole training paradigm later

Human Motion Retargeting

- Solution 3: RL-based Motion Tracking

  The physical simulator can ‘filter’ the artifacts in raw motions, and provide more physically plausible motions

  Note: We will introduce the whole training paradigm later

- However, it’s not easy to train a motion tracking model, especially using RL. Sometimes you need to manually tune the hyper-parameters

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Where can we get the data?

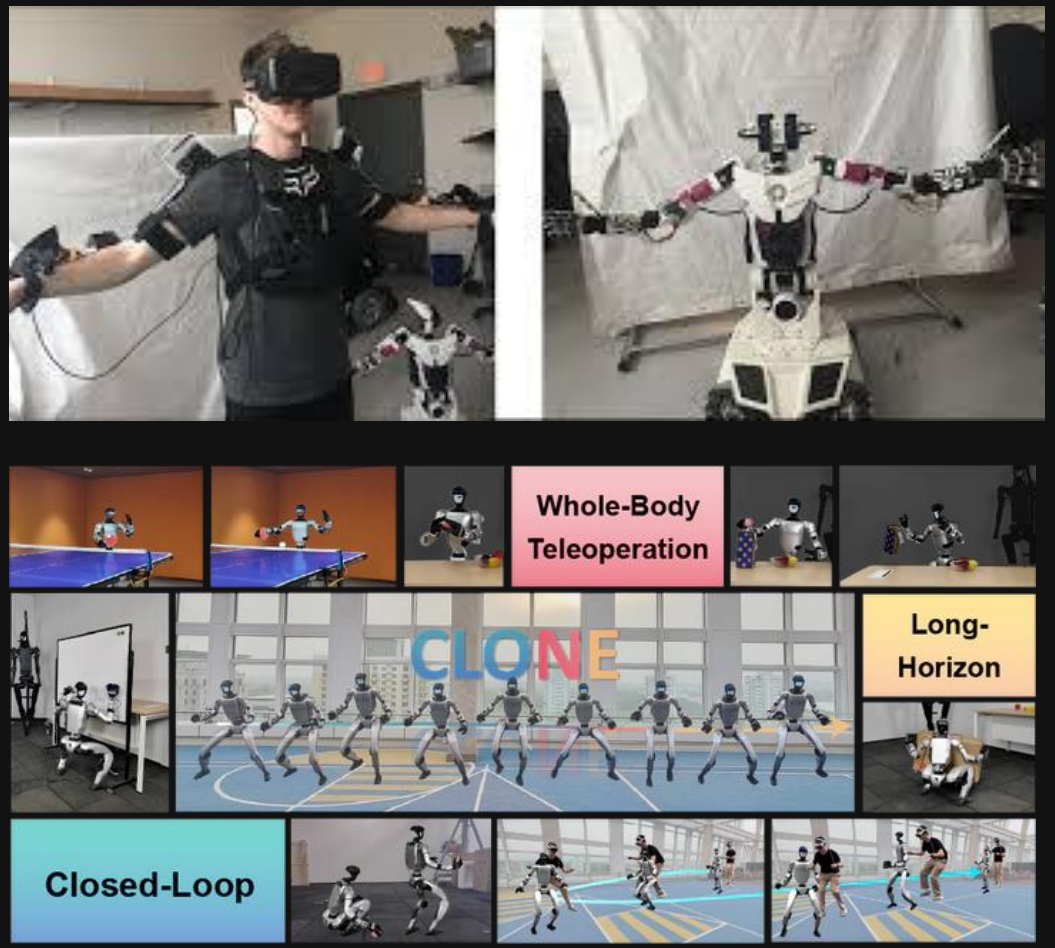
- From real-world robot

  Collected by humanoid teleoperation system



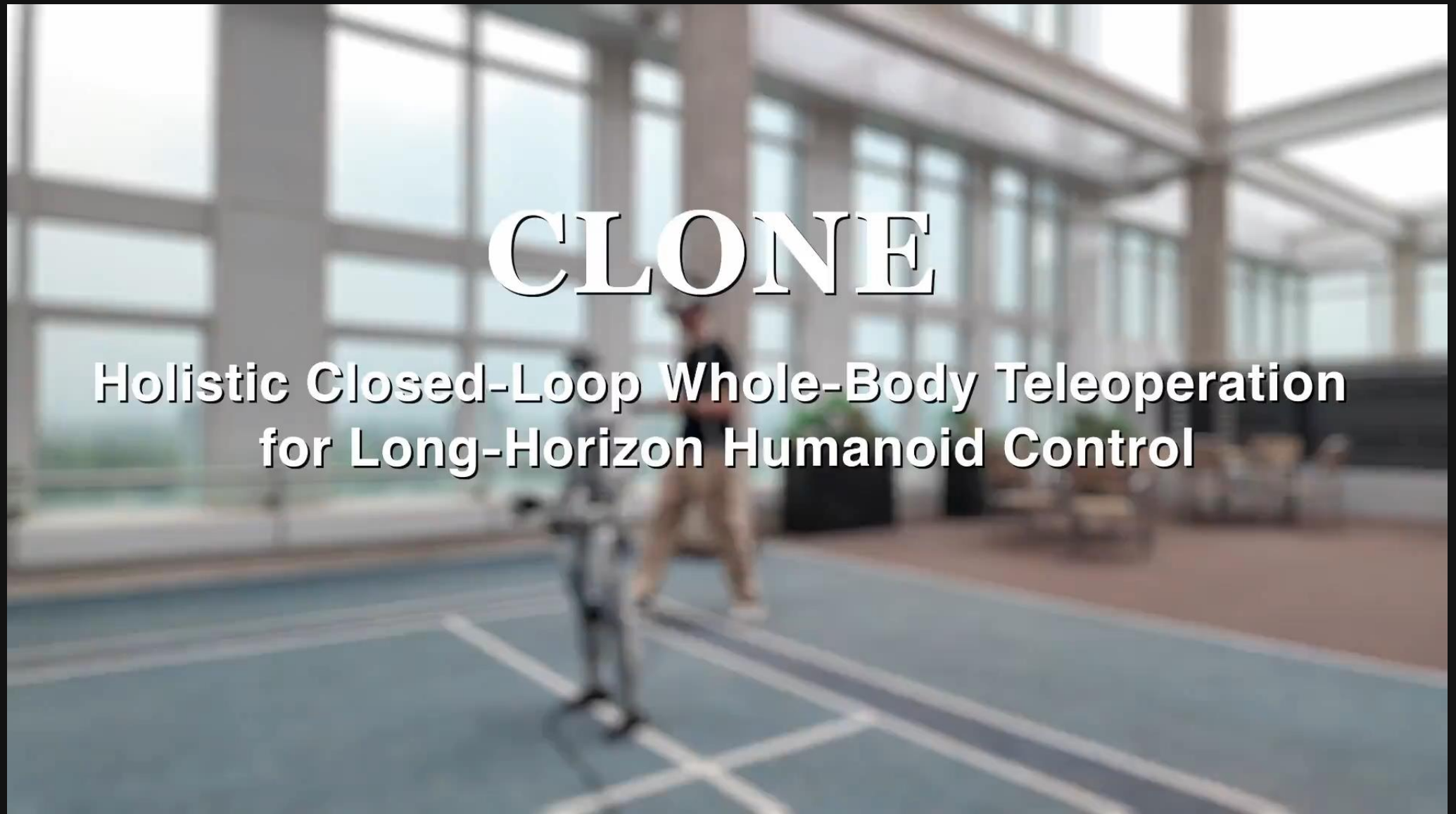
MoCap-based tracking

Exo-skeleton-based tracking



VR-based tracking

What is whole-body teleoperation?



Where can we get the data?

- From real-world robot

We can build a robust humanoid teleoperation system to collect real-world robot data

Where can we get the data?

- From real-world robot

We can build a robust humanoid teleoperation system to collect real-world robot data

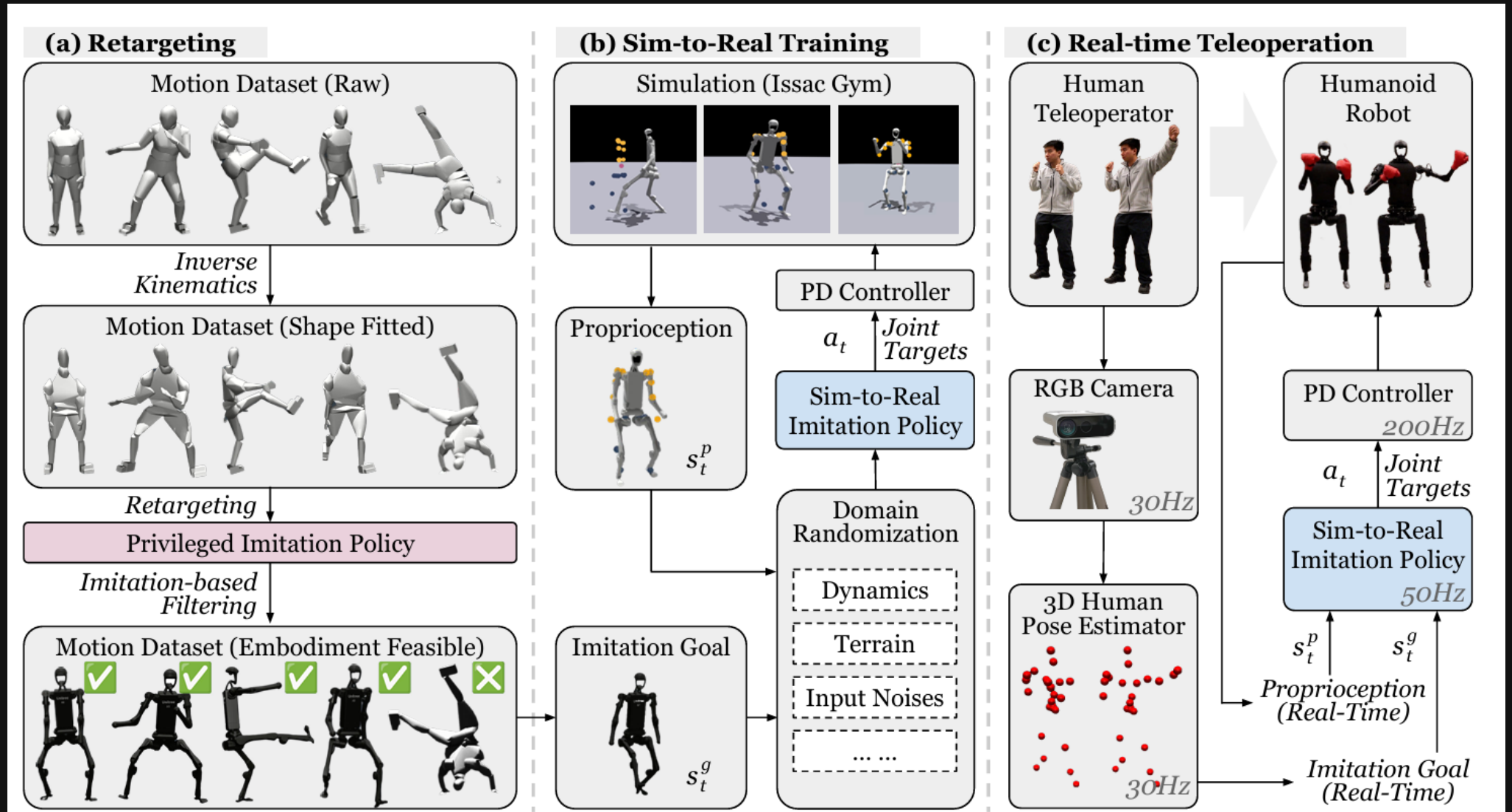
Learning Human-to-Humanoid Real-Time Whole-Body Teleoperation

Tairan He† Zhengyi Luo† Wenli Xiao Chong Zhang Kris Kitani Changliu Liu Guanya Shi
Carnegie Mellon University
†Equal Contributions https://human2humanoid.com

IROS 2024

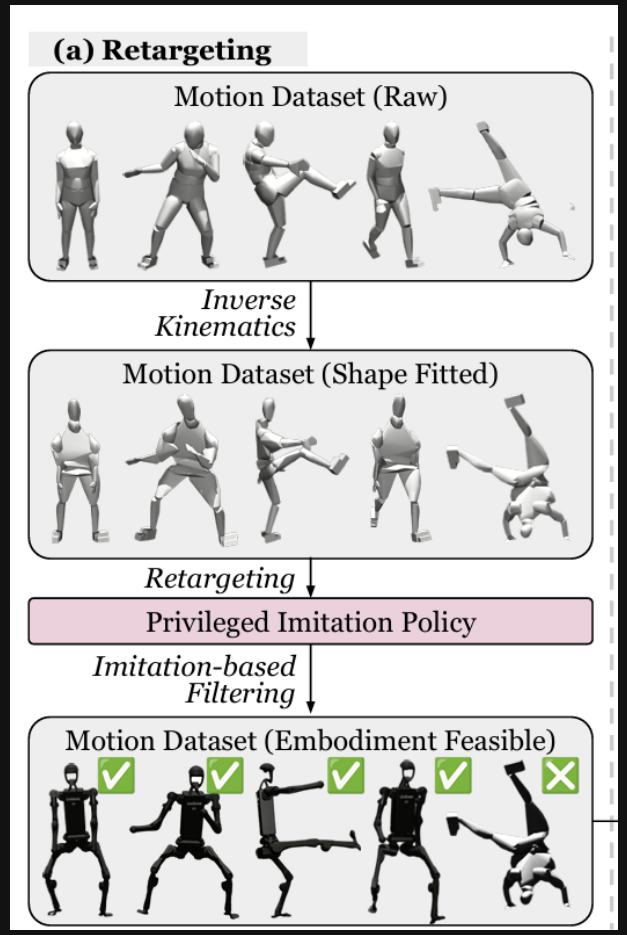
Case Study: Whole-Body Teleoperation

- Three stages to build this system

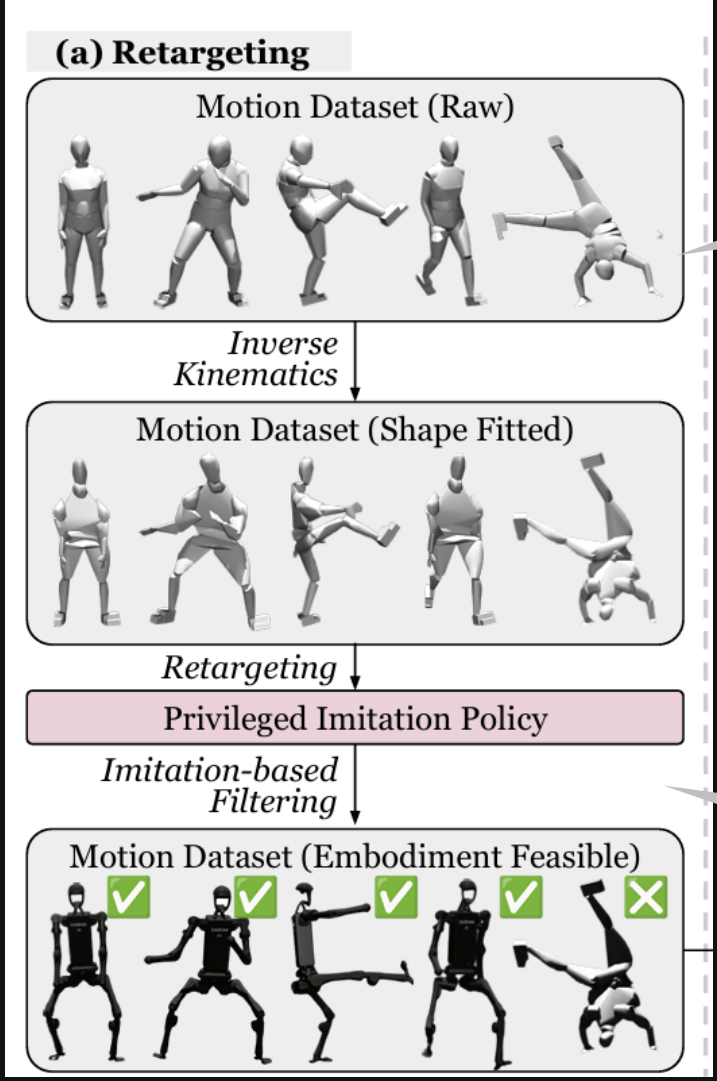


Case Study: Whole-Body Teleoperation

- Stage 1: Motion data retargeting



Case Study: Whole-Body Teleoperation

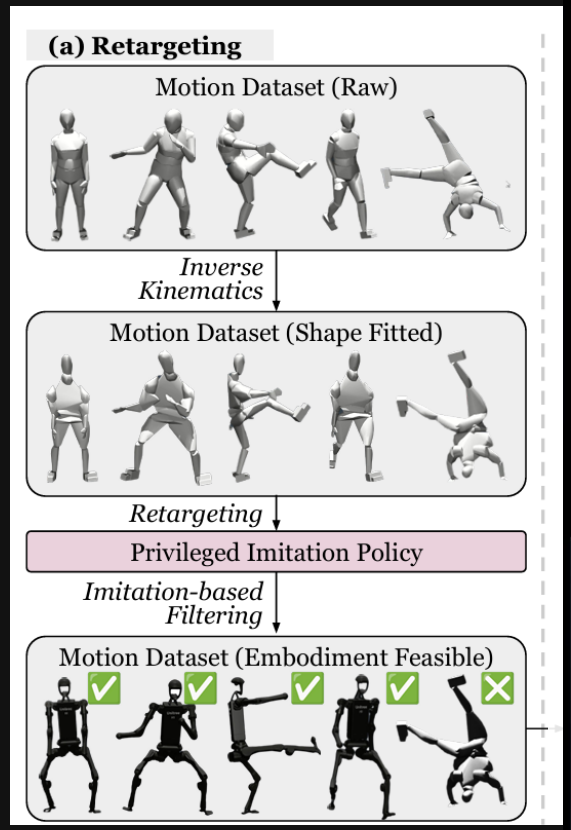


Apply inverse kinematics for motion retargeting (solution 2)

Find new SMPL shape parameters to better fit humanoid morphology (last lecture)

Train motion tracking policy for better motion quality (solution 3)

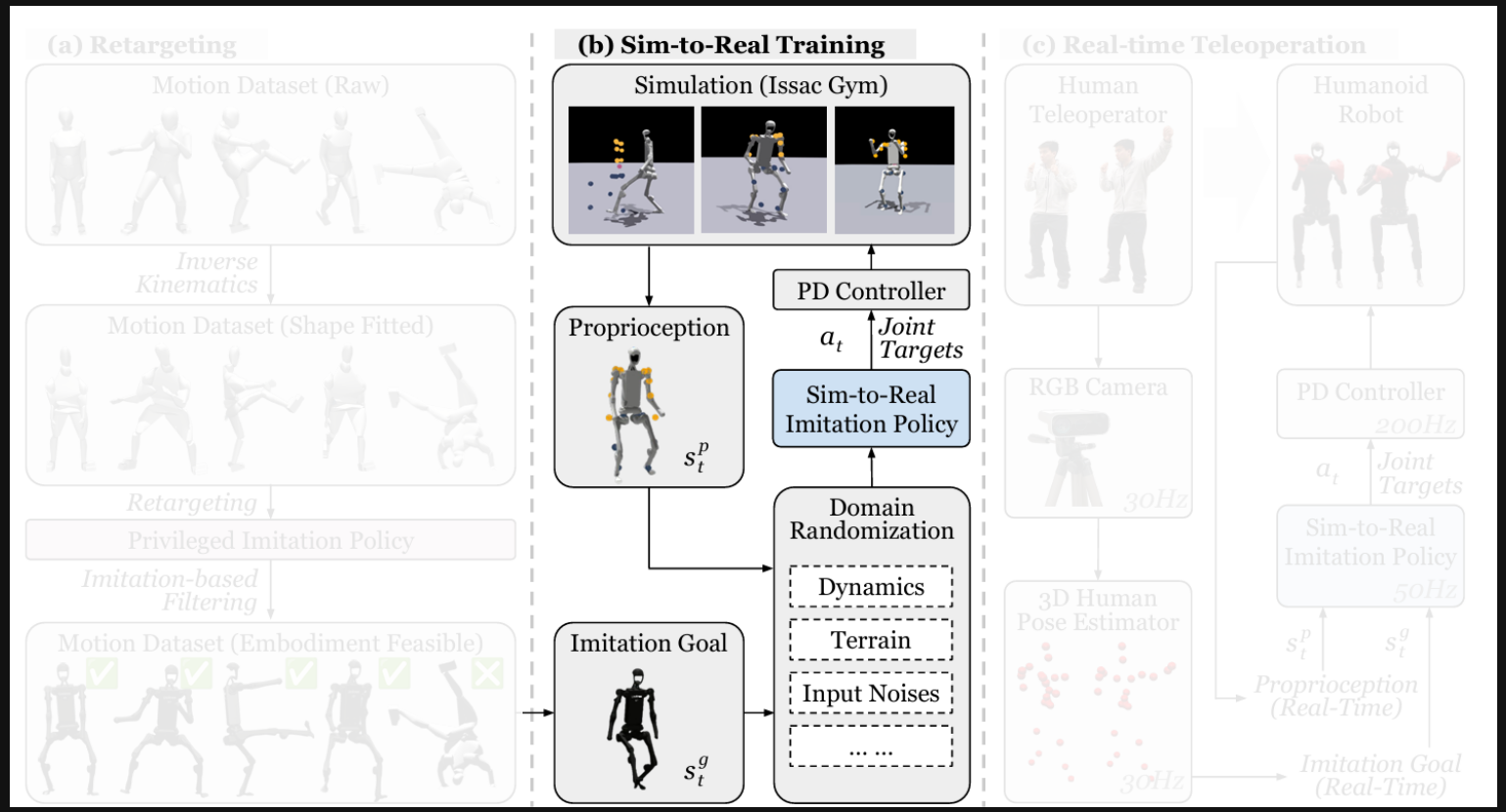
Human Motion Retargeting





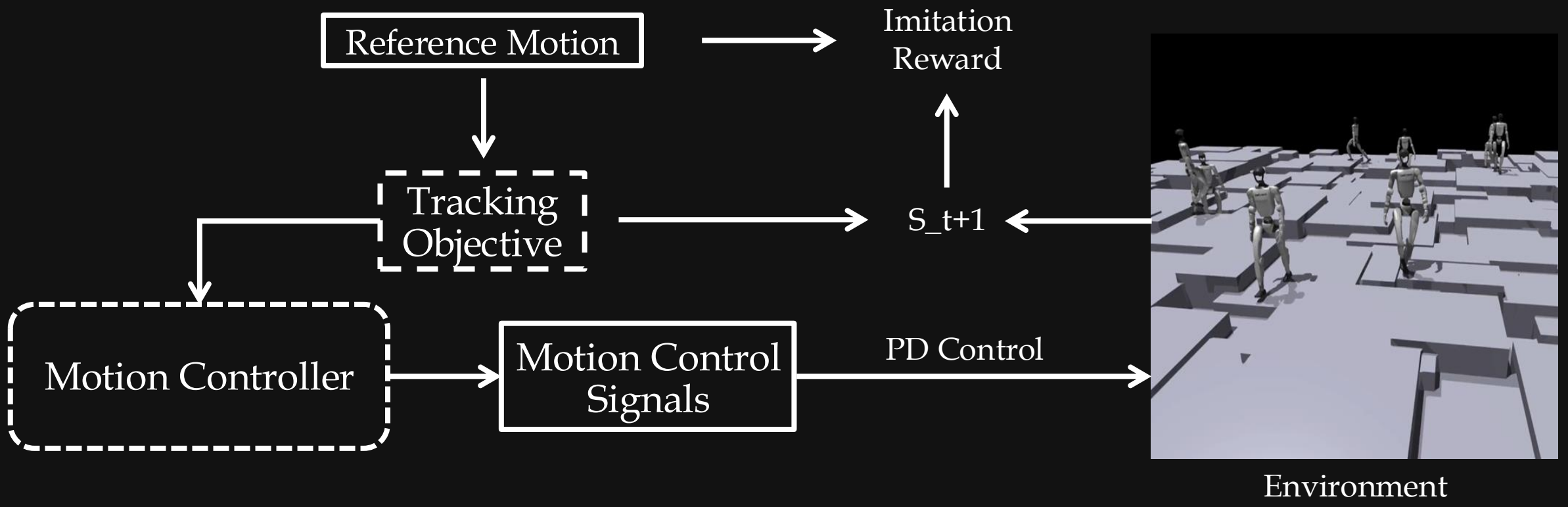
Case Study: Whole-Body Teleoperation

- Stage 2: Training in simulator

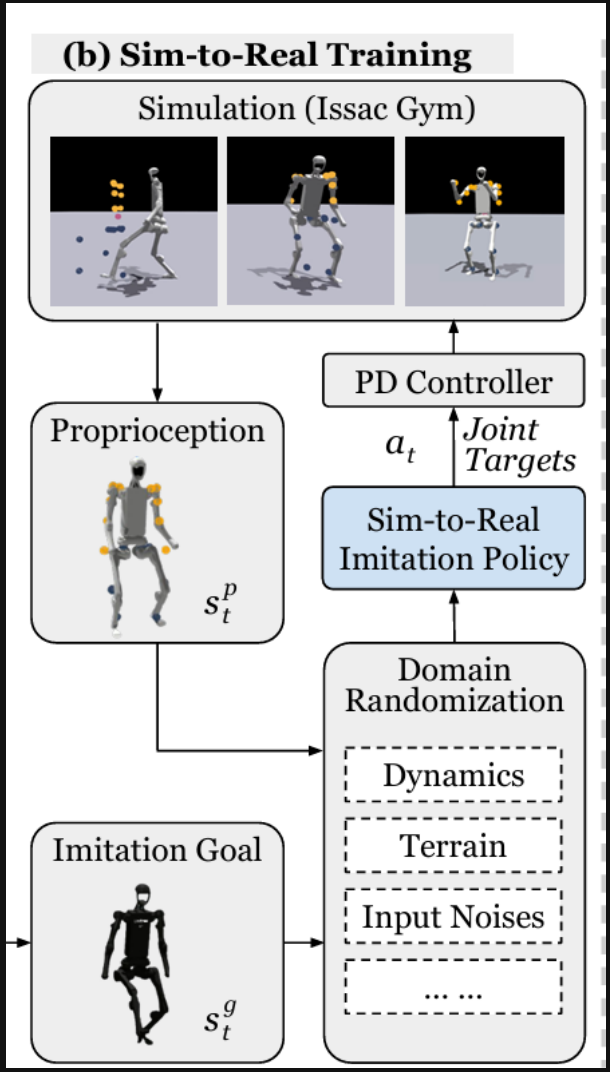


Typical Training Pipeline

- RL paradigm



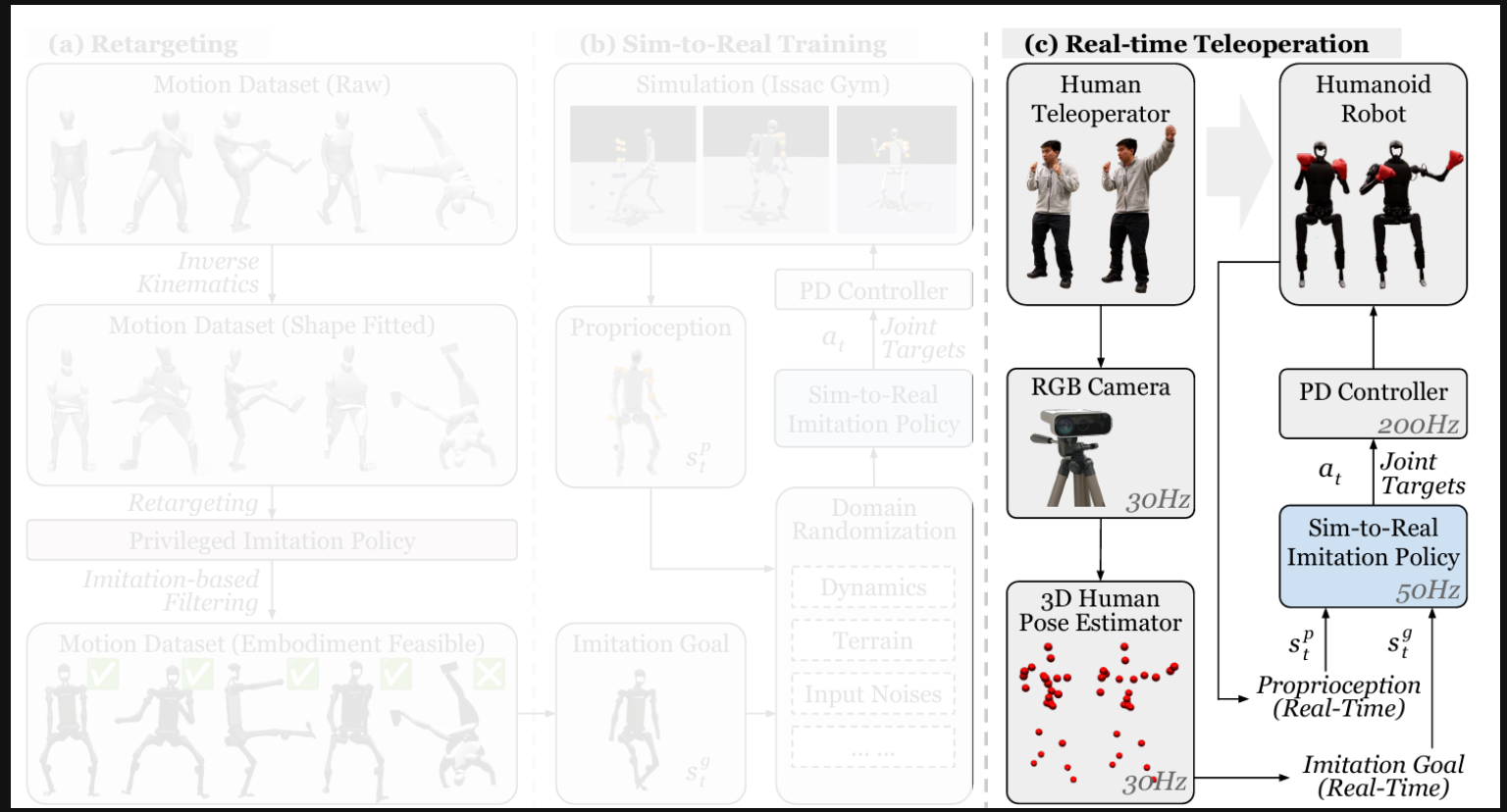
Case Study: Whole-Body Teleoperation



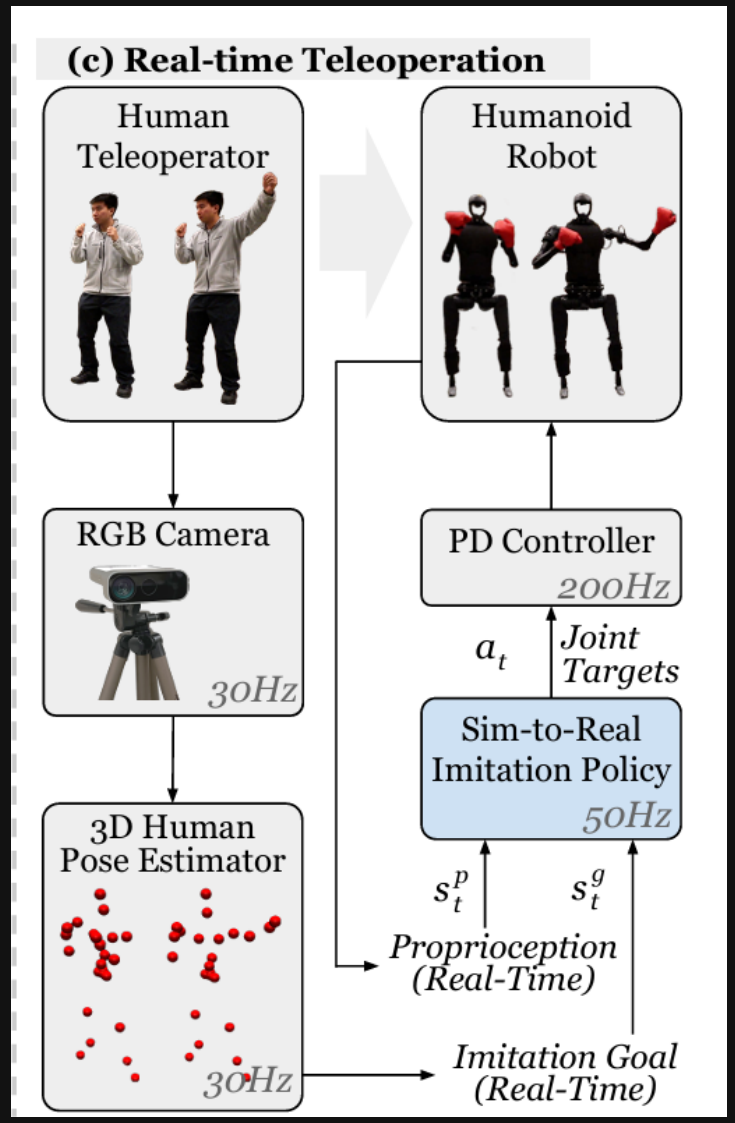
- During training, the policy learns to mimic the reference motion, while keeping "survival"
- Add domain randomization for sim-to-real transfer
  - We will discuss sim-to-real later

Case Study: Whole-Body Teleoperation

- Stage 3: Deployment in real world

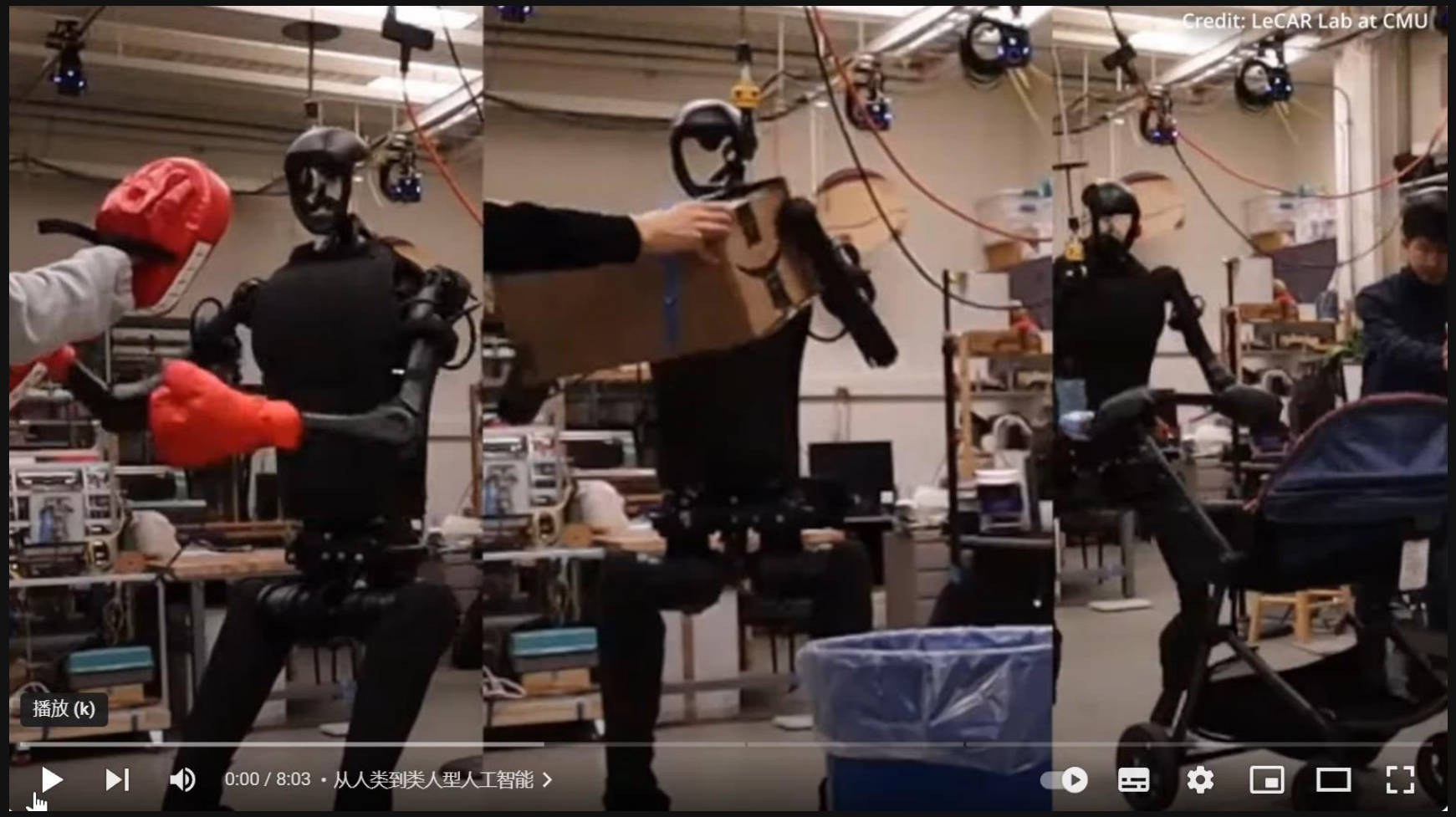


Case Study: Whole-Body Teleoperation



- Capture 3D human pose from a camera
- During inference, the policy takes robot states and human poses as input, and outputs control targets in real-time

Real Robot Results



The Agenda

- Section 1: Introduction

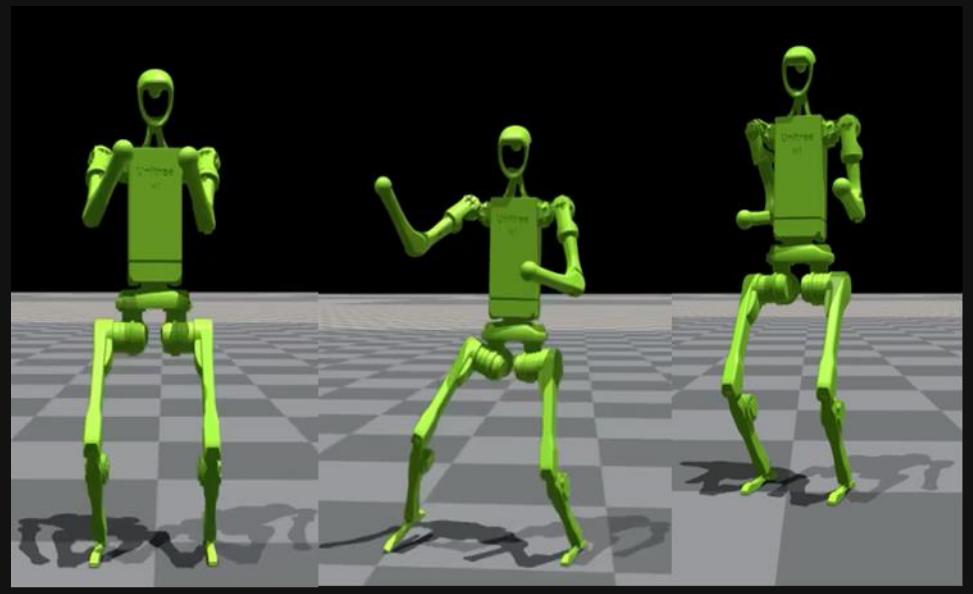
- Section 2: Traditional Humanoid Robot
  - Passive Dynamic Walking (PDW)
  - Zero Moment Point (ZMP) and Linear Inverted Pendulum Model (LIPM)

- Section 3: Learning-based Humanoid Robot
  - Reinforcement Learning & Imitation Learning
  - Motion Retargeting
  - Humanoid Teleoperation
  - Sim-to-real Transfer

Existing Problem: Sim-to-real Gap

- Why do we choose physical simulation …

  — —When there's always a gap between simulator and real world



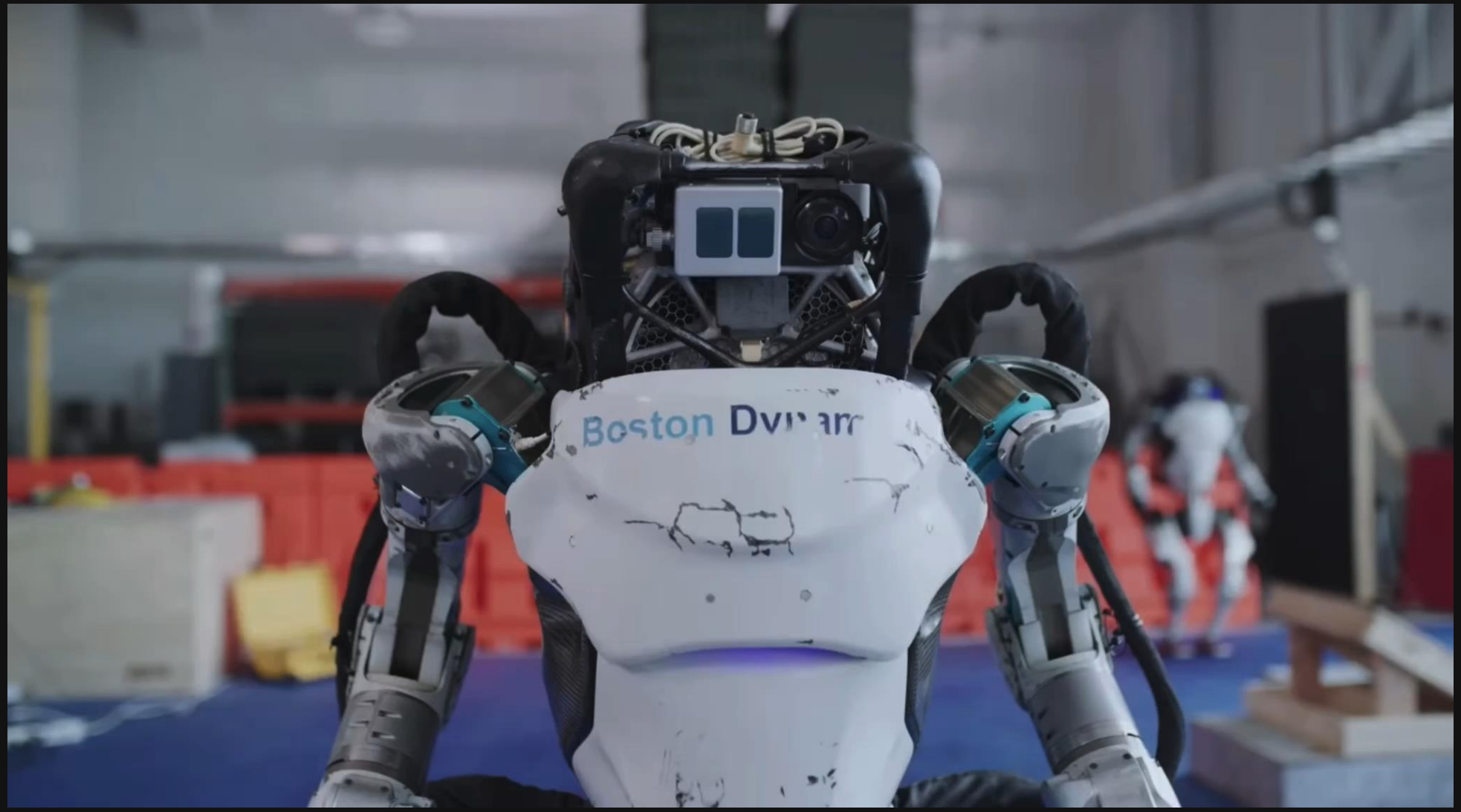
Simulator

Deploy



Real World

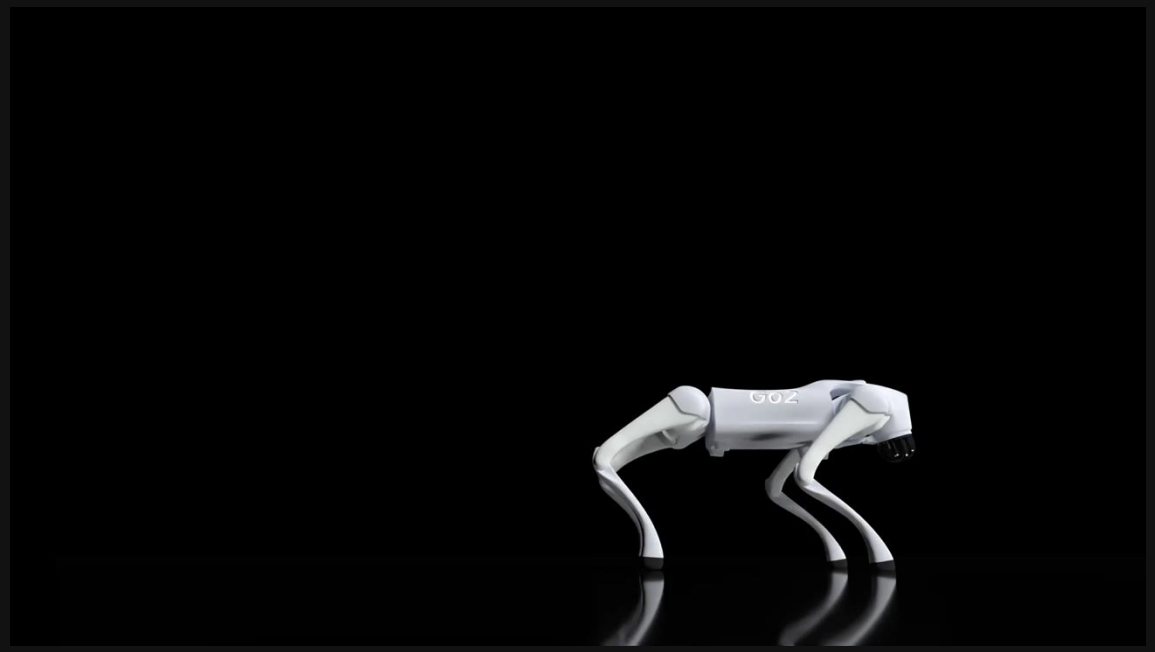
What is sim-to-real gap ?



What is sim-to-real gap ?

Hard to replicate the real world exactly

- Different terrain, mass, and friction
- Assume linear or idealized physics





What is the common cause of sim-to-real gap ?

Control Ideal, fixed frequencies (e.g., 1 kHz) are often used in simulations, but real machines have discrete implementations and delays, which lead to phase lag. This, in turn, can cause Kp and Kd to be very good in simulation, but overshoot/oscillation on the real machine

Actuation Friction, backlash and compliance break the ideal torque law, especially at low speed and reversal

Sensing Filtering & timing in the sensor chain reduce effective damping and add lag

Ontolog Inertial parameter errors and unmodeled flexible modes cause dynamic/static torque mismatch and resonances (URDF)

Numeric In reality, energy change stems from physical dissipation; discrete stepping may add numerical damping/drift. -> different dt will change the result

……

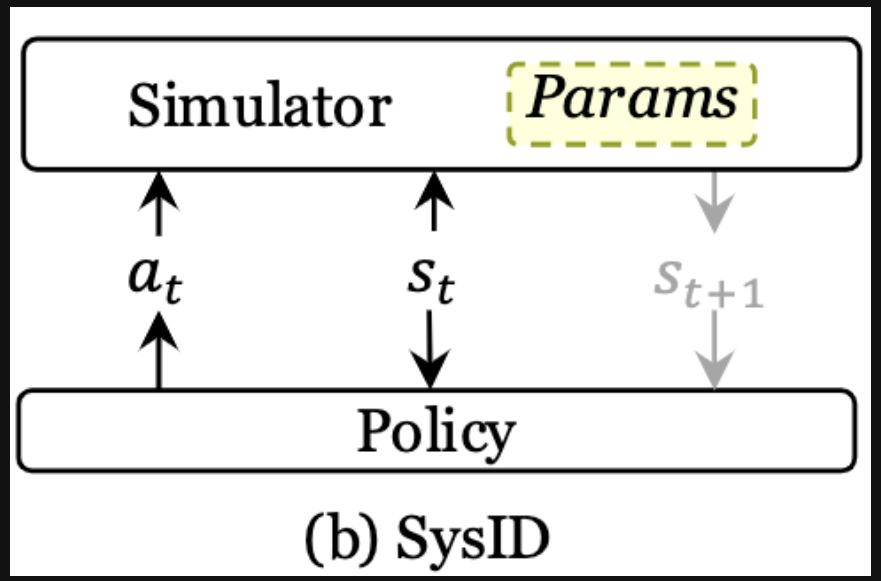
What Is sim-to-real Gap

- TLDR: Sim-to-real gap comes from
  - Environment physical parameter error
  - Motor error
  - Robot physical parameter error
  - Noisy observation
  - ……

Limited model capability + Sim-to-real gap = Failed real-world experiment

How to fix sim-to-real gap

- System Identification



Friction

$$\hat{\theta} = \arg\min_{\theta} \sum_{t \in \mathcal{W}_{\text{ls}}} [\tau(t) - \tau_{\text{fric}}(\dot{q}(t); \theta)]^2$$

Center of mass

$$\hat{\theta} = \arg\min_{\theta} \sum_{t \in \mathcal{W}_{\text{static}}} \|\tau(t) - g(q(t); \theta)\|^2$$

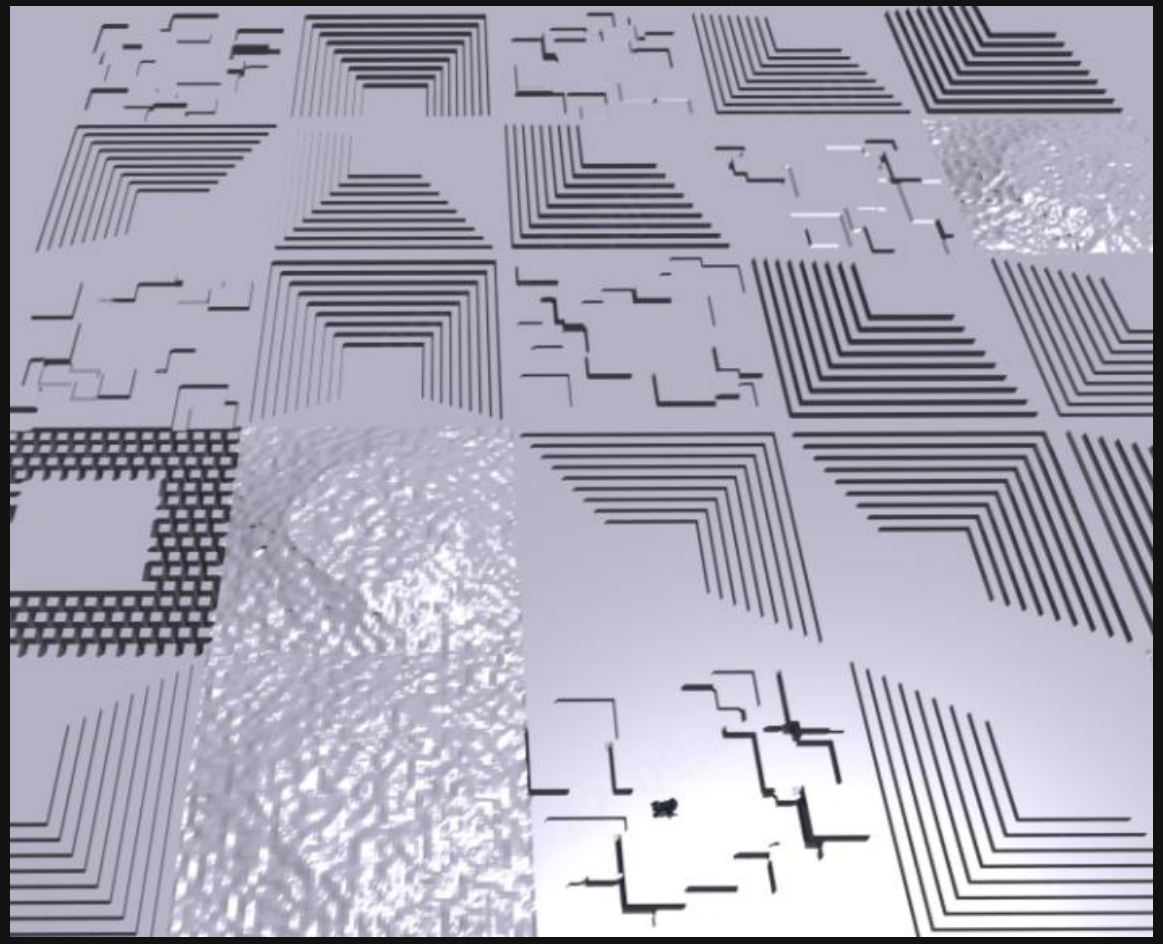
Delay

$$\hat{d} = \arg\max_{\Delta} \ \text{xcorr}\{u(t), y(t + \Delta)\}, \quad \text{delay} = \hat{d} T_s$$

Estimate physical parameters, such as
actuation/contact/inertial via experiments

How to fix sim-to-real gap

- Dynamics Randomization





May lead to inefficient training and limited scenarios

How to fix sim-to-real gap

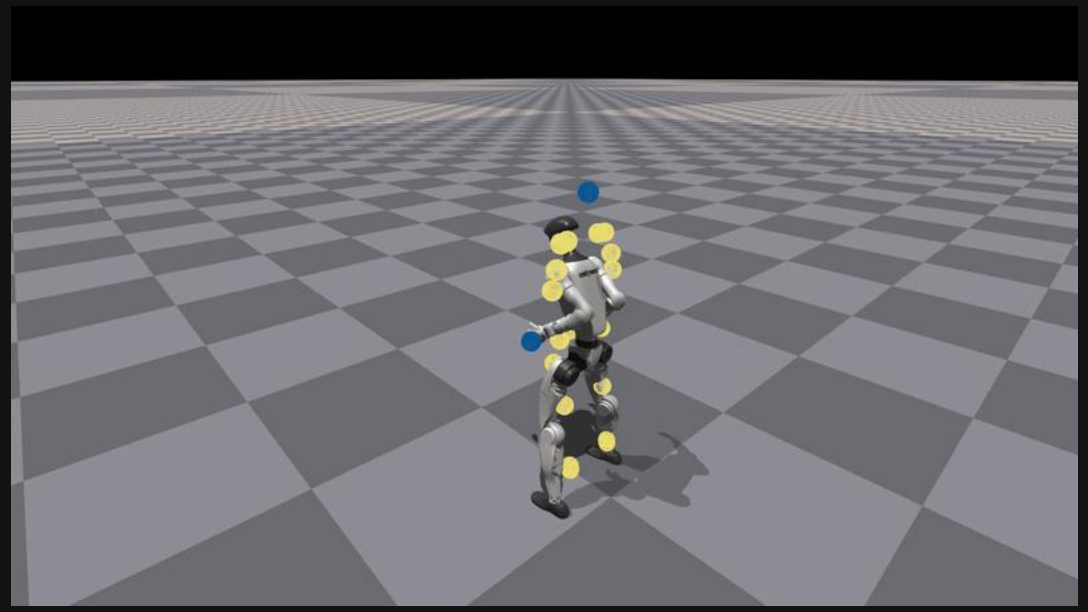
- Training a delta action model

ASAP: Aligning Simulation and Real-World Physics for Learning Agile Humanoid Whole-Body Skills

Tairan He[†1,2] Jiawei Gao[†1] Wenli Xiao[†1,2] Yuanhang Zhang[†1] Zi Wang[1] Jiashun Wang[1]
Zhengyi Luo[1,2] Guanqi He[1] Nikhil Sobanbab[1] Chaoyi Pan[1] Zeji Yi[1] Guannan Qu[1]
Kris Kitani[1] Jessica Hodgins[1] Linxi "Jim" Fan[2] Yuke Zhu[2] Changliu Liu[1] Guanya Shi[1]
[1]Carnegie Mellon University [2]NVIDIA [†]Equal Contributions

Page: https://agile.human2humanoid.com Code: https://github.com/LeCAR-Lab/ASAP

Robotics: Science and Systems (RSS) 2025



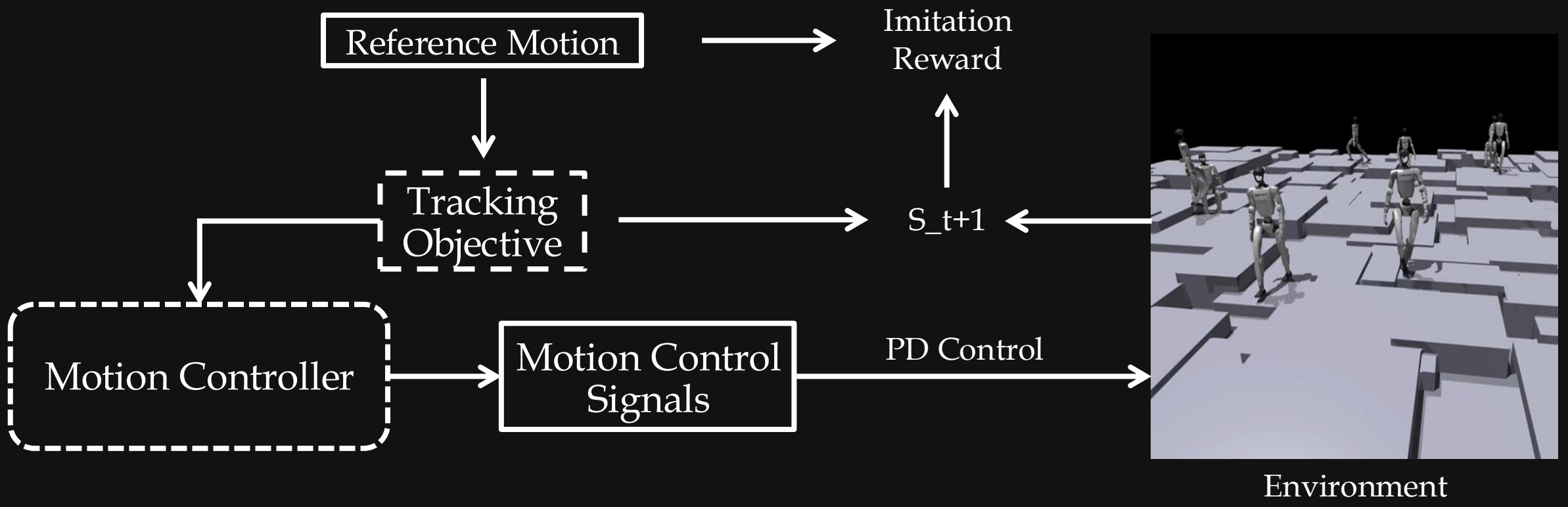

LAKERS
23


Emirates
FLY BETTER

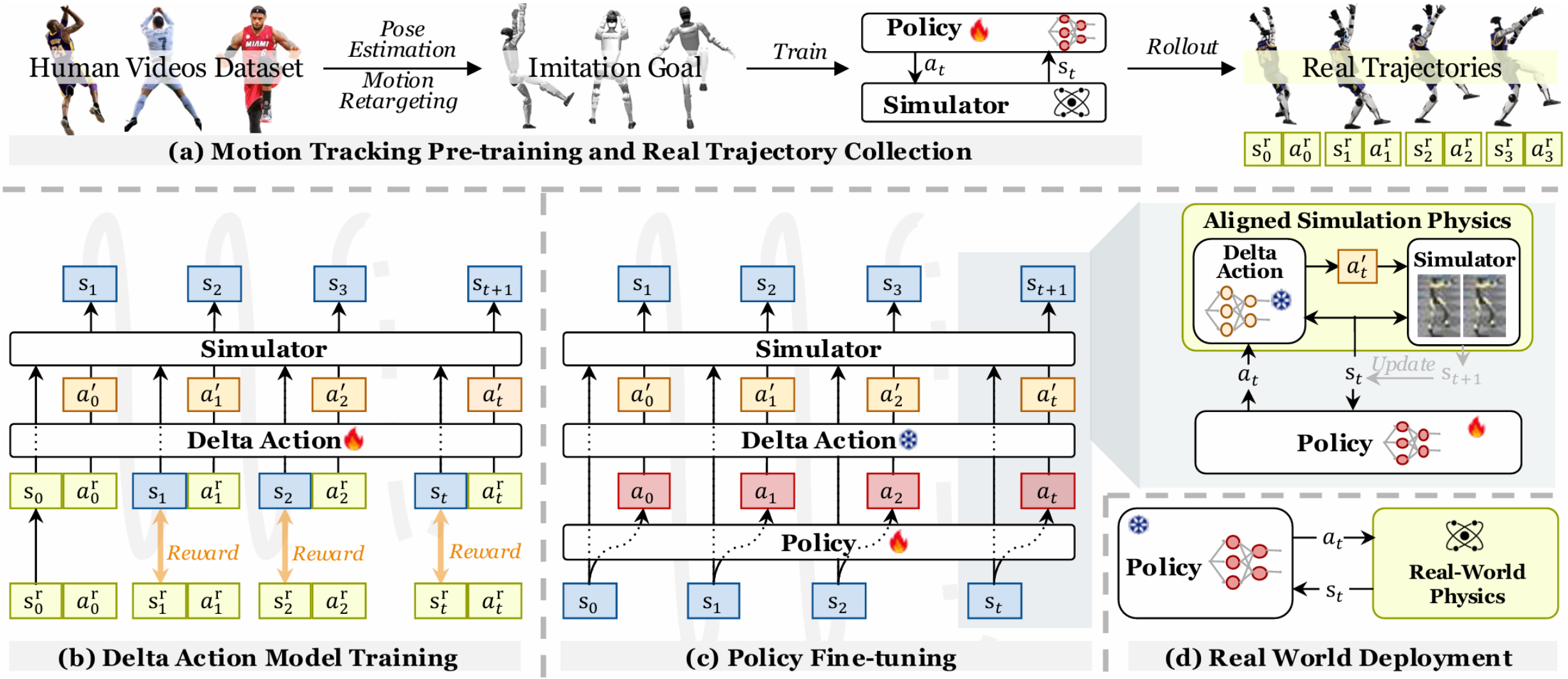

LAKERS
24

Recap: What Is Motion Controller

- RL paradigm



Method: Pretraining + Post-training



Pretraining: Dataset Collection

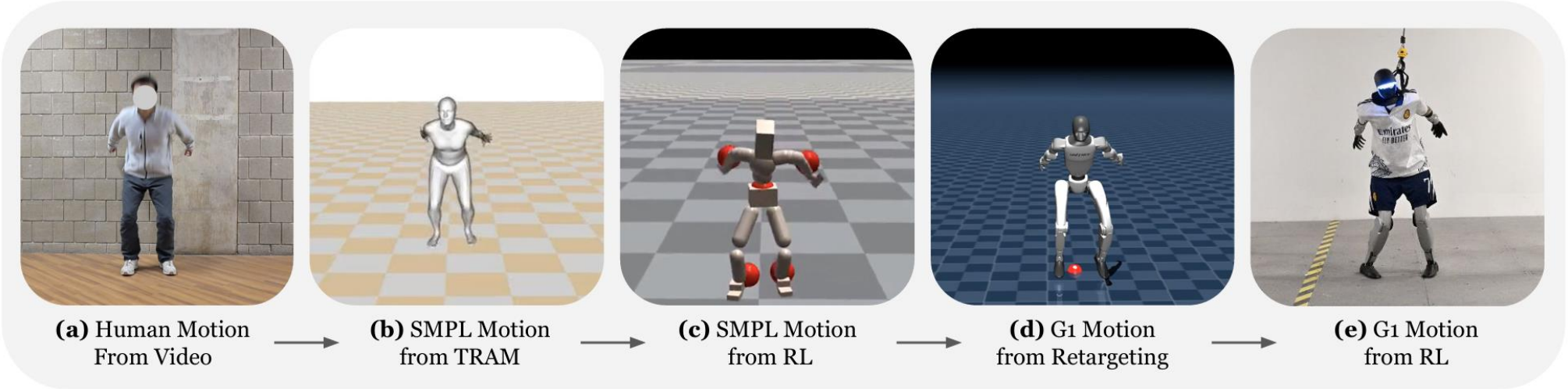


Fig. 3. Retargeting Human Video Motions to Robot Motions: (a) Human motions are captured from video. (b) Using TRAM [93], 3D human motion is reconstructed in the SMPL parameter format. (c) A reinforcement learning (RL) policy is trained in simulation to track the SMPL motion. (d) The learned SMPL motion is retargeted to the Unitree G1 humanoid robot in simulation. (e) The trained RL policy is deployed on the real robot, executing the final motion in the physical world. This pipeline ensures the retargeted motions remain physically feasible and suitable for real-world deployment.

- Reconstruct human motion from videos
- Retarget human motion on G1 robot

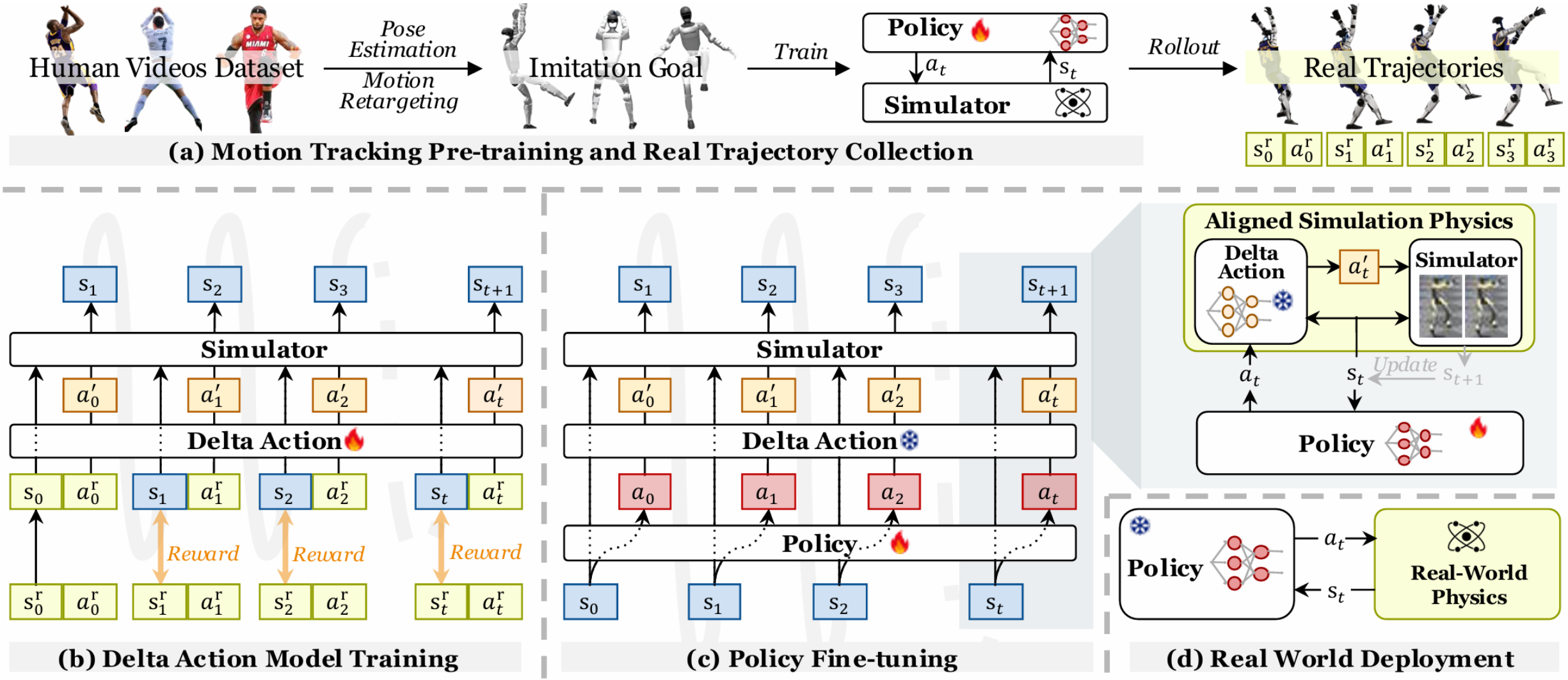
Pretraining: RL training

- Some training tricks
  - Asymmetric Actor-Critic Training (add privileged information in critic obs)
  - Termination Curriculum of Tracking Tolerance
  - Reference State Initialization
  - Domain Randomization
  - Reward Design

REWARD TERMS FOR PRETRAINING

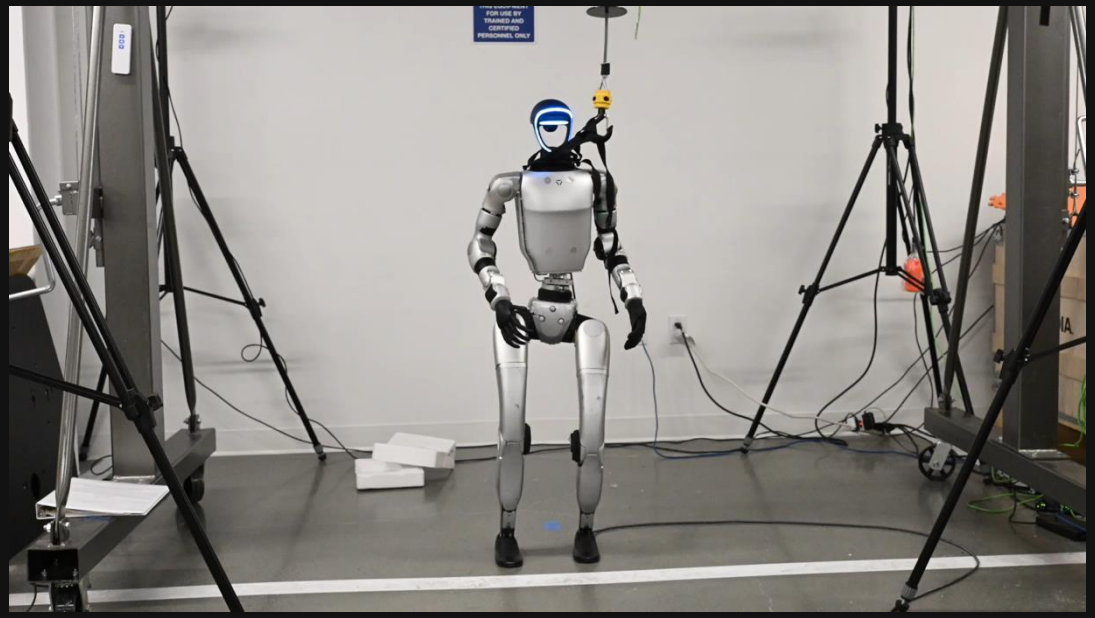
Term	Weight	Term	Weight
Penalty			
DoF position limits	-10.0	DoF velocity limits	-5.0
Torque limits	-5.0	Termination	-200.0
Regularization			
Torques	-1×10^{-6}	Action rate	-0.5
Feet orientation	-2.0	Feet heading	-0.1
Slippage	-1.0		
Task Reward			
Body position	1.0	VR 3-point	1.6
Body position (feet)	2.1	Body rotation	0.5
Body angular velocity	0.5	Body velocity	0.5
DoF position	0.75	DoF velocity	0.5

Post-training

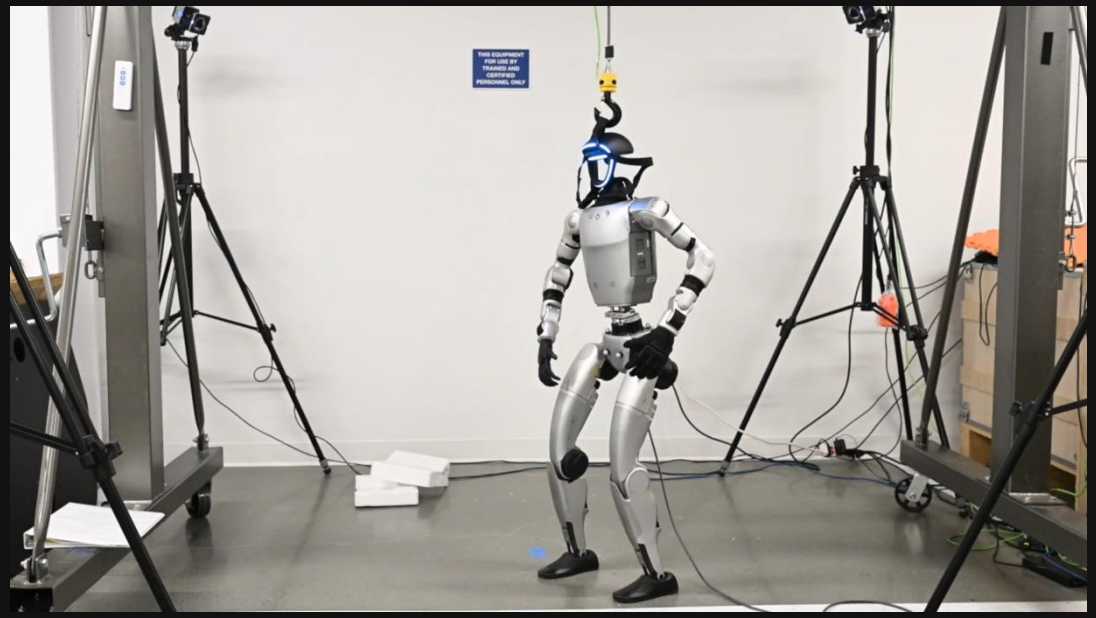


Post-training: Problem of Pretrained Policy

- The policy trained in the first stage can track the reference motion in the real-world, but does not achieve high motion quality

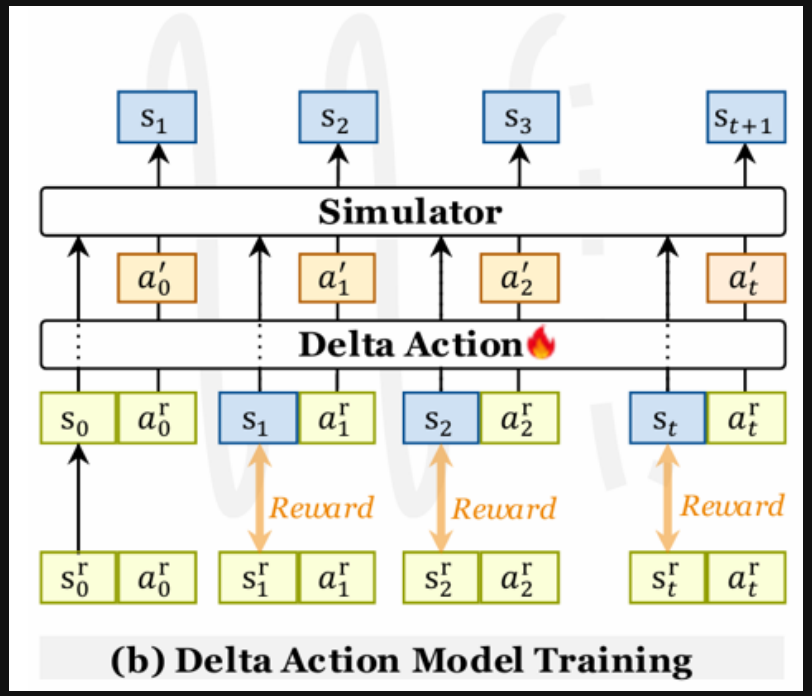


Pretraining-only

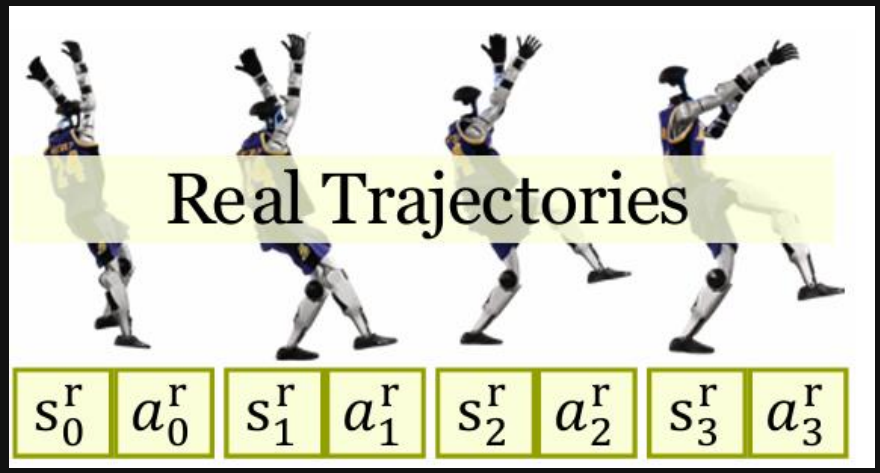


After post-training

Post-training: Training Delta Action Model



The goal is to minimize the discrepancy between simulated state and real-world state



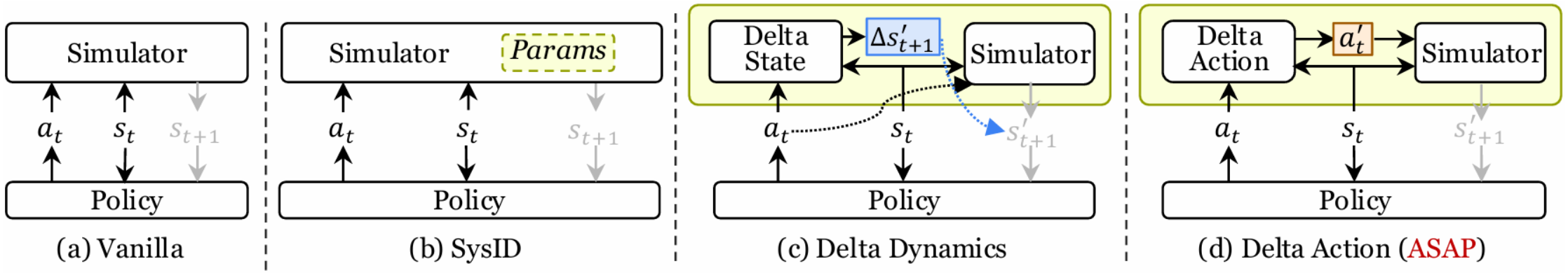
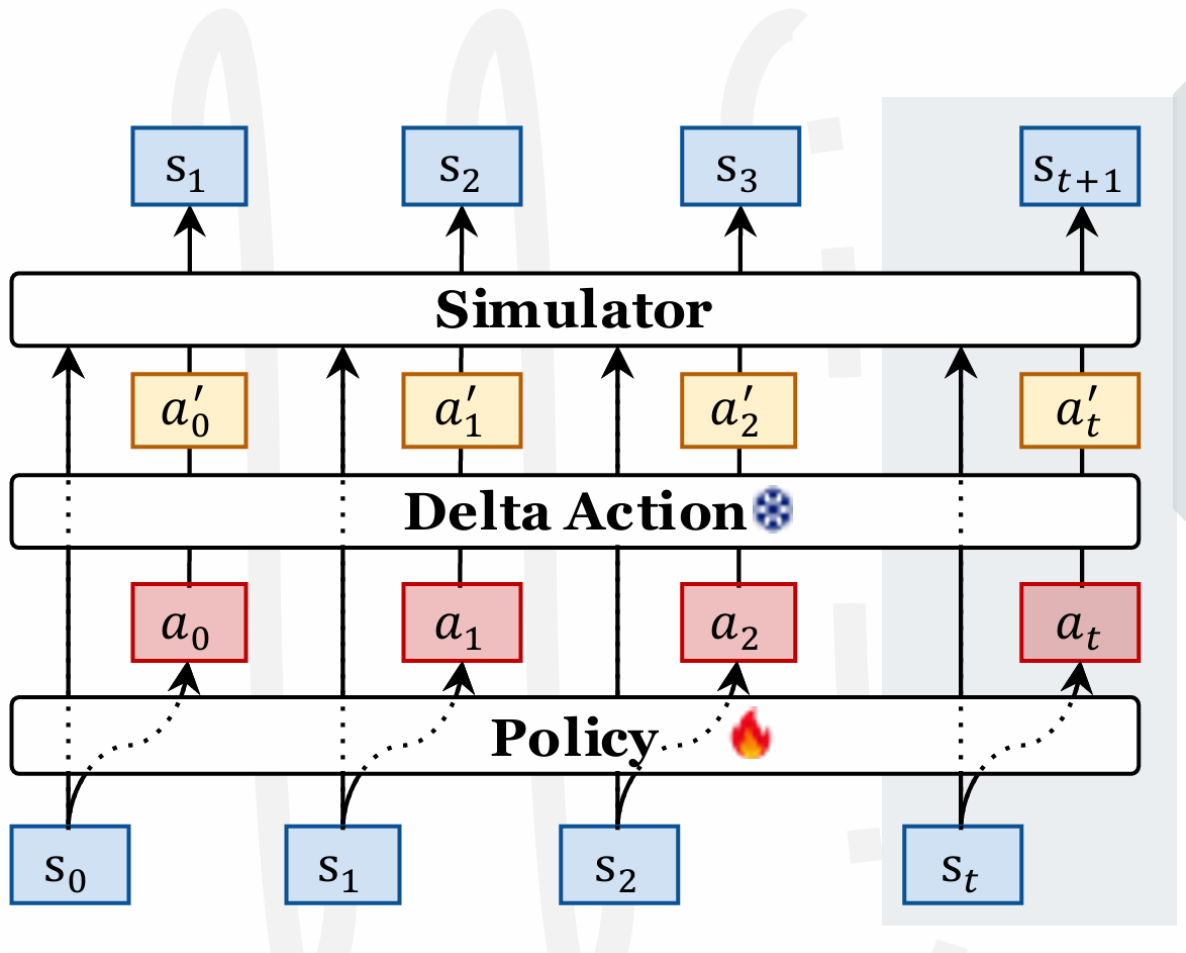


Fig. 4. Baselines of ASAP. (a) Model-free RL training. (b) System ID from real to sim using real-world data. (c) Learning delta dynamics model using real-world data. (d) Our proposed method, learning delta action model using real-world data.

Post-training: Fine-tuning Motion Tracking Policy



(c) Policy Fine-tuning

Freeze the delta action model and incorporate it into the simulator to align the real-world physics

Use this 'aligned' physical simulator to finetune motion tracking policy

Experiments

- **Q1**: Can ASAP outperform other baseline methods to compensate for the dynamics mismatch?
- **Q2**: Can ASAP finetune policy to outperform SysID and Delta Dynamics methods?
- **Q3**: Does ASAP work for sim-to-real transfer?

Experiments

- **Q1**: Can ASAP outperform other baseline methods to compensate for the dynamics mismatch?

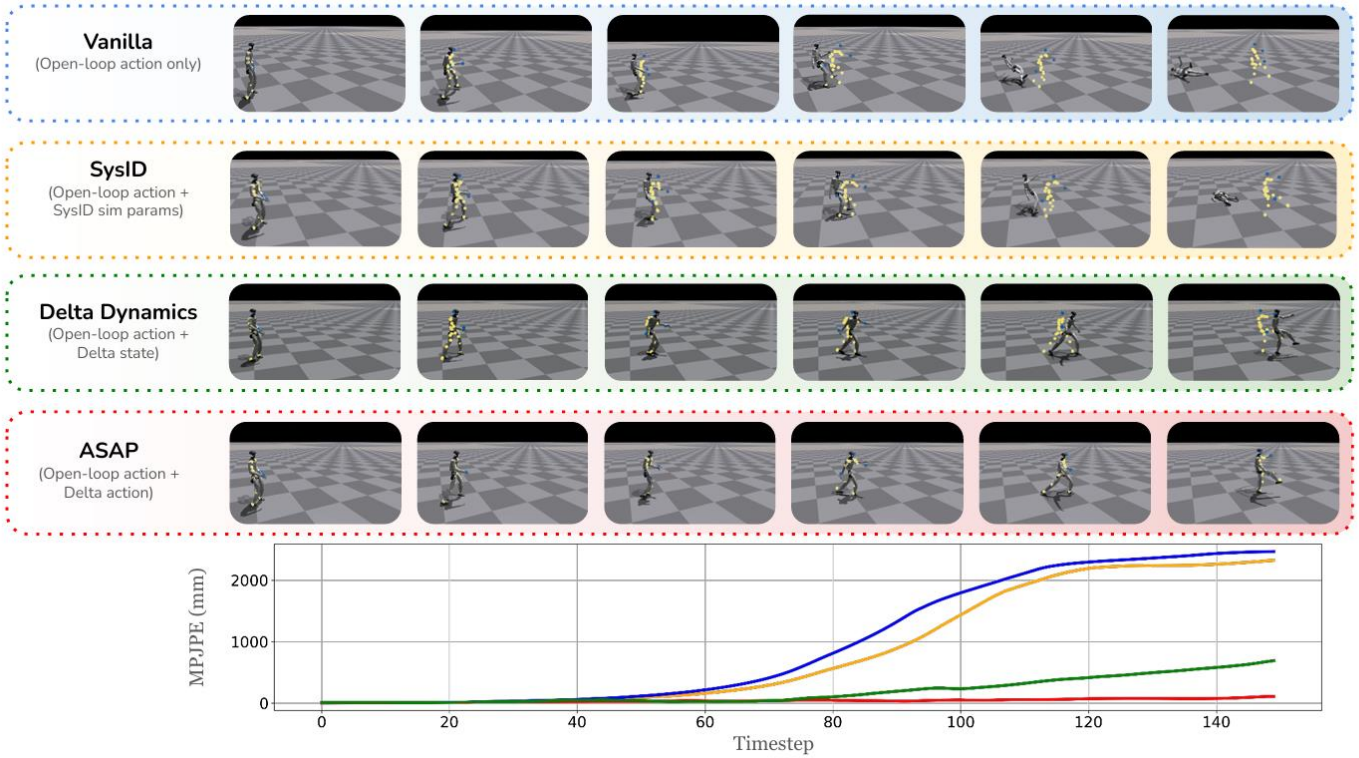


TABLE III
OPEN-LOOP PERFORMANCE COMPARISON ACROSS SIMULATORS AND MOTION LENGTHS.

Simulator & Length		IsaacSim				Genesis			
Length	Method	$E_{\text{g-mpjpe}}$	E_{mpjpe}	E_{acc}	E_{vel}	$E_{\text{g-mpjpe}}$	E_{mpjpe}	E_{acc}	E_{vel}
0.25s	OpenLoop	**19.5**	15.1	**6.44**	**5.80**	19.8	15.3	6.53	5.88
	SysID	**19.4**	15.0	**6.43**	**5.74**	19.3	15.0	6.42	5.73
	DeltaDynamics	24.4	**13.6**	9.43	7.85	20.0	**12.4**	8.42	6.89
	ASAP	**19.9**	15.6	**6.48**	**5.86**	**19.0**	14.9	**6.19**	**5.59**
0.5s	OpenLoop	33.3	23.2	6.80	6.84	33.1	23.0	6.78	6.82
	SysID	32.1	22.2	6.57	6.56	32.2	22.3	6.57	6.57
	DeltaDynamics	36.5	**16.4**	8.89	7.98	27.8	**14.0**	7.63	6.74
	ASAP	**26.8**	19.2	**5.09**	**5.36**	**25.9**	18.4	**4.93**	**5.19**
1.0s	OpenLoop	80.8	43.5	10.6	11.1	82.5	44.5	10.8	11.4
	SysID	77.6	41.5	10.2	10.7	76.5	41.6	10.0	10.5
	DeltaDynamics	68.1	**21.5**	9.61	9.14	50.2	**17.2**	8.19	7.62
	ASAP	**37.9**	22.9	**4.38**	**5.26**	**36.9**	22.6	**4.23**	**5.10**

Delta action model is better than other sim-to-real transfer methods

Experiments

- **Q2**: Can ASAP finetune policy to outperform SysID and Delta Dynamics methods?

TABLE IV

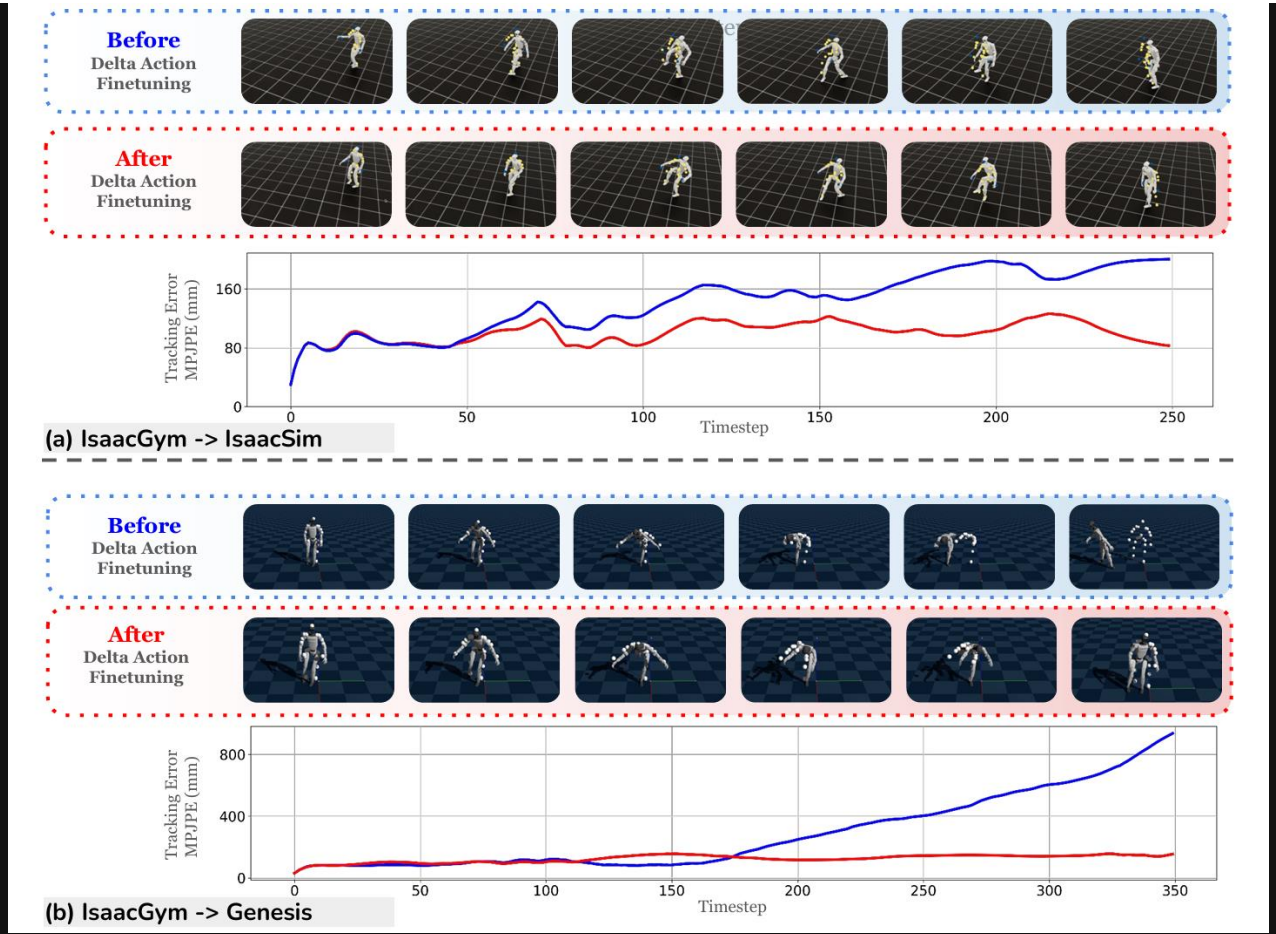
CLOSED-LOOP MOTION IMITATION EVALUATION ACROSS DIFFERENT SIMULATORS. ALL VARIANTS ARE TRAINED WITH IDENTICAL REWARDS.

	Test Environment	IsaacSim					Genesis				
Level	Method	Succ ↑	$E_{\text{g-mpjpe}}$ ↓	E_{mpjpe} ↓	E_{acc} ↓	E_{vel} ↓	Succ ↑	$E_{\text{g-mpjpe}}$ ↓	E_{mpjpe} ↓	E_{acc} ↓	E_{vel} ↓
Easy	Oracle (IsaacGym → IsaacGym)	100%±0.000%	97.5±0.605	43.2±0.112	2.56±0.024	4.48±0.023	100%±0.000%	97.5±0.605	43.2±0.112	2.56±0.024	4.48±0.023
	Vanilla (IsaacGym → TestEnv)	100%±0.000%	**107** ±0.578	45.4±0.169	2.83±0.012	4.59±0.021	100%±0.000%	140±1.85	**70.1**±0.626	2.68±0.042	4.65±0.046
	SysID	100%±0.000%	**105** ±1.35	47.8±0.970	3.09±0.011	4.98±0.020	100%±0.000%	**127**±0.233	79.9±0.330	2.99±0.035	4.95±0.012
	DeltaDynamics	100%±0.000%	127±2.97	56.7±0.390	3.50±0.028	5.56±0.031	83.3%±0.000%	168±7.62	87.0±1.51	3.08±0.18	5.39±0.34
	ASAP	100%±0.000%	**106** ±0.498	**44.3** ±0.103	**2.74** ±0.025	**4.46** ±0.020	100%±0.000%	**125**±4.75	73.5±0.570	**2.10**±0.083	**4.11**±0.133
Medium	Oracle (IsaacGym → IsaacGym)	100%±0.000%	111±0.635	48.8±0.133	2.63±0.017	4.82±0.019	100%±0.000%	111±0.635	48.8±0.133	2.63±0.017	4.82±0.019
	Vanilla (IsaacGym → TestEnv)	100%±0.000%	**114**±0.720	**49.2**±0.104	2.92±0.021	5.07±0.016	94.3%±7.00%	169±5.76	72.0±0.692	3.26±0.076	5.86±0.101
	SysID	100%±0.000%	**115**±1.256	**49.1**±0.560	3.43±0.021	5.01±0.017	100%±0.000%	138±2.70	75.4±1.18	3.14±0.042	5.50±0.058
	DeltaDynamics	83.3%±0.000%	151±2.62	68.0±0.364	2.90±0.047	5.90±0.107	83.3%±0.000%	190±1.46	89.4±0.50	3.44±0.16	7.49±0.11
	ASAP	100%±0.000%	**112** ±1.648	**49.3** ±0.574	**2.53** ±0.019	**4.45** ±0.026	100%±0.000%	**126**±1.63	**71.2**±0.163	**2.81**±0.037	**5.13**±0.066
Hard	Oracle (IsaacGym → IsaacGym)	100%±0.000%	116±0.711	52.5±0.298	3.40±0.027	6.16 ±0.028	100%±0.000%	116±0.711	52.5±0.298	3.40±0.027	6.16 ±0.028
	Vanilla (IsaacGym → TestEnv)	100%±0.000%	148±0.845	**51.6**±0.137	4.41±0.055	6.88±0.064	82.9%±5.70%	175±9.77	80.7±1.69	3.87±0.175	7.19±0.199
	SysID	100%±0.000%	165±3.83	58.4±0.229	4.87±0.197	7.13±0.131	100%±0.000%	186±3.84	93.0±1.49	4.98±0.245	8.98±0.119
	DeltaDynamics	66.7%±0.000%	137±2.59	60.2±0.477	4.20±0.041	7.10±0.024	60.0%±5.70%	190±14.0	89.6±9.34	4.29±1.16	8.70±2.33
	ASAP	100%±0.000%	**129** ±1.57	56.5 ±1.15	**3.72** ±0.036	**6.52** ±0.042	100%±0.000%	**129**±2.31	**77.0**±1.07	**2.69**±0.040	**5.65**±0.073

The policy performs better after finetuned with delta action model

Experiments

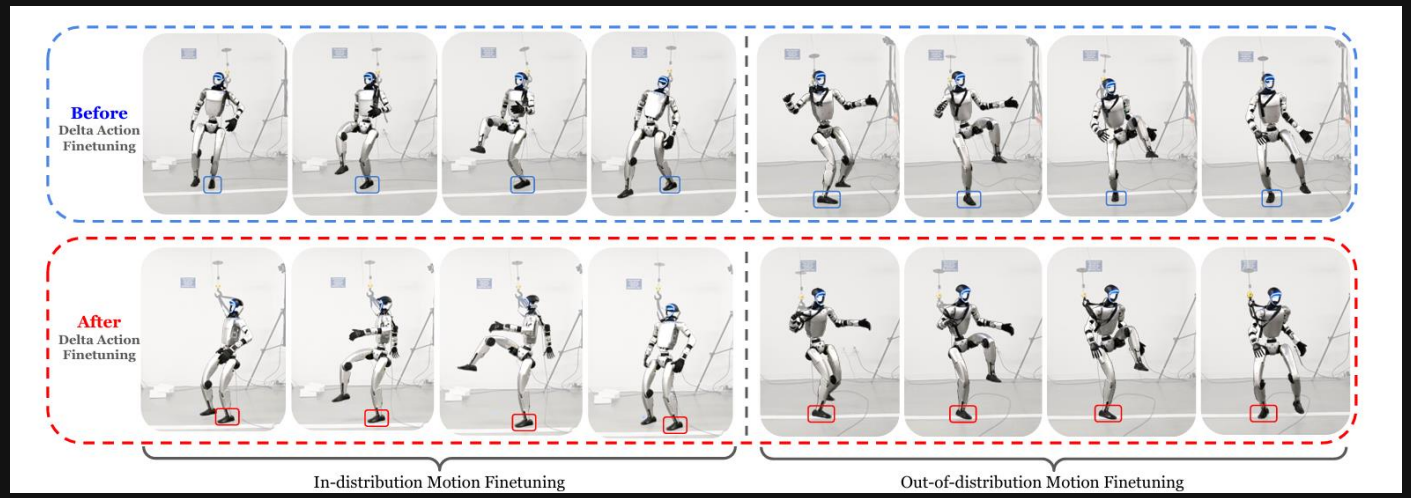
- **Q2**: Can ASAP finetune policy to outperform SysID and Delta Dynamics methods?



The tracking error of the policy is not accumulated after delta action finetuning

Experiments

- **Q3**: Does ASAP work for sim-to-real transfer?



The motion quality generally enhanced with ASAP

TABLE V

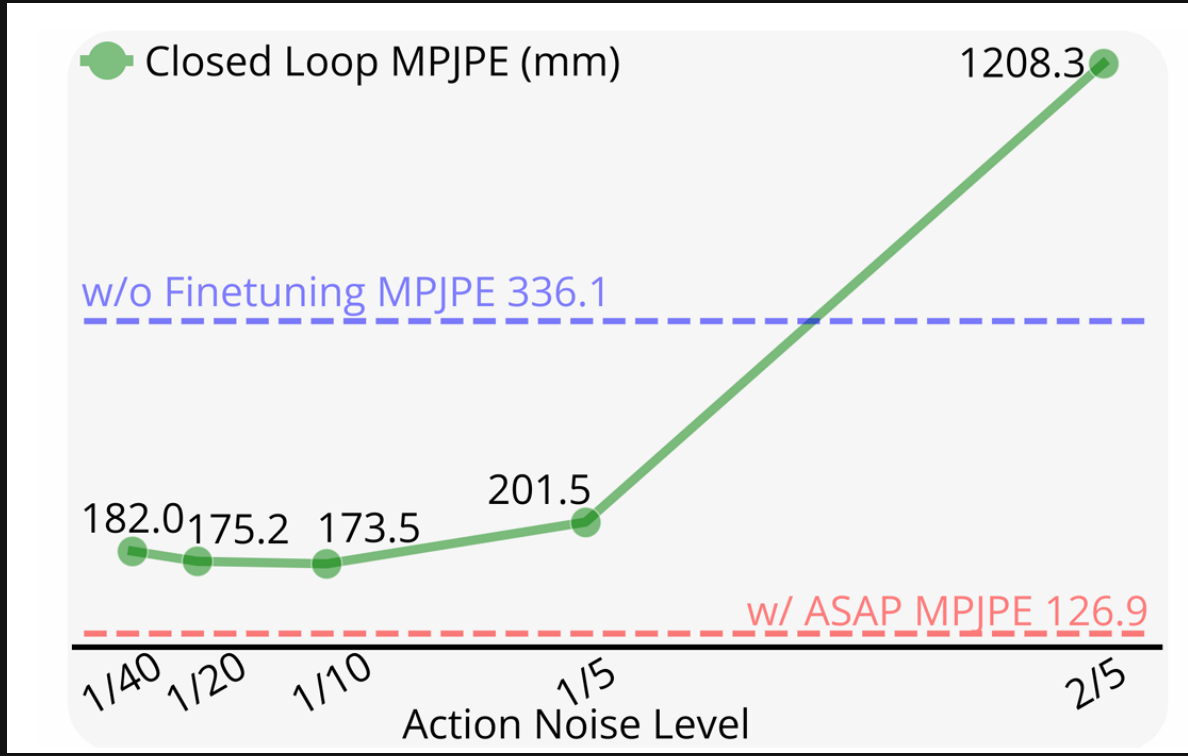
REAL-WORLD CLOSED-LOOP PERFORMANCE COMPARING WITH AND WITHOUT ASAP FINETUNING ON ONE IN-DISTRIBUTION MOTION AND ONE OUT-OF-DISTRIBUTION MOTION.

Motion	Real-World-Kick				Real-World-LeBron (OOD)			
Method	$E_{\text{g-mpjpe}}$	E_{mpjpe}	E_{acc}	E_{vel}	$E_{\text{g-mpjpe}}$	E_{mpjpe}	E_{acc}	E_{vel}
Vanilla	61.2	43.5	2.96	2.91	159	55.3	3.43	6.43
ASAP	**50.2**	**40.1**	**2.46**	**2.70**	**112**	**47.5**	**2.84**	**5.94**

The motion is more stable and the tracking error is lower

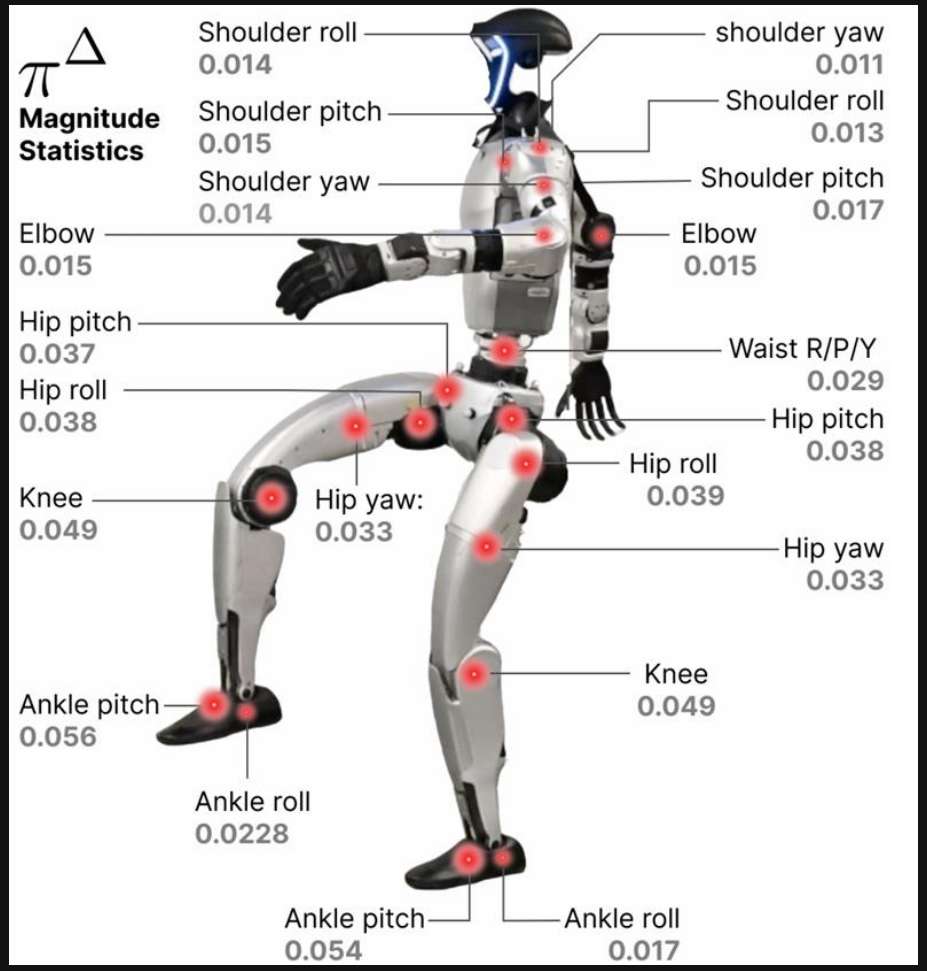
Experiments

- Delta action model is more than random action noise
  - Similar to what we have done in domain randomization, we can add random noise on actions instead of using delta action model
  - Adding random noise during training can enhance the robustness of the policy, but not as good as finetuning with ASAP



Experiments

- Delta action model is more than random action noise



For the vulnerable joints such as ankles, the action delta model learns a larger ‘delta’ value, which indicates these joints suffer a larger sim-to-real gap

Discussion: Factors Affecting Humanoid Performance

Humanoid Performance = max[0, (Robot Ontology - 1)] × Control Algorithm × Engineering Capability
where:

Robot Ontology = Hardware Reliability × Mechanical Design
Robot Ontology == 1 indicates that the robot can be used
Control Algorithm = Algorithmic Framework × Computational Capability

Discussion: Factors Affecting Humanoid Performance

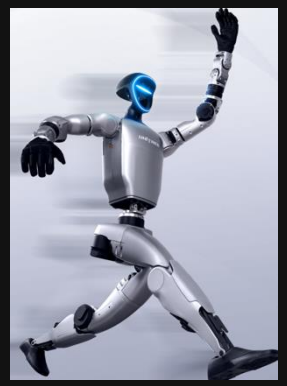


= (Excellent Robot Ontology - 1) × Unremarkable Control Algorithm × Excellent Engineering Capability



= (Excellent Robot Ontology - 1) × Good Control Algorithm × Unbeatable Engineering Capability

- When evaluating the performance of a robot, attention should be paid to the robot ontology, algorithm and engineering effort.
- The performance of the demo alone is not enough to judge whether a robot is good or bad.



= (Excellent Robot Ontology - 1) × Excellent Control Algorithm × Good Engineering Capability

Thanks for your listening and sleeping